

DTU

Danmarks Tekniske Universitet

Department of Civil Engineering

Multi-Sample Comparison Report

Hydraulic Conductivity Analysis - 51 Samples

Date: July 10, 2026

Multi-Sample Comparison Report

Report Date: July 10, 2026 at 10:46

Number of Samples: 51

Temperature: 20 °C

Porosity: Varies by sample (0.267 to 0.434)

Executive Summary

Comparison Analysis: This report compares 51 soil samples based on grain size distribution and hydraulic conductivity estimates.

Key Findings:

- Highest permeability: B102_21 (7.93e-04 m/s)
- Lowest permeability: B101_13 (4.96e-05 m/s)
- Permeability range: 16.0x difference

Methodology

Comparative Analysis Approach

This comparison report presents a side-by-side analysis of multiple soil samples to identify patterns, variations, and relationships between grain size characteristics and hydraulic conductivity estimates.

Analysis Components

Grain Size Comparison: Overlapping distribution curves allow visual assessment of particle size variations between samples.

K-Value Comparison: Box plots and statistical summaries reveal the range and reliability of hydraulic conductivity estimates across samples.

Method Reliability: A reliability matrix shows which empirical methods are applicable for each sample, helping identify the most suitable estimation approaches.

Results & Analysis

Sample Overview

Sample	Temp (?C)	Porosity	D10 (mm)	D50 (mm)	D60 (mm)	Cu
B106_1	20.00	0.417	0.141	0.295	0.341	2.42
B106_2	20.00	0.417	0.149	0.317	0.360	2.42
B106_3	20.00	0.419	0.079	0.164	0.186	2.36
B106_4	20.00	0.417	0.148	0.322	0.358	2.42
B106_5	20.00	0.419	0.145	0.303	0.343	2.37
B106_6	20.00	0.417	0.082	0.180	0.201	2.44
B106_7	20.00	0.4	N/A	0.140	0.165	N/A

Sample	Temp (°C)	Porosity	D10 (mm)	D50 (mm)	D60 (mm)	Cu
B101_1	20.00	0.395	0.095	0.229	0.305	3.20
B101_3	20.00	0.4	N/A	0.211	0.244	N/A
B101_5	20.00	0.425	0.092	0.182	0.199	2.17
B101_7	20.00	0.4	N/A	0.220	0.330	N/A
B101_9	20.00	0.4	N/A	0.181	0.206	N/A
B101_11	20.00	0.434	0.119	0.204	0.225	1.89
B101_13	20.00	0.4	N/A	0.094	0.103	N/A
B101_15	20.00	0.4	N/A	0.096	0.110	N/A
B101_17	20.00	0.4	N/A	0.082	0.097	N/A
B101_19	20.00	0.418	0.067	0.137	0.161	2.41
B101_21	20.00	0.426	0.085	0.165	0.182	2.15
B101_23	20.00	0.424	0.078	0.153	0.173	2.21
B101_25	20.00	0.429	0.093	0.175	0.191	2.06
B101_27	20.00	0.428	0.098	0.186	0.204	2.08
B102_13	20.00	0.407	0.074	0.186	0.206	2.77
B102_15	20.00	0.407	0.173	0.420	0.479	2.77
B102_17	20.00	0.406	0.147	0.371	0.415	2.82
B102_19	20.00	0.407	0.085	0.211	0.238	2.79
B102_21	20.00	0.394	0.260	0.730	0.849	3.26
B102_23	20.00	0.396	0.186	0.490	0.591	3.18
B102_25	20.00	0.434	0.127	0.220	0.243	1.91
B103_8	20.00	0.4	N/A	0.205	0.229	N/A
B103_10	20.00	0.408	0.131	0.323	0.360	2.75
B103_12	20.00	0.316	0.135	0.615	1.035	7.67
B103_14	20.00	0.385	0.084	0.248	0.303	3.63
B103_16	20.00	0.432	0.121	0.215	0.238	1.97
B103_18	20.00	0.415	0.142	0.321	0.358	2.51
B104_9	20.00	0.429	0.198	0.369	0.404	2.04
B104_11	20.00	0.425	0.170	0.340	0.372	2.18
B104_13	20.00	0.408	0.132	0.327	0.362	2.75
B104_15	20.00	0.423	0.166	0.339	0.372	2.23
B104_17	20.00	0.418	0.138	0.287	0.330	2.40
B104_19	20.00	0.407	0.109	0.252	0.302	2.78
B104_21	20.00	0.411	0.129	0.295	0.339	2.63
B105_10	20.00	0.379	0.105	0.358	0.408	3.88
B105_12	20.00	0.267	0.238	2.969	3.858	16.24

Sample	Temp (?C)	Porosity	D10 (mm)	D50 (mm)	D60 (mm)	Cu
B105_14	20.00	0.407	0.132	0.325	0.364	2.76
B105_16	20.00	0.41	0.172	0.410	0.461	2.68
B105_18	20.00	0.417	0.151	0.332	0.368	2.44
B105_20	20.00	0.414	0.139	0.314	0.352	2.52
B105_22	20.00	0.414	0.134	0.302	0.342	2.55
B105_24	20.00	0.421	0.131	0.253	0.303	2.31
B105_26	20.00	0.42	0.153	0.323	0.359	2.35
B105_28	20.00	0.418	0.144	0.310	0.348	2.42

Sample	Soil type	Descriptor	K geometric mean (m/s)
B106_1	Poorly-graded sand	Moderately well sorted sand low in fines	2.41e-04
B106_2	Poorly-graded sand	Moderately well sorted sand low in fines	2.72e-04
B106_3	Poorly-graded sand	Moderately well sorted sand low in fines	7.68e-05
B106_4	Poorly-graded sand	Moderately well sorted sand low in fines	2.74e-04
B106_5	Poorly-graded sand	Moderately well sorted sand low in fines	2.55e-04
B106_6	Poorly-graded sand	Moderately well sorted sand low in fines	8.90e-05
B106_7	Sand	Clayey sand with fines	7.99e-05
B101_1	Poorly-graded sand	Moderately well sorted gravelly sand low in fines	1.13e-04
B101_3	Sand	Clayey gravelly sand with fines	1.51e-04
B101_5	Poorly-graded sand	Moderately well sorted sand with fines	1.14e-04
B101_7	Sand	Clayey gravelly sand with fines	1.33e-04
B101_9	Sand	Clayey sand with fines	1.14e-04
B101_11	Poorly-graded sand	Uniform sand low in fines	1.74e-04
B101_13	Sand	Clayey sand with fines	4.96e-05
B101_15	Sand	Clayey sand with fines	8.65e-05
B101_17	Sand	Clayey sand with fines	9.37e-05

Sample	Soil type	Descriptor	K geometric mean (m/s)
B101_19	Poorly-graded sand	Moderately well sorted sand with fines	5.86e-05
B101_21	Poorly-graded sand	Moderately well sorted sand low in fines	8.80e-05
B101_23	Poorly-graded sand	Moderately well sorted sand low in fines	7.46e-05
B101_25	Poorly-graded sand	Moderately well sorted sand low in fines	1.08e-04
B101_27	Poorly-graded sand	Moderately well sorted sand low in fines	1.21e-04
B102_13	Poorly-graded sand	Moderately well sorted gravelly sand with fines	8.37e-05
B102_15	Poorly-graded sand	Moderately well sorted gravelly sand low in fines	3.79e-04
B102_17	Poorly-graded sand	Moderately well sorted sand low in fines	2.86e-04
B102_19	Poorly-graded sand	Moderately well sorted sand low in fines	1.01e-04
B102_21	Poorly-graded sand	Moderately well sorted gravelly sand low in fines	7.93e-04
B102_23	Poorly-graded sand	Moderately well sorted sand low in fines	4.27e-04
B102_25	Poorly-graded sand	Uniform sand low in fines	1.97e-04
B103_8	Sand	Clayey gravelly sand with fines	1.88e-04
B103_10	Poorly-graded sand	Moderately well sorted sand with fines	2.23e-04
B103_12	Well-graded sand	Poorly sorted gravelly sand low in fines	1.26e-04
B103_14	Poorly-graded sand	Moderately well sorted sand with fines	9.91e-05
B103_16	Poorly-graded sand	Uniform sand low in fines	1.80e-04
B103_18	Poorly-graded sand	Moderately well sorted sand low in fines	2.55e-04
B104_9	Poorly-graded sand	Moderately well sorted sand low in fines	4.99e-04
B104_11	Poorly-graded sand	Moderately well sorted sand low in fines	3.72e-04

Sample	Soil type	Descriptor	K geometric mean (m/s)
B104_13	Poorly-graded sand	Moderately well sorted sand low in fines	2.21e-04
B104_15	Poorly-graded sand	Moderately well sorted sand low in fines	3.53e-04
B104_17	Poorly-graded sand	Moderately well sorted sand low in fines	2.31e-04
B104_19	Poorly-graded sand	Moderately well sorted sand low in fines	1.47e-04
B104_21	Poorly-graded sand	Moderately well sorted sand low in fines	2.06e-04
B105_10	Poorly-graded sand	Moderately well sorted gravelly sand with fines	1.42e-04
B105_12	Well-graded gravel	Poorly sorted sandy gravel low in fines	4.10e-04
B105_14	Poorly-graded sand	Moderately well sorted sand low in fines	2.19e-04
B105_16	Poorly-graded sand	Moderately well sorted sand low in fines	3.82e-04
B105_18	Poorly-graded sand	Moderately well sorted sand low in fines	2.90e-04
B105_20	Poorly-graded sand	Moderately well sorted sand low in fines	2.42e-04
B105_22	Poorly-graded sand	Moderately well sorted sand low in fines	2.23e-04
B105_24	Poorly-graded sand	Moderately well sorted sand low in fines	2.07e-04
B105_26	Poorly-graded sand	Moderately well sorted sand low in fines	2.92e-04
B105_28	Poorly-graded sand	Moderately well sorted sand low in fines	2.55e-04

Grain Parameters Comparison

Parameter	Sample	Value
D10 (mm)	B106_1	0.141
D10 (mm)	B106_2	0.149
D10 (mm)	B106_3	0.079
D10 (mm)	B106_4	0.148

Parameter	Sample	Value
D10 (mm)	B106_5	0.145
D10 (mm)	B106_6	0.082
D10 (mm)	B106_7	-
D10 (mm)	B101_1	0.095
D10 (mm)	B101_3	-
D10 (mm)	B101_5	0.092
D10 (mm)	B101_7	-
D10 (mm)	B101_9	-
D10 (mm)	B101_11	0.119
D10 (mm)	B101_13	-
D10 (mm)	B101_15	-
D10 (mm)	B101_17	-
D10 (mm)	B101_19	0.067
D10 (mm)	B101_21	0.085
D10 (mm)	B101_23	0.078
D10 (mm)	B101_25	0.093
D10 (mm)	B101_27	0.098
D10 (mm)	B102_13	0.074
D10 (mm)	B102_15	0.173
D10 (mm)	B102_17	0.147
D10 (mm)	B102_19	0.085
D10 (mm)	B102_21	0.260
D10 (mm)	B102_23	0.186
D10 (mm)	B102_25	0.127
D10 (mm)	B103_8	-
D10 (mm)	B103_10	0.131
D10 (mm)	B103_12	0.135
D10 (mm)	B103_14	0.084
D10 (mm)	B103_16	0.121
D10 (mm)	B103_18	0.142
D10 (mm)	B104_9	0.198
D10 (mm)	B104_11	0.170
D10 (mm)	B104_13	0.132
D10 (mm)	B104_15	0.166
D10 (mm)	B104_17	0.138
D10 (mm)	B104_19	0.109

Parameter	Sample	Value
D10 (mm)	B104_21	0.129
D10 (mm)	B105_10	0.105
D10 (mm)	B105_12	0.238
D10 (mm)	B105_14	0.132
D10 (mm)	B105_16	0.172
D10 (mm)	B105_18	0.151
D10 (mm)	B105_20	0.139
D10 (mm)	B105_22	0.134
D10 (mm)	B105_24	0.131
D10 (mm)	B105_26	0.153
D10 (mm)	B105_28	0.144
D10 (mm) summary	Included samples	Mean: 0.132 Std: 0.041
D50 (mm)	B106_1	0.295
D50 (mm)	B106_2	0.317
D50 (mm)	B106_3	0.164
D50 (mm)	B106_4	0.322
D50 (mm)	B106_5	0.303
D50 (mm)	B106_6	0.180
D50 (mm)	B106_7	0.140
D50 (mm)	B101_1	0.229
D50 (mm)	B101_3	0.211
D50 (mm)	B101_5	0.182
D50 (mm)	B101_7	0.220
D50 (mm)	B101_9	0.181
D50 (mm)	B101_11	0.204
D50 (mm)	B101_13	0.094
D50 (mm)	B101_15	0.096
D50 (mm)	B101_17	0.082
D50 (mm)	B101_19	0.137
D50 (mm)	B101_21	0.165
D50 (mm)	B101_23	0.153
D50 (mm)	B101_25	0.175
D50 (mm)	B101_27	0.186
D50 (mm)	B102_13	0.186
D50 (mm)	B102_15	0.420

Parameter	Sample	Value
D50 (mm)	B102_17	0.371
D50 (mm)	B102_19	0.211
D50 (mm)	B102_21	0.730
D50 (mm)	B102_23	0.490
D50 (mm)	B102_25	0.220
D50 (mm)	B103_8	0.205
D50 (mm)	B103_10	0.323
D50 (mm)	B103_12	0.615
D50 (mm)	B103_14	0.248
D50 (mm)	B103_16	0.215
D50 (mm)	B103_18	0.321
D50 (mm)	B104_9	0.369
D50 (mm)	B104_11	0.340
D50 (mm)	B104_13	0.327
D50 (mm)	B104_15	0.339
D50 (mm)	B104_17	0.287
D50 (mm)	B104_19	0.252
D50 (mm)	B104_21	0.295
D50 (mm)	B105_10	0.358
D50 (mm)	B105_12	2.969
D50 (mm)	B105_14	0.325
D50 (mm)	B105_16	0.410
D50 (mm)	B105_18	0.332
D50 (mm)	B105_20	0.314
D50 (mm)	B105_22	0.302
D50 (mm)	B105_24	0.253
D50 (mm)	B105_26	0.323
D50 (mm)	B105_28	0.310
D50 (mm) summary	Included samples	Mean: 0.327 Std: 0.392
D60 (mm)	B106_1	0.341
D60 (mm)	B106_2	0.360
D60 (mm)	B106_3	0.186
D60 (mm)	B106_4	0.358
D60 (mm)	B106_5	0.343
D60 (mm)	B106_6	0.201

Parameter	Sample	Value
D60 (mm)	B106_7	0.165
D60 (mm)	B101_1	0.305
D60 (mm)	B101_3	0.244
D60 (mm)	B101_5	0.199
D60 (mm)	B101_7	0.330
D60 (mm)	B101_9	0.206
D60 (mm)	B101_11	0.225
D60 (mm)	B101_13	0.103
D60 (mm)	B101_15	0.110
D60 (mm)	B101_17	0.097
D60 (mm)	B101_19	0.161
D60 (mm)	B101_21	0.182
D60 (mm)	B101_23	0.173
D60 (mm)	B101_25	0.191
D60 (mm)	B101_27	0.204
D60 (mm)	B102_13	0.206
D60 (mm)	B102_15	0.479
D60 (mm)	B102_17	0.415
D60 (mm)	B102_19	0.238
D60 (mm)	B102_21	0.849
D60 (mm)	B102_23	0.591
D60 (mm)	B102_25	0.243
D60 (mm)	B103_8	0.229
D60 (mm)	B103_10	0.360
D60 (mm)	B103_12	1.035
D60 (mm)	B103_14	0.303
D60 (mm)	B103_16	0.238
D60 (mm)	B103_18	0.358
D60 (mm)	B104_9	0.404
D60 (mm)	B104_11	0.372
D60 (mm)	B104_13	0.362
D60 (mm)	B104_15	0.372
D60 (mm)	B104_17	0.330
D60 (mm)	B104_19	0.302
D60 (mm)	B104_21	0.339
D60 (mm)	B105_10	0.408

Parameter	Sample	Value
D60 (mm)	B105_12	3.858
D60 (mm)	B105_14	0.364
D60 (mm)	B105_16	0.461
D60 (mm)	B105_18	0.368
D60 (mm)	B105_20	0.352
D60 (mm)	B105_22	0.342
D60 (mm)	B105_24	0.303
D60 (mm)	B105_26	0.359
D60 (mm)	B105_28	0.348
D60 (mm) summary	Included samples	Mean: 0.390 Std: 0.516
Cu	B106_1	2.42
Cu	B106_2	2.42
Cu	B106_3	2.36
Cu	B106_4	2.42
Cu	B106_5	2.37
Cu	B106_6	2.44
Cu	B106_7	-
Cu	B101_1	3.20
Cu	B101_3	-
Cu	B101_5	2.17
Cu	B101_7	-
Cu	B101_9	-
Cu	B101_11	1.89
Cu	B101_13	-
Cu	B101_15	-
Cu	B101_17	-
Cu	B101_19	2.41
Cu	B101_21	2.15
Cu	B101_23	2.21
Cu	B101_25	2.06
Cu	B101_27	2.08
Cu	B102_13	2.77
Cu	B102_15	2.77
Cu	B102_17	2.82
Cu	B102_19	2.79

Parameter	Sample	Value
Cu	B102_21	3.26
Cu	B102_23	3.18
Cu	B102_25	1.91
Cu	B103_8	-
Cu	B103_10	2.75
Cu	B103_12	7.67
Cu	B103_14	3.63
Cu	B103_16	1.97
Cu	B103_18	2.51
Cu	B104_9	2.04
Cu	B104_11	2.18
Cu	B104_13	2.75
Cu	B104_15	2.23
Cu	B104_17	2.40
Cu	B104_19	2.78
Cu	B104_21	2.63
Cu	B105_10	3.88
Cu	B105_12	16.24
Cu	B105_14	2.76
Cu	B105_16	2.68
Cu	B105_18	2.44
Cu	B105_20	2.52
Cu	B105_22	2.55
Cu	B105_24	2.31
Cu	B105_26	2.35
Cu	B105_28	2.42
Cu summary	Included samples	Mean: 2.97 Std: 2.23
Cc	B106_1	0.94
Cc	B106_2	1.01
Cc	B106_3	0.96
Cc	B106_4	1.18
Cc	B106_5	1.01
Cc	B106_6	1.15
Cc	B106_7	-
Cc	B101_1	0.96

Parameter	Sample	Value
Cc	B101_3	-
Cc	B101_5	1.22
Cc	B101_7	-
Cc	B101_9	-
Cc	B101_11	1.01
Cc	B101_13	-
Cc	B101_15	-
Cc	B101_17	-
Cc	B101_19	0.93
Cc	B101_21	1.10
Cc	B101_23	0.97
Cc	B101_25	1.14
Cc	B101_27	1.12
Cc	B102_13	1.38
Cc	B102_15	1.11
Cc	B102_17	1.31
Cc	B102_19	1.19
Cc	B102_21	1.09
Cc	B102_23	1.13
Cc	B102_25	0.97
Cc	B103_8	-
Cc	B103_10	1.33
Cc	B103_12	0.39
Cc	B103_14	1.23
Cc	B103_16	1.00
Cc	B103_18	1.19
Cc	B104_9	1.13
Cc	B104_11	1.19
Cc	B104_13	1.36
Cc	B104_15	1.22
Cc	B104_17	0.96
Cc	B104_19	1.03
Cc	B104_21	1.02
Cc	B105_10	1.57
Cc	B105_12	1.73
Cc	B105_14	1.25

Parameter	Sample	Value
Cc	B105_16	1.18
Cc	B105_18	1.21
Cc	B105_20	1.13
Cc	B105_22	1.05
Cc	B105_24	0.92
Cc	B105_26	1.16
Cc	B105_28	1.06
Cc summary	Included samples	Mean: 1.12 Std: 0.20

K-Value Aggregate Summary

Scope	Datasets	Geometric mean (m/s)	Arithmetic mean (m/s)	Median (m/s)	ln(K) std dev
Overall	51	1.88e-04	2.78e-04	2.01e-04	0.902
B106_1	1	2.41e-04	2.86e-04	2.28e-04	0.562
B106_2	1	2.72e-04	3.23e-04	2.58e-04	0.562
B106_3	1	7.68e-05	9.70e-05	7.43e-05	0.666
B106_4	1	2.74e-04	3.22e-04	2.64e-04	0.547
B106_5	1	2.55e-04	3.01e-04	2.41e-04	0.552
B106_6	1	8.90e-05	1.13e-04	8.55e-05	0.675
B106_7	1	7.99e-05	1.11e-04	1.26e-04	0.969
B101_1	1	1.13e-04	1.57e-04	9.32e-05	0.778
B101_3	1	1.51e-04	2.18e-04	2.08e-04	0.966
B101_5	1	1.14e-04	1.39e-04	1.01e-04	0.619
B101_7	1	1.33e-04	1.98e-04	2.07e-04	1.041
B101_9	1	1.14e-04	1.72e-04	1.52e-04	1.066
B101_11	1	1.74e-04	2.06e-04	1.60e-04	0.577
B101_13	1	4.96e-05	6.02e-05	6.98e-05	0.718
B101_15	1	8.65e-05	1.05e-04	1.26e-04	0.685
B101_17	1	9.37e-05	1.27e-04	1.55e-04	0.870
B101_19	1	5.86e-05	7.58e-05	5.28e-05	0.720
B101_21	1	8.80e-05	1.07e-04	8.32e-05	0.612
B101_23	1	7.46e-05	9.21e-05	7.30e-05	0.635
B101_25	1	1.08e-04	1.30e-04	9.62e-05	0.583
B101_27	1	1.21e-04	1.47e-04	1.06e-04	0.597
B102_13	1	8.37e-05	1.19e-04	7.45e-05	0.819
B102_15	1	3.79e-04	4.97e-04	3.48e-04	0.683

Scope	Datasets	Geometric mean (m/s)	Arithmetic mean (m/s)	Median (m/s)	ln(K) std dev
B102_17	1	2.86e-04	3.50e-04	2.83e-04	0.618
B102_19	1	1.01e-04	1.34e-04	9.21e-05	0.744
B102_21	1	7.93e-04	9.77e-04	7.41e-04	0.618
B102_23	1	4.27e-04	5.17e-04	4.14e-04	0.606
B102_25	1	1.97e-04	2.35e-04	1.81e-04	0.586
B103_8	1	1.88e-04	2.60e-04	2.24e-04	0.869
B103_10	1	2.23e-04	2.72e-04	2.14e-04	0.616
B103_12	1	1.26e-04	2.32e-04	1.23e-04	1.229
B103_14	1	9.91e-05	1.45e-04	8.38e-05	0.885
B103_16	1	1.80e-04	2.14e-04	1.62e-04	0.583
B103_18	1	2.55e-04	3.02e-04	2.54e-04	0.562
B104_9	1	4.99e-04	5.77e-04	4.54e-04	0.518
B104_11	1	3.72e-04	4.31e-04	3.44e-04	0.518
B104_13	1	2.21e-04	2.66e-04	2.20e-04	0.587
B104_15	1	3.53e-04	4.11e-04	3.24e-04	0.519
B104_17	1	2.31e-04	2.71e-04	2.19e-04	0.555
B104_19	1	1.47e-04	1.80e-04	1.40e-04	0.613
B104_21	1	2.06e-04	2.47e-04	1.95e-04	0.590
B105_10	1	1.42e-04	2.19e-04	1.31e-04	0.919
B105_12	1	4.10e-04	7.74e-04	4.34e-04	1.253
B105_14	1	2.19e-04	2.66e-04	2.12e-04	0.601
B105_16	1	3.82e-04	4.60e-04	3.70e-04	0.581
B105_18	1	2.90e-04	3.41e-04	2.84e-04	0.548
B105_20	1	2.42e-04	2.86e-04	2.43e-04	0.560
B105_22	1	2.23e-04	2.64e-04	2.20e-04	0.567
B105_24	1	2.07e-04	2.44e-04	1.89e-04	0.555
B105_26	1	2.92e-04	3.43e-04	2.77e-04	0.542
B105_28	1	2.55e-04	3.01e-04	2.44e-04	0.551
Scope	Included K cells		Warning cells	Permeability class	
Overall	630 / 816		186	High (Clean Sand)	
B106_1	14 / 16		2	High (Clean Sand)	
B106_2	14 / 16		2	High (Clean Sand)	
B106_3	12 / 16		4	Moderate (Fine Sand)	
B106_4	14 / 16		2	High (Clean Sand)	

Scope	Included K cells	Warning cells	Permeability class
B106_5	14 / 16	2	High (Clean Sand)
B106_6	12 / 16	4	Moderate (Fine Sand)
B106_7	8 / 16	8	Moderate (Fine Sand)
B101_1	12 / 16	4	High (Clean Sand)
B101_3	8 / 16	8	High (Clean Sand)
B101_5	12 / 16	4	High (Clean Sand)
B101_7	8 / 16	8	High (Clean Sand)
B101_9	8 / 16	8	High (Clean Sand)
B101_11	13 / 16	3	High (Clean Sand)
B101_13	8 / 16	8	Moderate (Fine Sand)
B101_15	7 / 16	9	Moderate (Fine Sand)
B101_17	7 / 16	9	Moderate (Fine Sand)
B101_19	12 / 16	4	Moderate (Fine Sand)
B101_21	12 / 16	4	Moderate (Fine Sand)
B101_23	12 / 16	4	Moderate (Fine Sand)
B101_25	12 / 16	4	High (Clean Sand)
B101_27	12 / 16	4	High (Clean Sand)
B102_13	12 / 16	4	Moderate (Fine Sand)
B102_15	14 / 16	2	High (Clean Sand)
B102_17	14 / 16	2	High (Clean Sand)
B102_19	12 / 16	4	High (Clean Sand)
B102_21	14 / 16	2	High (Clean Sand)
B102_23	14 / 16	2	High (Clean Sand)
B102_25	13 / 16	3	High (Clean Sand)
B103_8	8 / 16	8	High (Clean Sand)
B103_10	14 / 16	2	High (Clean Sand)
B103_12	12 / 16	4	High (Clean Sand)
B103_14	12 / 16	4	Moderate (Fine Sand)
B103_16	13 / 16	3	High (Clean Sand)
B103_18	14 / 16	2	High (Clean Sand)
B104_9	14 / 16	2	High (Clean Sand)

Scope	Included K cells	Warning cells	Permeability class
B104_11	14 / 16	2	High (Clean Sand)
B104_13	14 / 16	2	High (Clean Sand)
B104_15	14 / 16	2	High (Clean Sand)
B104_17	14 / 16	2	High (Clean Sand)
B104_19	14 / 16	2	High (Clean Sand)
B104_21	14 / 16	2	High (Clean Sand)
B105_10	14 / 16	2	High (Clean Sand)
B105_12	9 / 16	7	High (Clean Sand)
B105_14	14 / 16	2	High (Clean Sand)
B105_16	14 / 16	2	High (Clean Sand)
B105_18	14 / 16	2	High (Clean Sand)
B105_20	14 / 16	2	High (Clean Sand)
B105_22	14 / 16	2	High (Clean Sand)
B105_24	14 / 16	2	High (Clean Sand)
B105_26	14 / 16	2	High (Clean Sand)
B105_28	14 / 16	2	High (Clean Sand)

K-Value Calculations by Dataset and Method

Sample	Method	K (m/s)	Status	Included	Notes
B106_1	Hazen	3.04e-04	OK	Yes	OK
B106_1	Hazen_1892	1.99e-04	Error	No	Empirical mm ² variant (10°C reference)
B106_1	Slichter	1.12e-04	OK	Yes	OK
B106_1	Terzaghi	1.96e-04	OK	Yes	Reference 10°C calibration
B106_1	Beyer	2.37e-04	OK	Yes	Within recommended range
B106_1	Sauerbrei	2.32e-04	OK	Yes	OK
B106_1	Kruger	3.58e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B106_1	Kozeny-Carman	8.46e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B106_1	Zunker	4.55e-04	OK	Yes	OK
B106_1	Zamarin	4.84e-04	OK	Yes	OK
B106_1	USBR	8.89e-05	OK	Yes	OK
B106_1	Barr	1.70e-04	OK	Yes	OK
B106_1	Alyamani-Sen	1.70e-04	OK	Yes	OK
B106_1	Chapuis	2.63e-04	OK	Yes	OK
B106_1	Shepherd	2.21e-04	OK	Yes	OK
B106_1	Krumbein-Monk	2.23e-04	OK	Yes	OK
B106_2	Hazen	3.40e-04	OK	Yes	OK
B106_2	Hazen_1892	2.22e-04	Error	No	Empirical mm ² variant (10°C reference)
B106_2	Slichter	1.25e-04	OK	Yes	OK
B106_2	Terzaghi	2.19e-04	OK	Yes	Reference 10°C calibration
B106_2	Beyer	2.65e-04	OK	Yes	Within recommended range
B106_2	Sauerbrei	2.67e-04	OK	Yes	OK
B106_2	Kruger	4.10e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B106_2	Kozeny-Carman	9.61e-04	OK	Yes	OK
B106_2	Zunker	5.21e-04	OK	Yes	OK
B106_2	Zamarin	5.53e-04	OK	Yes	OK
B106_2	USBR	1.05e-04	OK	Yes	OK
B106_2	Barr	1.90e-04	OK	Yes	OK
B106_2	Alyamani-Sen	1.85e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B106_2	Chapuis	2.97e-04	OK	Yes	OK
B106_2	Shepherd	2.49e-04	OK	Yes	OK
B106_2	Krumbein-Monk	2.51e-04	OK	Yes	OK
B106_3	Hazen	9.60e-05	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B106_3	Hazen_1892	6.22e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B106_3	Slichter	3.54e-05	OK	Yes	OK
B106_3	Terzaghi	6.23e-05	OK	Yes	Reference 10°C calibration
B106_3	Beyer	7.47e-05	OK	Yes	Within recommended range
B106_3	Sauerbrei	7.39e-05	OK	Yes	OK
B106_3	Kruger	1.27e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B106_3	Kozeny-Carman	2.95e-04	OK	Yes	OK
B106_3	Zunker	1.62e-04	OK	Yes	OK
B106_3	Zamarin	1.71e-04	OK	Yes	OK
B106_3	USBR	2.33e-05	OK	Yes	OK
B106_3	Barr	5.42e-05	OK	Yes	OK
B106_3	Alyamani-Sen	5.38e-05	OK	Yes	OK
B106_3	Chapuis	7.65e-05	Warning	No	D10 > 0.10 mm not satisfied
B106_3	Shepherd	8.33e-05	OK	Yes	OK
B106_3	Krumbein-Monk	7.46e-05	OK	Yes	OK
B106_4	Hazen	3.35e-04	OK	Yes	OK
B106_4	Hazen_1892	2.19e-04	Error	No	Empirical mm ² variant (10°C reference)

Sample	Method	K (m/s)	Status	Included	Notes
B106_4	Slichter	1.23e-04	OK	Yes	OK
B106_4	Terzaghi	2.16e-04	OK	Yes	Reference 10°C calibration
B106_4	Beyer	2.61e-04	OK	Yes	Within recommended range
B106_4	Sauerbrei	2.83e-04	OK	Yes	OK
B106_4	Kruger	4.11e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B106_4	Kozeny-Carman	9.13e-04	OK	Yes	OK
B106_4	Zunker	5.23e-04	OK	Yes	OK
B106_4	Zamarin	5.55e-04	OK	Yes	OK
B106_4	USBR	1.16e-04	OK	Yes	OK
B106_4	Barr	1.87e-04	OK	Yes	OK
B106_4	Alyamani-Sen	1.78e-04	OK	Yes	OK
B106_4	Chapuis	2.92e-04	OK	Yes	OK
B106_4	Shepherd	2.54e-04	OK	Yes	OK
B106_4	Krumbein-Monk	2.67e-04	OK	Yes	OK
B106_5	Hazen	3.23e-04	OK	Yes	OK
B106_5	Hazen_1892	2.09e-04	Error	No	Empirical mm ² variant (10°C reference)
B106_5	Slichter	1.19e-04	OK	Yes	OK
B106_5	Terzaghi	2.09e-04	OK	Yes	Reference 10°C calibration
B106_5	Beyer	2.51e-04	OK	Yes	Within recommended range
B106_5	Sauerbrei	2.53e-04	OK	Yes	OK
B106_5	Kruger	3.54e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)

Sample	Method	K (m/s)	Status	Included	Notes
B106_5	Kozeny-Carman	9.28e-04	OK	Yes	OK
B106_5	Zunker	4.52e-04	OK	Yes	OK
B106_5	Zamarin	4.78e-04	OK	Yes	OK
B106_5	USBR	9.71e-05	OK	Yes	OK
B106_5	Barr	1.82e-04	OK	Yes	OK
B106_5	Alyamani-Sen	1.79e-04	OK	Yes	OK
B106_5	Chapuis	2.86e-04	OK	Yes	OK
B106_5	Shepherd	2.30e-04	OK	Yes	OK
B106_5	Krumbein-Monk	2.30e-04	OK	Yes	OK
B106_6	Hazen	1.04e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B106_6	Hazen_1892	6.78e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B106_6	Slichter	3.79e-05	OK	Yes	OK
B106_6	Terzaghi	6.67e-05	OK	Yes	Reference 10°C calibration
B106_6	Beyer	8.09e-05	OK	Yes	Within recommended range
B106_6	Sauerbrei	9.00e-05	OK	Yes	OK
B106_6	Kruger	1.60e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B106_6	Kozeny-Carman	3.25e-04	OK	Yes	OK
B106_6	Zunker	2.03e-04	OK	Yes	OK
B106_6	Zamarin	2.16e-04	OK	Yes	OK
B106_6	USBR	3.17e-05	OK	Yes	OK
B106_6	Barr	5.76e-05	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B106_6	Alyamani-Sen	5.49e-05	OK	Yes	OK
B106_6	Chapuis	8.02e-05	Warning	No	D10 > 0.10 mm not satisfied
B106_6	Shepherd	9.75e-05	OK	Yes	OK
B106_6	Krumbein-Monk	9.55e-05	OK	Yes	OK
B106_7	Hazen	N/A	Warning	No	D10 required
B106_7	Hazen_1892	N/A	Warning	No	D10 required
B106_7	Slichter	N/A	Warning	No	D10 required
B106_7	Terzaghi	N/A	Warning	No	D10 required
B106_7	Beyer	N/A	Warning	No	D10 and D60 required
B106_7	Sauerbrei	2.83e-05	OK	Yes	OK
B106_7	Kruger	1.49e-04	OK	Yes	OK
B106_7	Kozeny-Carman	1.37e-04	OK	Yes	OK
B106_7	Zunker	1.82e-04	OK	Yes	OK
B106_7	Zamarin	2.00e-04	OK	Yes	OK
B106_7	USBR	1.07e-05	OK	Yes	OK
B106_7	Barr	N/A	Warning	No	D10 required
B106_7	Alyamani-Sen	N/A	Warning	No	D10 and D50 required
B106_7	Chapuis	N/A	Warning	No	D10 required
B106_7	Shepherd	6.42e-05	OK	Yes	OK
B106_7	Krumbein-Monk	1.16e-04	OK	Yes	OK
B101_1	Hazen	1.27e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B101_1	Hazen_1892	9.05e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B101_1	Slichter	4.26e-05	OK	Yes	OK
B101_1	Terzaghi	7.44e-05	OK	Yes	Reference 10°C calibration
B101_1	Beyer	1.03e-04	OK	Yes	Within recommended range

Sample	Method	K (m/s)	Status	Included	Notes
B101_1	Sauerbrei	1.01e-04	OK	Yes	OK
B101_1	Kruger	2.70e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B101_1	Kozeny-Carman	4.79e-04	OK	Yes	OK
B101_1	Zunker	3.25e-04	OK	Yes	OK
B101_1	Zamarin	3.61e-04	OK	Yes	OK
B101_1	USBR	4.83e-05	OK	Yes	OK
B101_1	Barr	6.11e-05	OK	Yes	OK
B101_1	Alyamani-Sen	6.37e-05	OK	Yes	OK
B101_1	Chapuis	7.40e-05	Warning	No	D10 > 0.10 mm not satisfied
B101_1	Shepherd	1.45e-04	OK	Yes	OK
B101_1	Krumbein-Monk	8.51e-05	OK	Yes	OK
B101_3	Hazen	N/A	Warning	No	D10 required
B101_3	Hazen_1892	N/A	Warning	No	D10 required
B101_3	Slichter	N/A	Warning	No	D10 required
B101_3	Terzaghi	N/A	Warning	No	D10 required
B101_3	Beyer	N/A	Warning	No	D10 and D60 required
B101_3	Sauerbrei	5.52e-05	OK	Yes	OK
B101_3	Kruger	3.25e-04	OK	Yes	OK
B101_3	Kozeny-Carman	2.89e-04	OK	Yes	OK
B101_3	Zunker	3.97e-04	OK	Yes	OK
B101_3	Zamarin	4.37e-04	OK	Yes	OK
B101_3	USBR	2.62e-05	OK	Yes	OK
B101_3	Barr	N/A	Warning	No	D10 required
B101_3	Alyamani-Sen	N/A	Warning	No	D10 and D50 required
B101_3	Chapuis	N/A	Warning	No	D10 required

Sample	Method	K (m/s)	Status	Included	Notes
B101_3	Shepherd	1.26e-04	OK	Yes	OK
B101_3	Krumbein-Monk	8.89e-05	OK	Yes	OK
B101_5	Hazen	1.33e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B101_5	Hazen_1892	8.40e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B101_5	Slichter	5.02e-05	OK	Yes	OK
B101_5	Terzaghi	8.84e-05	OK	Yes	Reference 10°C calibration
B101_5	Beyer	1.03e-04	OK	Yes	Within recommended range
B101_5	Sauerbrei	1.32e-04	OK	Yes	OK
B101_5	Kruger	1.93e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B101_5	Kozeny-Carman	3.65e-04	OK	Yes	OK
B101_5	Zunker	2.50e-04	OK	Yes	OK
B101_5	Zamarin	2.61e-04	OK	Yes	OK
B101_5	USBR	4.54e-05	OK	Yes	OK
B101_5	Barr	7.82e-05	OK	Yes	OK
B101_5	Alyamani-Sen	7.66e-05	OK	Yes	OK
B101_5	Chapuis	1.20e-04	Warning	No	D10 > 0.10 mm not satisfied
B101_5	Shepherd	9.93e-05	OK	Yes	OK
B101_5	Krumbein-Monk	1.22e-04	OK	Yes	OK
B101_7	Hazen	N/A	Warning	No	D10 required
B101_7	Hazen_1892	N/A	Warning	No	D10 required
B101_7	Slichter	N/A	Warning	No	D10 required
B101_7	Terzaghi	N/A	Warning	No	D10 required
B101_7	Beyer	N/A	Warning	No	D10 and D60 required

Sample	Method	K (m/s)	Status	Included	Notes
B101_7	Sauerbrei	4.41e-05	OK	Yes	OK
B101_7	Kruger	2.88e-04	OK	Yes	OK
B101_7	Kozeny-Carman	2.78e-04	OK	Yes	OK
B101_7	Zunker	3.51e-04	OK	Yes	OK
B101_7	Zamarin	3.86e-04	OK	Yes	OK
B101_7	USBR	1.79e-05	OK	Yes	OK
B101_7	Barr	N/A	Warning	No	D10 required
B101_7	Alyamani-Sen	N/A	Warning	No	D10 and D50 required
B101_7	Chapuis	N/A	Warning	No	D10 required
B101_7	Shepherd	1.35e-04	OK	Yes	OK
B101_7	Krumbein-Monk	8.51e-05	OK	Yes	OK
B101_9	Hazen	N/A	Warning	No	D10 required
B101_9	Hazen_1892	N/A	Warning	No	D10 required
B101_9	Slichter	N/A	Warning	No	D10 required
B101_9	Terzaghi	N/A	Warning	No	D10 required
B101_9	Beyer	N/A	Warning	No	D10 and D60 required
B101_9	Sauerbrei	3.06e-05	OK	Yes	OK
B101_9	Kruger	2.61e-04	OK	Yes	OK
B101_9	Kozeny-Carman	1.93e-04	OK	Yes	OK
B101_9	Zunker	3.19e-04	OK	Yes	OK
B101_9	Zamarin	3.51e-04	OK	Yes	OK
B101_9	USBR	1.52e-05	OK	Yes	OK
B101_9	Barr	N/A	Warning	No	D10 required
B101_9	Alyamani-Sen	N/A	Warning	No	D10 and D50 required
B101_9	Chapuis	N/A	Warning	No	D10 required
B101_9	Shepherd	9.84e-05	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B101_9	Krumbein-Monk	1.11e-04	OK	Yes	OK
B101_11	Hazen	2.32e-04	OK	Yes	OK
B101_11	Hazen_1892	1.42e-04	Error	No	Empirical mm ² variant (10°C reference)
B101_11	Slichter	9.10e-05	OK	Yes	OK
B101_11	Terzaghi	1.60e-04	OK	Yes	Reference 10°C calibration
B101_11	Beyer	1.78e-04	OK	Yes	Within recommended range
B101_11	Sauerbrei	1.87e-04	OK	Yes	OK
B101_11	Kruger	2.54e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B101_11	Kozeny-Carman	5.52e-04	OK	Yes	OK
B101_11	Zunker	3.37e-04	OK	Yes	OK
B101_11	Zamarin	3.43e-04	OK	Yes	OK
B101_11	USBR	5.51e-05	OK	Yes	OK
B101_11	Barr	1.45e-04	OK	Yes	OK
B101_11	Alyamani-Sen	1.50e-04	OK	Yes	OK
B101_11	Chapuis	2.49e-04	Warning	No	Uniformity coefficient outside 2–12
B101_11	Shepherd	1.20e-04	OK	Yes	OK
B101_11	Krumbein-Monk	1.32e-04	OK	Yes	OK
B101_13	Hazen	N/A	Warning	No	D10 required
B101_13	Hazen_1892	N/A	Warning	No	D10 required
B101_13	Slichter	N/A	Warning	No	D10 required
B101_13	Terzaghi	N/A	Warning	No	D10 required
B101_13	Beyer	N/A	Warning	No	D10 and D60 required
B101_13	Sauerbrei	3.11e-05	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B101_13	Kruger	6.73e-05	OK	Yes	OK
B101_13	Kozeny-Carman	9.45e-05	OK	Yes	OK
B101_13	Zunker	8.22e-05	OK	Yes	OK
B101_13	Zamarin	9.04e-05	OK	Yes	OK
B101_13	USBR	1.03e-05	OK	Yes	OK
B101_13	Barr	N/A	Warning	No	D10 required
B101_13	Alyamani-Sen	N/A	Warning	No	D10 and D50 required
B101_13	Chapuis	N/A	Warning	No	D10 required
B101_13	Shepherd	3.36e-05	OK	Yes	OK
B101_13	Krumbein-Monk	7.24e-05	OK	Yes	OK
B101_15	Hazen	N/A	Warning	No	D10 required
B101_15	Hazen_1892	N/A	Warning	No	D10 required
B101_15	Slichter	N/A	Warning	No	D10 required
B101_15	Terzaghi	N/A	Warning	No	D10 required
B101_15	Beyer	N/A	Warning	No	D10 and D60 required
B101_15	Sauerbrei	2.77e-05	OK	Yes	OK
B101_15	Kruger	1.26e-04	OK	Yes	OK
B101_15	Kozeny-Carman	8.32e-05	OK	Yes	OK
B101_15	Zunker	1.54e-04	OK	Yes	OK
B101_15	Zamarin	1.70e-04	OK	Yes	OK
B101_15	USBR	N/A	Warning	No	D20 required
B101_15	Barr	N/A	Warning	No	D10 required
B101_15	Alyamani-Sen	N/A	Warning	No	D10 and D50 required
B101_15	Chapuis	N/A	Warning	No	D10 required
B101_15	Shepherd	3.46e-05	OK	Yes	OK
B101_15	Krumbein-Monk	1.37e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B101_17	Hazen	N/A	Warning	No	D10 required
B101_17	Hazen_1892	N/A	Warning	No	D10 required
B101_17	Slichter	N/A	Warning	No	D10 required
B101_17	Terzaghi	N/A	Warning	No	D10 required
B101_17	Beyer	N/A	Warning	No	D10 and D60 required
B101_17	Sauerbrei	2.77e-05	OK	Yes	OK
B101_17	Kruger	1.55e-04	OK	Yes	OK
B101_17	Kozeny-Carman	6.46e-05	OK	Yes	OK
B101_17	Zunker	1.89e-04	OK	Yes	OK
B101_17	Zamarin	2.08e-04	OK	Yes	OK
B101_17	USBR	N/A	Warning	No	D20 required
B101_17	Barr	N/A	Warning	No	D10 required
B101_17	Alyamani-Sen	N/A	Warning	No	D10 and D50 required
B101_17	Chapuis	N/A	Warning	No	D10 required
B101_17	Shepherd	2.67e-05	OK	Yes	OK
B101_17	Krumbein-Monk	2.18e-04	OK	Yes	OK
B101_19	Hazen	6.89e-05	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B101_19	Hazen_1892	4.48e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B101_19	Slichter	2.53e-05	OK	Yes	OK
B101_19	Terzaghi	4.44e-05	OK	Yes	Reference 10°C calibration
B101_19	Beyer	5.37e-05	OK	Yes	Within recommended range
B101_19	Sauerbrei	5.20e-05	OK	Yes	OK
B101_19	Kruger	1.15e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B101_19	Kozeny-Carman	1.98e-04	OK	Yes	OK
B101_19	Zunker	1.46e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B101_19	Zamarin	1.55e-04	OK	Yes	OK
B101_19	USBR	1.58e-05	OK	Yes	OK
B101_19	Barr	3.85e-05	OK	Yes	OK
B101_19	Alyamani-Sen	3.95e-05	OK	Yes	OK
B101_19	Chapuis	5.20e-05	Warning	No	D10 > 0.10 mm not satisfied
B101_19	Shepherd	6.22e-05	OK	Yes	OK
B101_19	Krumbein-Monk	7.87e-05	OK	Yes	OK
B101_21	Hazen	1.13e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B101_21	Hazen_1892	7.15e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B101_21	Slichter	4.29e-05	OK	Yes	OK
B101_21	Terzaghi	7.55e-05	OK	Yes	Reference 10°C calibration
B101_21	Beyer	8.73e-05	OK	Yes	Within recommended range
B101_21	Sauerbrei	9.42e-05	OK	Yes	OK
B101_21	Kruger	1.27e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B101_21	Kozeny-Carman	3.26e-04	OK	Yes	OK
B101_21	Zunker	1.64e-04	OK	Yes	OK
B101_21	Zamarin	1.71e-04	OK	Yes	OK
B101_21	USBR	2.88e-05	OK	Yes	OK
B101_21	Barr	6.68e-05	OK	Yes	OK
B101_21	Alyamani-Sen	6.65e-05	OK	Yes	OK
B101_21	Chapuis	1.01e-04	Warning	No	D10 > 0.10 mm not satisfied

Sample	Method	K (m/s)	Status	Included	Notes
B101_21	Shepherd	8.44e-05	OK	Yes	OK
B101_21	Krumbein-Monk	8.20e-05	OK	Yes	OK
B101_23	Hazen	9.67e-05	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B101_23	Hazen_1892	6.15e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B101_23	Slichter	3.64e-05	OK	Yes	OK
B101_23	Terzaghi	6.40e-05	OK	Yes	Reference 10°C calibration
B101_23	Beyer	7.48e-05	OK	Yes	Within recommended range
B101_23	Sauerbrei	7.48e-05	OK	Yes	OK
B101_23	Kruger	1.12e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B101_23	Kozeny-Carman	2.78e-04	OK	Yes	OK
B101_23	Zunker	1.44e-04	OK	Yes	OK
B101_23	Zamarin	1.51e-04	OK	Yes	OK
B101_23	USBR	2.20e-05	OK	Yes	OK
B101_23	Barr	5.64e-05	OK	Yes	OK
B101_23	Alyamani-Sen	5.70e-05	OK	Yes	OK
B101_23	Chapuis	8.31e-05	Warning	No	D10 > 0.10 mm not satisfied
B101_23	Shepherd	7.49e-05	OK	Yes	OK
B101_23	Krumbein-Monk	7.13e-05	OK	Yes	OK
B101_25	Hazen	1.37e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B101_25	Hazen_1892	8.56e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B101_25	Slichter	5.25e-05	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B101_25	Terzaghi	9.25e-05	OK	Yes	Reference 10°C calibration
B101_25	Beyer	1.05e-04	OK	Yes	Within recommended range
B101_25	Sauerbrei	1.26e-04	OK	Yes	OK
B101_25	Kruger	1.49e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B101_25	Kozeny-Carman	3.86e-04	OK	Yes	OK
B101_25	Zunker	1.94e-04	OK	Yes	OK
B101_25	Zamarin	2.01e-04	OK	Yes	OK
B101_25	USBR	4.00e-05	OK	Yes	OK
B101_25	Barr	8.24e-05	OK	Yes	OK
B101_25	Alyamani-Sen	8.26e-05	OK	Yes	OK
B101_25	Chapuis	1.30e-04	Warning	No	D10 > 0.10 mm not satisfied
B101_25	Shepherd	9.27e-05	OK	Yes	OK
B101_25	Krumbein-Monk	9.97e-05	OK	Yes	OK
B101_27	Hazen	1.54e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B101_27	Hazen_1892	9.62e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B101_27	Slichter	5.87e-05	OK	Yes	OK
B101_27	Terzaghi	1.03e-04	OK	Yes	Reference 10°C calibration
B101_27	Beyer	1.18e-04	OK	Yes	Within recommended range
B101_27	Sauerbrei	1.38e-04	OK	Yes	OK
B101_27	Kruger	1.76e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B101_27	Kozeny-Carman	4.40e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B101_27	Zunker	2.30e-04	OK	Yes	OK
B101_27	Zamarin	2.38e-04	OK	Yes	OK
B101_27	USBR	4.46e-05	OK	Yes	OK
B101_27	Barr	9.21e-05	OK	Yes	OK
B101_27	Alyamani-Sen	9.23e-05	OK	Yes	OK
B101_27	Chapuis	1.46e-04	Warning	No	D10 > 0.10 mm not satisfied
B101_27	Shepherd	1.03e-04	OK	Yes	OK
B101_27	Krumbein-Monk	1.08e-04	OK	Yes	OK
B102_13	Hazen	8.13e-05	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B102_13	Hazen_1892	5.52e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B102_13	Slichter	2.86e-05	OK	Yes	OK
B102_13	Terzaghi	5.01e-05	OK	Yes	Reference 10°C calibration
B102_13	Beyer	6.43e-05	OK	Yes	Within recommended range
B102_13	Sauerbrei	9.10e-05	OK	Yes	OK
B102_13	Kruger	2.25e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B102_13	Kozeny-Carman	3.09e-04	OK	Yes	OK
B102_13	Zunker	2.79e-04	OK	Yes	OK
B102_13	Zamarin	3.03e-04	OK	Yes	OK
B102_13	USBR	4.01e-05	OK	Yes	OK
B102_13	Barr	4.23e-05	OK	Yes	OK
B102_13	Alyamani-Sen	3.64e-05	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B102_13	Chapuis	5.28e-05	Warning	No	D10 > 0.10 mm not satisfied
B102_13	Shepherd	1.03e-04	OK	Yes	OK
B102_13	Krumbein-Monk	8.47e-05	OK	Yes	OK
B102_15	Hazen	4.40e-04	OK	Yes	OK
B102_15	Hazen_1892	2.98e-04	Error	No	Empirical mm ² variant (10°C reference)
B102_15	Slichter	1.55e-04	OK	Yes	OK
B102_15	Terzaghi	2.71e-04	OK	Yes	Reference 10°C calibration
B102_15	Beyer	3.48e-04	OK	Yes	Within recommended range
B102_15	Sauerbrei	3.71e-04	OK	Yes	OK
B102_15	Kruger	8.03e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B102_15	Kozeny-Carman	1.68e-03	OK	Yes	OK
B102_15	Zunker	9.96e-04	OK	Yes	OK
B102_15	Zamarin	1.08e-03	OK	Yes	OK
B102_15	USBR	1.84e-04	OK	Yes	OK
B102_15	Barr	2.29e-04	OK	Yes	OK
B102_15	Alyamani-Sen	2.06e-04	OK	Yes	OK
B102_15	Chapuis	3.47e-04	OK	Yes	OK
B102_15	Shepherd	3.95e-04	OK	Yes	OK
B102_15	Krumbein-Monk	2.50e-04	OK	Yes	OK
B102_17	Hazen	3.18e-04	OK	Yes	OK
B102_17	Hazen_1892	2.17e-04	Error	No	Empirical mm ² variant (10°C reference)

Sample	Method	K (m/s)	Status	Included	Notes
B102_17	Slichter	1.11e-04	OK	Yes	OK
B102_17	Terzaghi	1.95e-04	OK	Yes	Reference 10°C calibration
B102_17	Beyer	2.52e-04	OK	Yes	Within recommended range
B102_17	Sauerbrei	3.13e-04	OK	Yes	OK
B102_17	Kruger	5.66e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B102_17	Kozeny-Carman	9.11e-04	OK	Yes	OK
B102_17	Zunker	7.00e-04	OK	Yes	OK
B102_17	Zamarin	7.62e-04	OK	Yes	OK
B102_17	USBR	1.63e-04	OK	Yes	OK
B102_17	Barr	1.64e-04	OK	Yes	OK
B102_17	Alyamani-Sen	1.41e-04	OK	Yes	OK
B102_17	Chapuis	2.38e-04	OK	Yes	OK
B102_17	Shepherd	3.22e-04	OK	Yes	OK
B102_17	Krumbein-Monk	3.14e-04	OK	Yes	OK
B102_19	Hazen	1.07e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B102_19	Hazen_1892	7.30e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B102_19	Slichter	3.77e-05	OK	Yes	OK
B102_19	Terzaghi	6.60e-05	OK	Yes	Reference 10°C calibration
B102_19	Beyer	8.49e-05	OK	Yes	Within recommended range
B102_19	Sauerbrei	9.92e-05	OK	Yes	OK
B102_19	Kruger	2.17e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)

Sample	Method	K (m/s)	Status	Included	Notes
B102_19	Kozeny-Carman	3.66e-04	OK	Yes	OK
B102_19	Zunker	2.69e-04	OK	Yes	OK
B102_19	Zamarin	2.92e-04	OK	Yes	OK
B102_19	USBR	4.20e-05	OK	Yes	OK
B102_19	Barr	5.57e-05	OK	Yes	OK
B102_19	Alyamani-Sen	4.94e-05	OK	Yes	OK
B102_19	Chapuis	7.16e-05	Warning	No	D10 > 0.10 mm not satisfied
B102_19	Shepherd	1.26e-04	OK	Yes	OK
B102_19	Krumbein-Monk	1.21e-04	OK	Yes	OK
B102_21	Hazen	9.45e-04	OK	Yes	OK
B102_21	Hazen_1892	6.78e-04	Error	No	Empirical mm ² variant (10°C reference)
B102_21	Slichter	3.15e-04	OK	Yes	OK
B102_21	Terzaghi	5.50e-04	OK	Yes	Reference 10°C calibration
B102_21	Beyer	7.66e-04	OK	Yes	Within recommended range
B102_21	Sauerbrei	7.61e-04	OK	Yes	OK
B102_21	Kruger	1.44e-03	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B102_21	Kozeny-Carman	2.95e-03	OK	Yes	OK
B102_21	Zunker	1.73e-03	OK	Yes	OK
B102_21	Zamarin	1.93e-03	OK	Yes	OK
B102_21	USBR	5.03e-04	OK	Yes	OK
B102_21	Barr	4.51e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B102_21	Alyamani-Sen	3.61e-04	OK	Yes	OK
B102_21	Chapuis	7.22e-04	OK	Yes	OK
B102_21	Shepherd	9.83e-04	OK	Yes	OK
B102_21	Krumbein-Monk	7.14e-04	OK	Yes	OK
B102_23	Hazen	4.88e-04	OK	Yes	OK
B102_23	Hazen_1892	3.47e-04	Error	No	Empirical mm ² variant (10°C reference)
B102_23	Slichter	1.64e-04	OK	Yes	OK
B102_23	Terzaghi	2.87e-04	OK	Yes	Reference 10°C calibration
B102_23	Beyer	3.94e-04	OK	Yes	Within recommended range
B102_23	Sauerbrei	4.34e-04	OK	Yes	OK
B102_23	Kruger	8.55e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B102_23	Kozeny-Carman	1.22e-03	OK	Yes	OK
B102_23	Zunker	1.03e-03	OK	Yes	OK
B102_23	Zamarin	1.14e-03	OK	Yes	OK
B102_23	USBR	2.64e-04	OK	Yes	OK
B102_23	Barr	2.36e-04	OK	Yes	OK
B102_23	Alyamani-Sen	2.09e-04	OK	Yes	OK
B102_23	Chapuis	3.47e-04	OK	Yes	OK
B102_23	Shepherd	5.10e-04	OK	Yes	OK
B102_23	Krumbein-Monk	5.16e-04	OK	Yes	OK
B102_25	Hazen	2.63e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B102_25	Hazen_1892	1.62e-04	Error	No	Empirical mm ² variant (10°C reference)
B102_25	Slichter	1.03e-04	OK	Yes	OK
B102_25	Terzaghi	1.81e-04	OK	Yes	Reference 10°C calibration
B102_25	Beyer	2.02e-04	OK	Yes	Within recommended range
B102_25	Sauerbrei	2.05e-04	OK	Yes	OK
B102_25	Kruger	2.91e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B102_25	Kozeny-Carman	6.39e-04	OK	Yes	OK
B102_25	Zunker	3.85e-04	OK	Yes	OK
B102_25	Zamarin	3.93e-04	OK	Yes	OK
B102_25	USBR	6.09e-05	OK	Yes	OK
B102_25	Barr	1.64e-04	OK	Yes	OK
B102_25	Alyamani-Sen	1.70e-04	OK	Yes	OK
B102_25	Chapuis	2.81e-04	Warning	No	Uniformity coefficient outside 2–12
B102_25	Shepherd	1.36e-04	OK	Yes	OK
B102_25	Krumbein-Monk	1.52e-04	OK	Yes	OK
B103_8	Hazen	N/A	Warning	No	D10 required
B103_8	Hazen_1892	N/A	Warning	No	D10 required
B103_8	Slichter	N/A	Warning	No	D10 required
B103_8	Terzaghi	N/A	Warning	No	D10 required
B103_8	Beyer	N/A	Warning	No	D10 and D60 required
B103_8	Sauerbrei	1.11e-04	OK	Yes	OK
B103_8	Kruger	3.94e-04	OK	Yes	OK
B103_8	Kozeny-Carman	3.28e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B103_8	Zunker	4.80e-04	OK	Yes	OK
B103_8	Zamarin	5.28e-04	OK	Yes	OK
B103_8	USBR	4.65e-05	OK	Yes	OK
B103_8	Barr	N/A	Warning	No	D10 required
B103_8	Alyamani-Sen	N/A	Warning	No	D10 and D50 required
B103_8	Chapuis	N/A	Warning	No	D10 required
B103_8	Shepherd	1.21e-04	OK	Yes	OK
B103_8	Krumbein-Monk	7.54e-05	OK	Yes	OK
B103_10	Hazen	2.54e-04	OK	Yes	OK
B103_10	Hazen_1892	1.72e-04	Error	No	Empirical mm ² variant (10°C reference)
B103_10	Slichter	8.94e-05	OK	Yes	OK
B103_10	Terzaghi	1.57e-04	OK	Yes	Reference 10°C calibration
B103_10	Beyer	2.00e-04	OK	Yes	Within recommended range
B103_10	Sauerbrei	2.27e-04	OK	Yes	OK
B103_10	Kruger	4.64e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B103_10	Kozeny-Carman	6.17e-04	OK	Yes	OK
B103_10	Zunker	5.76e-04	OK	Yes	OK
B103_10	Zamarin	6.24e-04	OK	Yes	OK
B103_10	USBR	1.06e-04	OK	Yes	OK
B103_10	Barr	1.33e-04	OK	Yes	OK
B103_10	Alyamani-Sen	1.16e-04	OK	Yes	OK
B103_10	Chapuis	1.90e-04	OK	Yes	OK
B103_10	Shepherd	2.57e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B103_10	Krumbein-Monk	2.67e-04	OK	Yes	OK
B103_12	Hazen	1.69e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B103_12	Hazen_1892	1.82e-04	Error	No	Empirical mm ² variant (10°C reference)
B103_12	Slichter	4.10e-05	OK	Yes	OK
B103_12	Terzaghi	6.77e-05	OK	Yes	Reference 10°C calibration
B103_12	Beyer	1.70e-04	OK	Yes	Within recommended range
B103_12	Sauerbrei	7.60e-05	OK	Yes	OK
B103_12	Kruger	4.13e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B103_12	Kozeny-Carman	4.45e-04	OK	Yes	OK
B103_12	Zunker	3.98e-04	OK	Yes	OK
B103_12	Zamarin	4.87e-04	OK	Yes	OK
B103_12	USBR	9.75e-05	Warning	No	U ≥ 5 exceeds USBR limit
B103_12	Barr	4.90e-05	OK	Yes	OK
B103_12	Alyamani-Sen	1.09e-05	OK	Yes	OK
B103_12	Chapuis	4.92e-05	OK	Yes	OK
B103_12	Shepherd	7.41e-04	OK	Yes	OK
B103_12	Krumbein-Monk	2.53e-04	OK	Yes	OK
B103_14	Hazen	9.38e-05	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B103_14	Hazen_1892	7.00e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
B103_14	Slichter	3.01e-05	OK	Yes	OK
B103_14	Terzaghi	5.24e-05	OK	Yes	Reference 10°C calibration

Sample	Method	K (m/s)	Status	Included	Notes
B103_14	Beyer	7.74e-05	OK	Yes	Within recommended range
B103_14	Sauerbrei	9.02e-05	OK	Yes	OK
B103_14	Kruger	2.79e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B103_14	Kozeny-Carman	3.25e-04	OK	Yes	OK
B103_14	Zunker	3.27e-04	OK	Yes	OK
B103_14	Zamarin	3.70e-04	OK	Yes	OK
B103_14	USBR	5.24e-05	OK	Yes	OK
B103_14	Barr	4.21e-05	OK	Yes	OK
B103_14	Alyamani-Sen	3.28e-05	OK	Yes	OK
B103_14	Chapuis	4.54e-05	Warning	No	D10 > 0.10 mm not satisfied
B103_14	Shepherd	1.65e-04	OK	Yes	OK
B103_14	Krumbein-Monk	1.73e-04	OK	Yes	OK
B103_16	Hazen	2.36e-04	OK	Yes	OK
B103_16	Hazen_1892	1.46e-04	Error	No	Empirical mm ² variant (10°C reference)
B103_16	Slichter	9.17e-05	OK	Yes	OK
B103_16	Terzaghi	1.62e-04	OK	Yes	Reference 10°C calibration
B103_16	Beyer	1.81e-04	OK	Yes	Within recommended range
B103_16	Sauerbrei	1.89e-04	OK	Yes	OK
B103_16	Kruger	2.70e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B103_16	Kozeny-Carman	5.81e-04	OK	Yes	OK
B103_16	Zunker	3.54e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B103_16	Zamarin	3.64e-04	OK	Yes	OK
B103_16	USBR	5.76e-05	OK	Yes	OK
B103_16	Barr	1.45e-04	OK	Yes	OK
B103_16	Alyamani-Sen	1.49e-04	OK	Yes	OK
B103_16	Chapuis	2.44e-04	Warning	No	Uniformity coefficient outside 2–12
B103_16	Shepherd	1.31e-04	OK	Yes	OK
B103_16	Krumbein-Monk	1.41e-04	OK	Yes	OK
B103_18	Hazen	3.07e-04	OK	Yes	OK
B103_18	Hazen_1892	2.02e-04	Error	No	Empirical mm ² variant (10°C reference)
B103_18	Slichter	1.11e-04	OK	Yes	OK
B103_18	Terzaghi	1.95e-04	OK	Yes	Reference 10°C calibration
B103_18	Beyer	2.40e-04	OK	Yes	Within recommended range
B103_18	Sauerbrei	2.60e-04	OK	Yes	OK
B103_18	Kruger	4.26e-04	Warning	No	Best for medium sand (0.25 < D ₅₀ < 0.5 mm, UC > 5)
B103_18	Kozeny-Carman	7.91e-04	OK	Yes	OK
B103_18	Zunker	5.38e-04	OK	Yes	OK
B103_18	Zamarin	5.74e-04	OK	Yes	OK
B103_18	USBR	1.10e-04	OK	Yes	OK
B103_18	Barr	1.68e-04	OK	Yes	OK
B103_18	Alyamani-Sen	1.57e-04	OK	Yes	OK
B103_18	Chapuis	2.55e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B103_18	Shepherd	2.53e-04	OK	Yes	OK
B103_18	Krumbein-Monk	2.64e-04	OK	Yes	OK
B104_9	Hazen	6.28e-04	OK	Yes	OK
B104_9	Hazen_1892	3.91e-04	Error	No	Empirical mm ² variant (10°C reference)
B104_9	Slichter	2.41e-04	OK	Yes	OK
B104_9	Terzaghi	4.25e-04	OK	Yes	Reference 10°C calibration
B104_9	Beyer	4.83e-04	OK	Yes	Within recommended range
B104_9	Sauerbrei	5.72e-04	OK	Yes	OK
B104_9	Kruger	6.40e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B104_9	Kozeny-Carman	1.66e-03	OK	Yes	OK
B104_9	Zunker	8.36e-04	OK	Yes	OK
B104_9	Zamarin	8.64e-04	OK	Yes	OK
B104_9	USBR	2.25e-04	OK	Yes	OK
B104_9	Barr	3.80e-04	OK	Yes	OK
B104_9	Alyamani-Sen	3.82e-04	OK	Yes	OK
B104_9	Chapuis	6.68e-04	OK	Yes	OK
B104_9	Shepherd	3.19e-04	OK	Yes	OK
B104_9	Krumbein-Monk	4.02e-04	OK	Yes	OK
B104_11	Hazen	4.58e-04	OK	Yes	OK
B104_11	Hazen_1892	2.90e-04	Error	No	Empirical mm ² variant (10°C reference)
B104_11	Slichter	1.73e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B104_11	Terzaghi	3.04e-04	OK	Yes	Reference 10°C calibration
B104_11	Beyer	3.54e-04	OK	Yes	Within recommended range
B104_11	Sauerbrei	4.23e-04	OK	Yes	OK
B104_11	Kruger	5.03e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B104_11	Kozeny-Carman	1.23e-03	OK	Yes	OK
B104_11	Zunker	6.51e-04	OK	Yes	OK
B104_11	Zamarin	6.80e-04	OK	Yes	OK
B104_11	USBR	1.71e-04	OK	Yes	OK
B104_11	Barr	2.68e-04	OK	Yes	OK
B104_11	Alyamani-Sen	2.63e-04	OK	Yes	OK
B104_11	Chapuis	4.50e-04	OK	Yes	OK
B104_11	Shepherd	2.78e-04	OK	Yes	OK
B104_11	Krumbein-Monk	3.35e-04	OK	Yes	OK
B104_13	Hazen	2.57e-04	OK	Yes	OK
B104_13	Hazen_1892	1.74e-04	Error	No	Empirical mm ² variant (10°C reference)
B104_13	Slichter	9.08e-05	OK	Yes	OK
B104_13	Terzaghi	1.59e-04	OK	Yes	Reference 10°C calibration
B104_13	Beyer	2.03e-04	OK	Yes	Within recommended range
B104_13	Sauerbrei	2.36e-04	OK	Yes	OK
B104_13	Kruger	3.66e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B104_13	Kozeny-Carman	7.71e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B104_13	Zunker	4.55e-04	OK	Yes	OK
B104_13	Zamarin	4.93e-04	OK	Yes	OK
B104_13	USBR	1.12e-04	OK	Yes	OK
B104_13	Barr	1.35e-04	OK	Yes	OK
B104_13	Alyamani-Sen	1.17e-04	OK	Yes	OK
B104_13	Chapuis	1.93e-04	OK	Yes	OK
B104_13	Shepherd	2.61e-04	OK	Yes	OK
B104_13	Krumbein-Monk	2.43e-04	OK	Yes	OK
B104_15	Hazen	4.35e-04	OK	Yes	OK
B104_15	Hazen_1892	2.77e-04	Error	No	Empirical mm ² variant (10°C reference)
B104_15	Slichter	1.63e-04	OK	Yes	OK
B104_15	Terzaghi	2.87e-04	OK	Yes	Reference 10°C calibration
B104_15	Beyer	3.36e-04	OK	Yes	Within recommended range
B104_15	Sauerbrei	4.06e-04	OK	Yes	OK
B104_15	Kruger	4.59e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B104_15	Kozeny-Carman	1.24e-03	OK	Yes	OK
B104_15	Zunker	5.91e-04	OK	Yes	OK
B104_15	Zamarin	6.20e-04	OK	Yes	OK
B104_15	USBR	1.68e-04	OK	Yes	OK
B104_15	Barr	2.52e-04	OK	Yes	OK
B104_15	Alyamani-Sen	2.45e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B104_15	Chapuis	4.17e-04	OK	Yes	OK
B104_15	Shepherd	2.78e-04	OK	Yes	OK
B104_15	Krumbein-Monk	3.12e-04	OK	Yes	OK
B104_17	Hazen	2.91e-04	OK	Yes	OK
B104_17	Hazen_1892	1.89e-04	Error	No	Empirical mm ² variant (10°C reference)
B104_17	Slichter	1.07e-04	OK	Yes	OK
B104_17	Terzaghi	1.88e-04	OK	Yes	Reference 10°C calibration
B104_17	Beyer	2.27e-04	OK	Yes	Within recommended range
B104_17	Sauerbrei	2.24e-04	OK	Yes	OK
B104_17	Kruger	3.50e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B104_17	Kozeny-Carman	7.58e-04	OK	Yes	OK
B104_17	Zunker	4.45e-04	OK	Yes	OK
B104_17	Zamarin	4.72e-04	OK	Yes	OK
B104_17	USBR	8.47e-05	OK	Yes	OK
B104_17	Barr	1.63e-04	OK	Yes	OK
B104_17	Alyamani-Sen	1.62e-04	OK	Yes	OK
B104_17	Chapuis	2.52e-04	OK	Yes	OK
B104_17	Shepherd	2.11e-04	OK	Yes	OK
B104_17	Krumbein-Monk	2.14e-04	OK	Yes	OK
B104_19	Hazen	1.74e-04	OK	Yes	OK
B104_19	Hazen_1892	1.18e-04	Error	No	Empirical mm ² variant (10°C reference)

Sample	Method	K (m/s)	Status	Included	Notes
B104_19	Slichter	6.10e-05	OK	Yes	OK
B104_19	Terzaghi	1.07e-04	OK	Yes	Reference 10°C calibration
B104_19	Beyer	1.37e-04	OK	Yes	Within recommended range
B104_19	Sauerbrei	1.43e-04	OK	Yes	OK
B104_19	Kruger	2.72e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B104_19	Kozeny-Carman	5.02e-04	OK	Yes	OK
B104_19	Zunker	3.37e-04	OK	Yes	OK
B104_19	Zamarin	3.66e-04	OK	Yes	OK
B104_19	USBR	6.12e-05	OK	Yes	OK
B104_19	Barr	9.03e-05	OK	Yes	OK
B104_19	Alyamani-Sen	8.78e-05	OK	Yes	OK
B104_19	Chapuis	1.23e-04	OK	Yes	OK
B104_19	Shepherd	1.70e-04	OK	Yes	OK
B104_19	Krumbein-Monk	1.62e-04	OK	Yes	OK
B104_21	Hazen	2.48e-04	OK	Yes	OK
B104_21	Hazen_1892	1.66e-04	Error	No	Empirical mm ² variant (10°C reference)
B104_21	Slichter	8.86e-05	OK	Yes	OK
B104_21	Terzaghi	1.56e-04	OK	Yes	Reference 10°C calibration
B104_21	Beyer	1.95e-04	OK	Yes	Within recommended range
B104_21	Sauerbrei	1.95e-04	OK	Yes	OK
B104_21	Kruger	3.75e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)

Sample	Method	K (m/s)	Status	Included	Notes
B104_21	Kozeny-Carman	6.21e-04	OK	Yes	OK
B104_21	Zunker	4.69e-04	OK	Yes	OK
B104_21	Zamarin	5.05e-04	OK	Yes	OK
B104_21	USBR	8.06e-05	OK	Yes	OK
B104_21	Barr	1.33e-04	OK	Yes	OK
B104_21	Alyamani-Sen	1.25e-04	OK	Yes	OK
B104_21	Chapuis	1.93e-04	OK	Yes	OK
B104_21	Shepherd	2.21e-04	OK	Yes	OK
B104_21	Krumbein-Monk	2.27e-04	OK	Yes	OK
B105_10	Hazen	1.44e-04	OK	Yes	OK
B105_10	Hazen_1892	1.11e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_10	Slichter	4.52e-05	OK	Yes	OK
B105_10	Terzaghi	7.85e-05	OK	Yes	Reference 10°C calibration
B105_10	Beyer	1.21e-04	OK	Yes	Within recommended range
B105_10	Sauerbrei	1.48e-04	OK	Yes	OK
B105_10	Kruger	5.27e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_10	Kozeny-Carman	5.13e-04	OK	Yes	OK
B105_10	Zunker	6.08e-04	OK	Yes	OK
B105_10	Zamarin	6.96e-04	OK	Yes	OK
B105_10	USBR	1.03e-04	OK	Yes	OK
B105_10	Barr	6.22e-05	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B105_10	Alyamani-Sen	3.49e-05	OK	Yes	OK
B105_10	Chapuis	6.98e-05	OK	Yes	OK
B105_10	Shepherd	3.04e-04	OK	Yes	OK
B105_10	Krumbein-Monk	1.42e-04	OK	Yes	OK
B105_12	Hazen	3.61e-04	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
B105_12	Hazen_1892	5.64e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_12	Slichter	7.33e-05	OK	Yes	OK
B105_12	Terzaghi	1.09e-04	OK	Yes	Reference 10°C calibration
B105_12	Beyer	4.34e-04	OK	Yes	Within recommended range
B105_12	Sauerbrei	2.77e-04	OK	Yes	OK
B105_12	Kruger	1.33e-03	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_12	Kozeny-Carman	1.26e-03	OK	Yes	OK
B105_12	Zunker	1.08e-03	OK	Yes	OK
B105_12	Zamarin	1.33e-03	OK	Yes	OK
B105_12	USBR	1.13e-03	Warning	No	U ≥ 5 exceeds USBR limit
B105_12	Barr	8.02e-05	OK	Yes	OK
B105_12	Alyamani-Sen	2.14e-03	Warning	No	Effective diameter non-positive
B105_12	Chapuis	1.25e-04	Warning	No	0.3 < n < 0.7 not satisfied; Uniformity coefficient outside 2–12
B105_12	Shepherd	9.95e-03	Warning	No	D50 outside 0.0063–2.0 mm range
B105_12	Krumbein-Monk	2.32e-03	OK	Yes	OK
B105_14	Hazen	2.56e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B105_14	Hazen_1892	1.74e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_14	Slichter	9.01e-05	OK	Yes	OK
B105_14	Terzaghi	1.58e-04	OK	Yes	Reference 10°C calibration
B105_14	Beyer	2.03e-04	OK	Yes	Within recommended range
B105_14	Sauerbrei	2.22e-04	OK	Yes	OK
B105_14	Kruger	3.75e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_14	Kozeny-Carman	7.70e-04	OK	Yes	OK
B105_14	Zunker	4.65e-04	OK	Yes	OK
B105_14	Zamarin	5.05e-04	OK	Yes	OK
B105_14	USBR	1.02e-04	OK	Yes	OK
B105_14	Barr	1.33e-04	OK	Yes	OK
B105_14	Alyamani-Sen	1.17e-04	OK	Yes	OK
B105_14	Chapuis	1.91e-04	OK	Yes	OK
B105_14	Shepherd	2.59e-04	OK	Yes	OK
B105_14	Krumbein-Monk	2.55e-04	OK	Yes	OK
B105_16	Hazen	4.43e-04	OK	Yes	OK
B105_16	Hazen_1892	2.97e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_16	Slichter	1.57e-04	OK	Yes	OK
B105_16	Terzaghi	2.76e-04	OK	Yes	Reference 10°C calibration
B105_16	Beyer	3.49e-04	OK	Yes	Within recommended range
B105_16	Sauerbrei	4.11e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B105_16	Kruger	6.65e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_16	Kozeny-Carman	1.30e-03	OK	Yes	OK
B105_16	Zunker	8.30e-04	OK	Yes	OK
B105_16	Zamarin	8.96e-04	OK	Yes	OK
B105_16	USBR	2.06e-04	OK	Yes	OK
B105_16	Barr	2.35e-04	OK	Yes	OK
B105_16	Alyamani-Sen	2.13e-04	OK	Yes	OK
B105_16	Chapuis	3.62e-04	OK	Yes	OK
B105_16	Shepherd	3.79e-04	OK	Yes	OK
B105_16	Krumbein-Monk	3.78e-04	OK	Yes	OK
B105_18	Hazen	3.50e-04	OK	Yes	OK
B105_18	Hazen_1892	2.28e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_18	Slichter	1.28e-04	OK	Yes	OK
B105_18	Terzaghi	2.25e-04	OK	Yes	Reference 10°C calibration
B105_18	Beyer	2.73e-04	OK	Yes	Within recommended range
B105_18	Sauerbrei	3.05e-04	OK	Yes	OK
B105_18	Kruger	4.45e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_18	Kozeny-Carman	9.57e-04	OK	Yes	OK
B105_18	Zunker	5.65e-04	OK	Yes	OK
B105_18	Zamarin	6.00e-04	OK	Yes	OK
B105_18	USBR	1.29e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B105_18	Barr	1.95e-04	OK	Yes	OK
B105_18	Alyamani-Sen	1.84e-04	OK	Yes	OK
B105_18	Chapuis	3.04e-04	OK	Yes	OK
B105_18	Shepherd	2.68e-04	OK	Yes	OK
B105_18	Krumbein-Monk	2.95e-04	OK	Yes	OK
B105_20	Hazen	2.95e-04	OK	Yes	OK
B105_20	Hazen_1892	1.94e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_20	Slichter	1.07e-04	OK	Yes	OK
B105_20	Terzaghi	1.87e-04	OK	Yes	Reference 10°C calibration
B105_20	Beyer	2.31e-04	OK	Yes	Within recommended range
B105_20	Sauerbrei	2.44e-04	OK	Yes	OK
B105_20	Kruger	3.86e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_20	Kozeny-Carman	7.89e-04	OK	Yes	OK
B105_20	Zunker	4.88e-04	OK	Yes	OK
B105_20	Zamarin	5.21e-04	OK	Yes	OK
B105_20	USBR	1.01e-04	OK	Yes	OK
B105_20	Barr	1.61e-04	OK	Yes	OK
B105_20	Alyamani-Sen	1.51e-04	OK	Yes	OK
B105_20	Chapuis	2.44e-04	OK	Yes	OK
B105_20	Shepherd	2.44e-04	OK	Yes	OK
B105_20	Krumbein-Monk	2.42e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B105_22	Hazen	2.73e-04	OK	Yes	OK
B105_22	Hazen_1892	1.81e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_22	Slichter	9.86e-05	OK	Yes	OK
B105_22	Terzaghi	1.73e-04	OK	Yes	Reference 10°C calibration
B105_22	Beyer	2.14e-04	OK	Yes	Within recommended range
B105_22	Sauerbrei	2.19e-04	OK	Yes	OK
B105_22	Kruger	3.64e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_22	Kozeny-Carman	7.17e-04	OK	Yes	OK
B105_22	Zunker	4.59e-04	OK	Yes	OK
B105_22	Zamarin	4.91e-04	OK	Yes	OK
B105_22	USBR	8.93e-05	OK	Yes	OK
B105_22	Barr	1.49e-04	OK	Yes	OK
B105_22	Alyamani-Sen	1.41e-04	OK	Yes	OK
B105_22	Chapuis	2.22e-04	OK	Yes	OK
B105_22	Shepherd	2.30e-04	OK	Yes	OK
B105_22	Krumbein-Monk	2.23e-04	OK	Yes	OK
B105_24	Hazen	2.67e-04	OK	Yes	OK
B105_24	Hazen_1892	1.72e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_24	Slichter	9.91e-05	OK	Yes	OK
B105_24	Terzaghi	1.74e-04	OK	Yes	Reference 10°C calibration
B105_24	Beyer	2.07e-04	OK	Yes	Within recommended range

Sample	Method	K (m/s)	Status	Included	Notes
B105_24	Sauerbrei	2.02e-04	OK	Yes	OK
B105_24	Kruger	3.00e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_24	Kozeny-Carman	7.00e-04	OK	Yes	OK
B105_24	Zunker	3.84e-04	OK	Yes	OK
B105_24	Zamarin	4.05e-04	OK	Yes	OK
B105_24	USBR	7.16e-05	OK	Yes	OK
B105_24	Barr	1.52e-04	OK	Yes	OK
B105_24	Alyamani-Sen	1.62e-04	OK	Yes	OK
B105_24	Chapuis	2.38e-04	OK	Yes	OK
B105_24	Shepherd	1.71e-04	OK	Yes	OK
B105_24	Krumbein-Monk	1.77e-04	OK	Yes	OK
B105_26	Hazen	3.62e-04	OK	Yes	OK
B105_26	Hazen_1892	2.34e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_26	Slichter	1.34e-04	OK	Yes	OK
B105_26	Terzaghi	2.36e-04	OK	Yes	Reference 10°C calibration
B105_26	Beyer	2.82e-04	OK	Yes	Within recommended range
B105_26	Sauerbrei	3.04e-04	OK	Yes	OK
B105_26	Kruger	4.15e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_26	Kozeny-Carman	1.02e-03	OK	Yes	OK
B105_26	Zunker	5.30e-04	OK	Yes	OK
B105_26	Zamarin	5.60e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
B105_26	USBR	1.22e-04	OK	Yes	OK
B105_26	Barr	2.05e-04	OK	Yes	OK
B105_26	Alyamani-Sen	1.98e-04	OK	Yes	OK
B105_26	Chapuis	3.27e-04	OK	Yes	OK
B105_26	Shepherd	2.56e-04	OK	Yes	OK
B105_26	Krumbein-Monk	2.73e-04	OK	Yes	OK
B105_28	Hazen	3.18e-04	OK	Yes	OK
B105_28	Hazen_1892	2.07e-04	Error	No	Empirical mm ² variant (10°C reference)
B105_28	Slichter	1.17e-04	OK	Yes	OK
B105_28	Terzaghi	2.05e-04	OK	Yes	Reference 10°C calibration
B105_28	Beyer	2.48e-04	OK	Yes	Within recommended range
B105_28	Sauerbrei	2.56e-04	OK	Yes	OK
B105_28	Kruger	3.72e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
B105_28	Kozeny-Carman	8.87e-04	OK	Yes	OK
B105_28	Zunker	4.73e-04	OK	Yes	OK
B105_28	Zamarin	5.02e-04	OK	Yes	OK
B105_28	USBR	1.01e-04	OK	Yes	OK
B105_28	Barr	1.78e-04	OK	Yes	OK
B105_28	Alyamani-Sen	1.72e-04	OK	Yes	OK
B105_28	Chapuis	2.77e-04	OK	Yes	OK
B105_28	Shepherd	2.39e-04	OK	Yes	OK

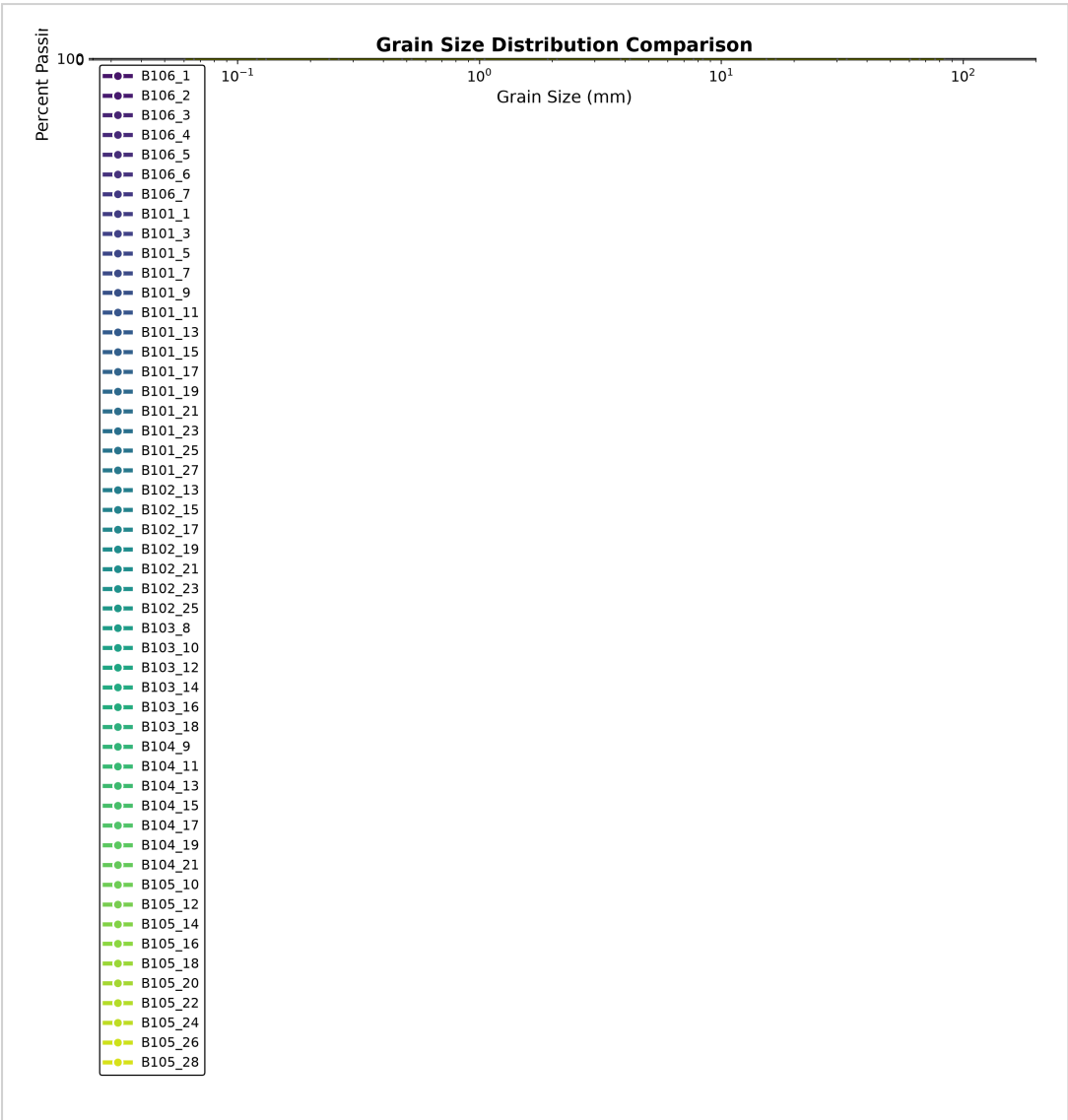
Sample	Method	K (m/s)	Status	Included	Notes
B105_28	Krumbein-Monk	2.39e-04	OK	Yes	OK

Permeability Classification Summary

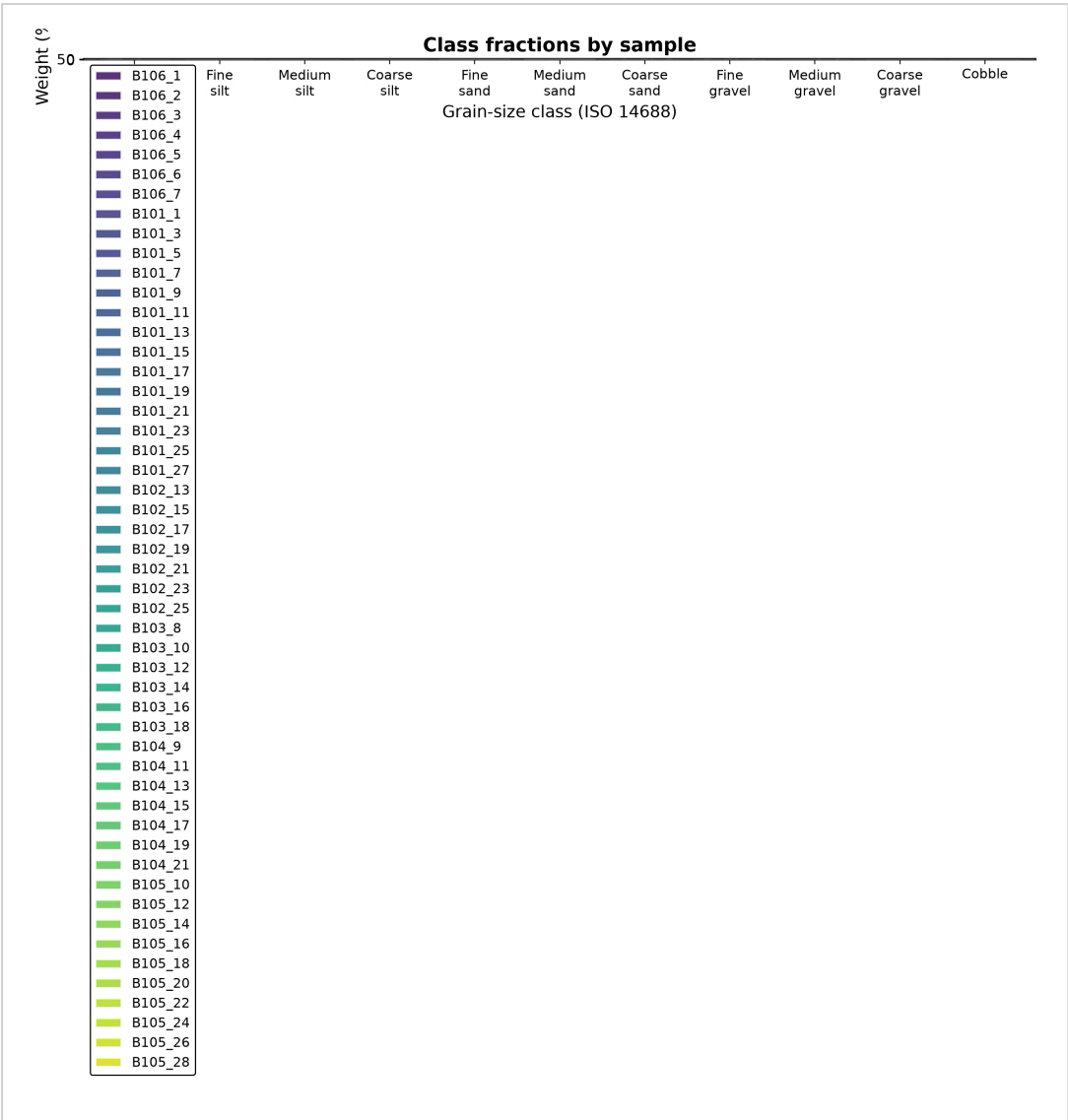
Sample Name	K geometric mean (m/s)	Classification
B106_1	2.41e-04	High (Clean Sand)
B106_2	2.72e-04	High (Clean Sand)
B106_3	7.68e-05	Moderate (Fine Sand)
B106_4	2.74e-04	High (Clean Sand)
B106_5	2.55e-04	High (Clean Sand)
B106_6	8.90e-05	Moderate (Fine Sand)
B106_7	7.99e-05	Moderate (Fine Sand)
B101_1	1.13e-04	High (Clean Sand)
B101_3	1.51e-04	High (Clean Sand)
B101_5	1.14e-04	High (Clean Sand)
B101_7	1.33e-04	High (Clean Sand)
B101_9	1.14e-04	High (Clean Sand)
B101_11	1.74e-04	High (Clean Sand)
B101_13	4.96e-05	Moderate (Fine Sand)
B101_15	8.65e-05	Moderate (Fine Sand)
B101_17	9.37e-05	Moderate (Fine Sand)
B101_19	5.86e-05	Moderate (Fine Sand)
B101_21	8.80e-05	Moderate (Fine Sand)
B101_23	7.46e-05	Moderate (Fine Sand)
B101_25	1.08e-04	High (Clean Sand)
B101_27	1.21e-04	High (Clean Sand)
B102_13	8.37e-05	Moderate (Fine Sand)
B102_15	3.79e-04	High (Clean Sand)
B102_17	2.86e-04	High (Clean Sand)
B102_19	1.01e-04	High (Clean Sand)
B102_21	7.93e-04	High (Clean Sand)

Sample Name	K geometric mean (m/s)	Classification
B102_23	4.27e-04	High (Clean Sand)
B102_25	1.97e-04	High (Clean Sand)
B103_8	1.88e-04	High (Clean Sand)
B103_10	2.23e-04	High (Clean Sand)
B103_12	1.26e-04	High (Clean Sand)
B103_14	9.91e-05	Moderate (Fine Sand)
B103_16	1.80e-04	High (Clean Sand)
B103_18	2.55e-04	High (Clean Sand)
B104_9	4.99e-04	High (Clean Sand)
B104_11	3.72e-04	High (Clean Sand)
B104_13	2.21e-04	High (Clean Sand)
B104_15	3.53e-04	High (Clean Sand)
B104_17	2.31e-04	High (Clean Sand)
B104_19	1.47e-04	High (Clean Sand)
B104_21	2.06e-04	High (Clean Sand)
B105_10	1.42e-04	High (Clean Sand)
B105_12	4.10e-04	High (Clean Sand)
B105_14	2.19e-04	High (Clean Sand)
B105_16	3.82e-04	High (Clean Sand)
B105_18	2.90e-04	High (Clean Sand)
B105_20	2.42e-04	High (Clean Sand)
B105_22	2.23e-04	High (Clean Sand)
B105_24	2.07e-04	High (Clean Sand)
B105_26	2.92e-04	High (Clean Sand)
B105_28	2.55e-04	High (Clean Sand)

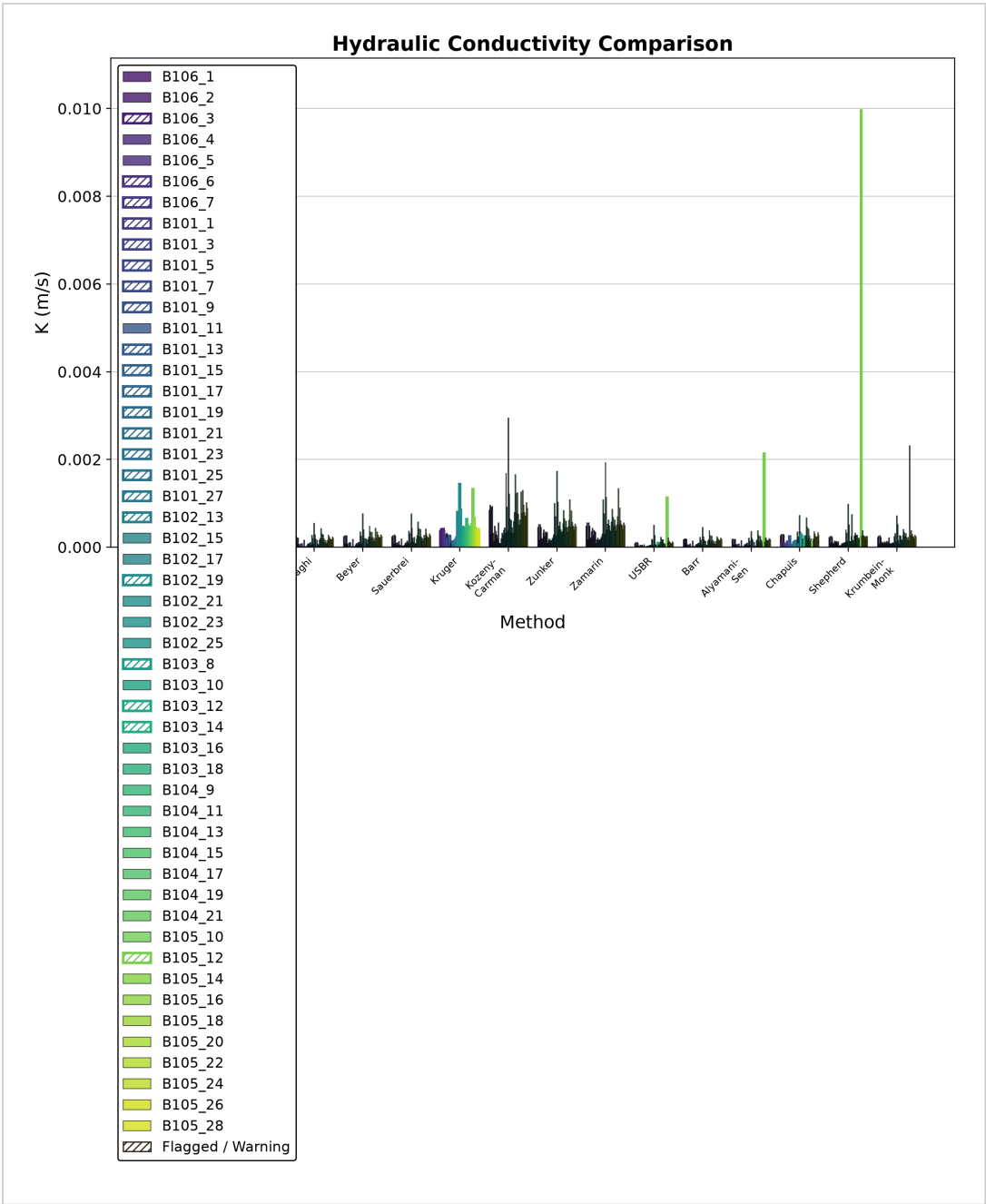
Grain Size Distribution Comparison



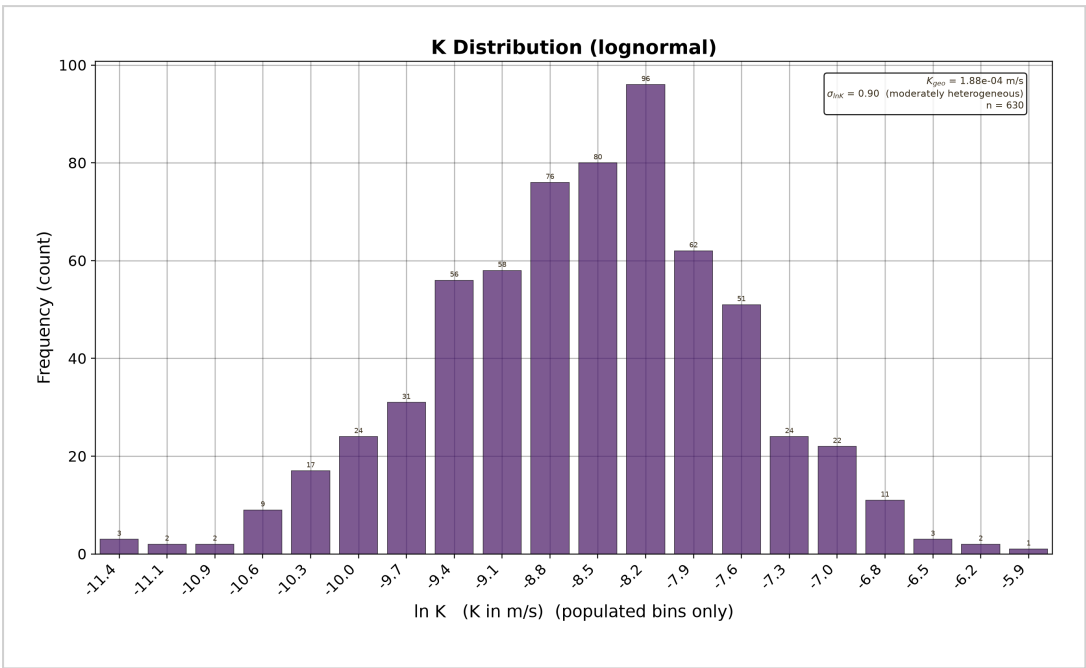
Grain-size Class Histogram



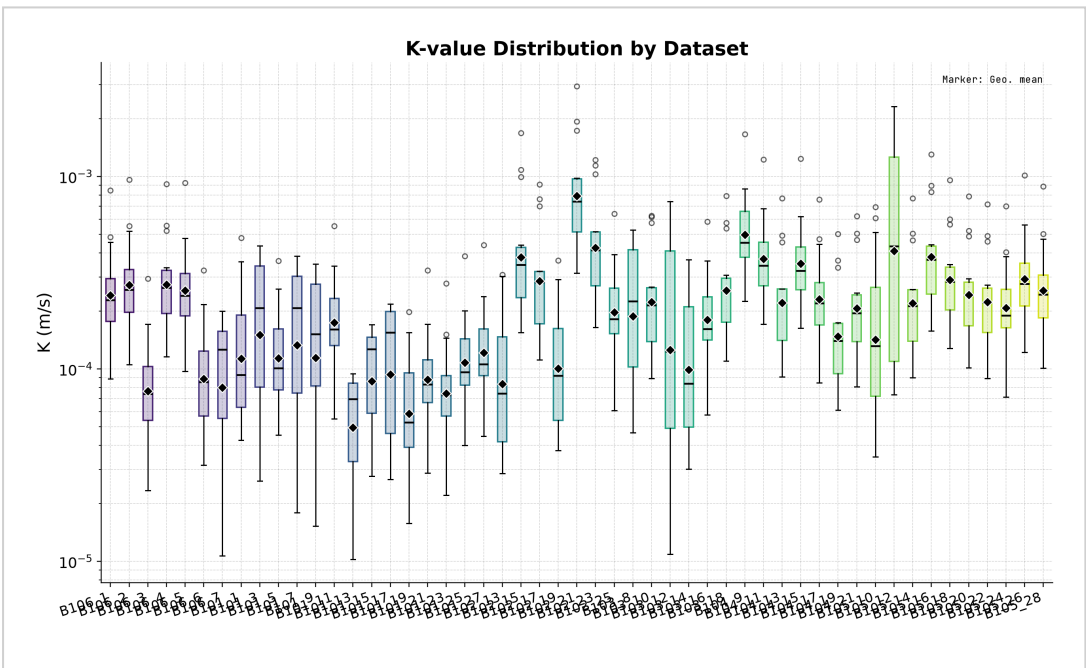
Hydraulic Conductivity by Method

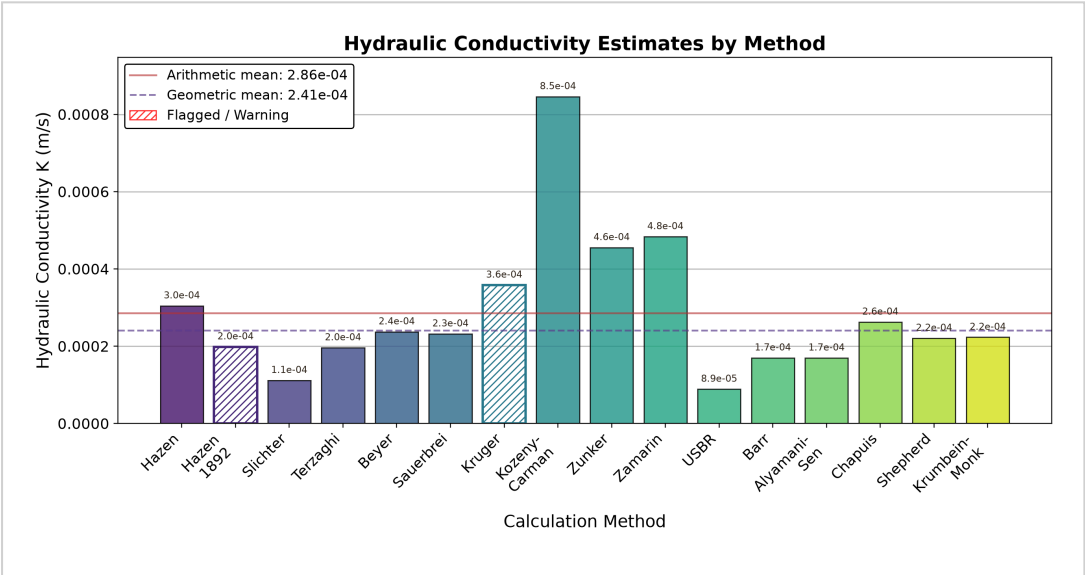
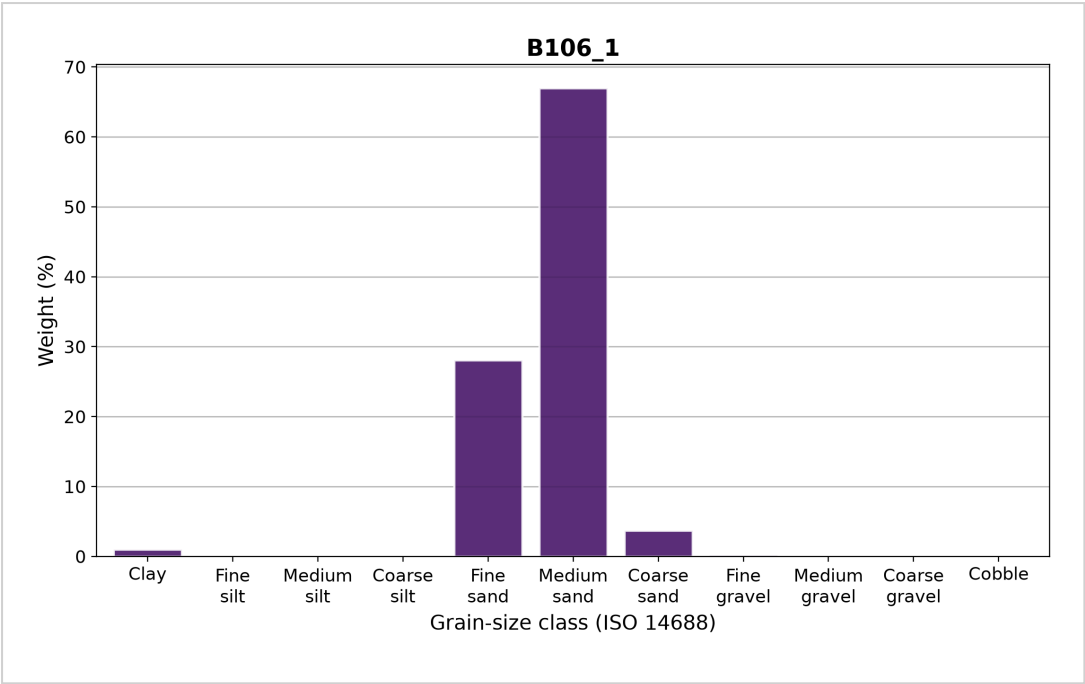


Hydraulic Conductivity Distribution (Lognormal)

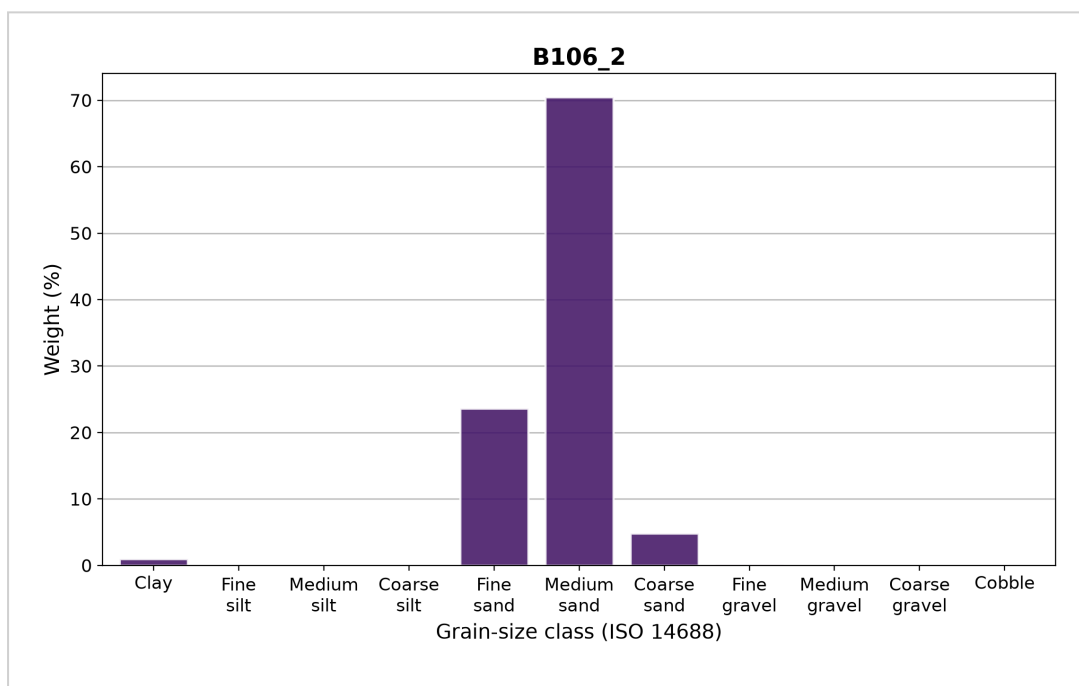
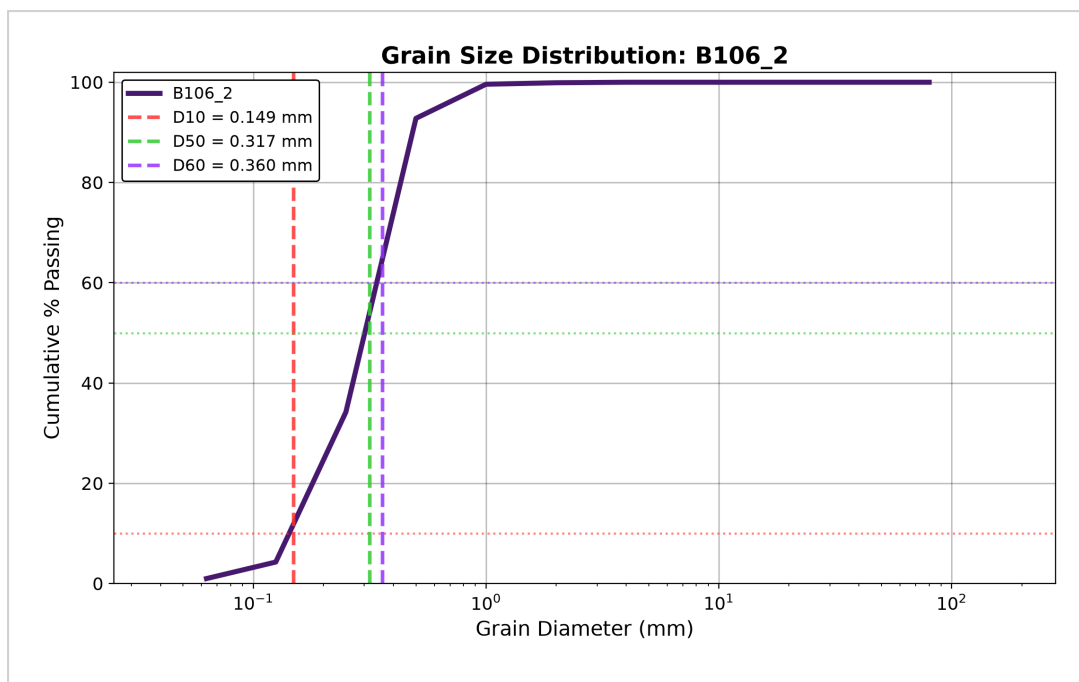


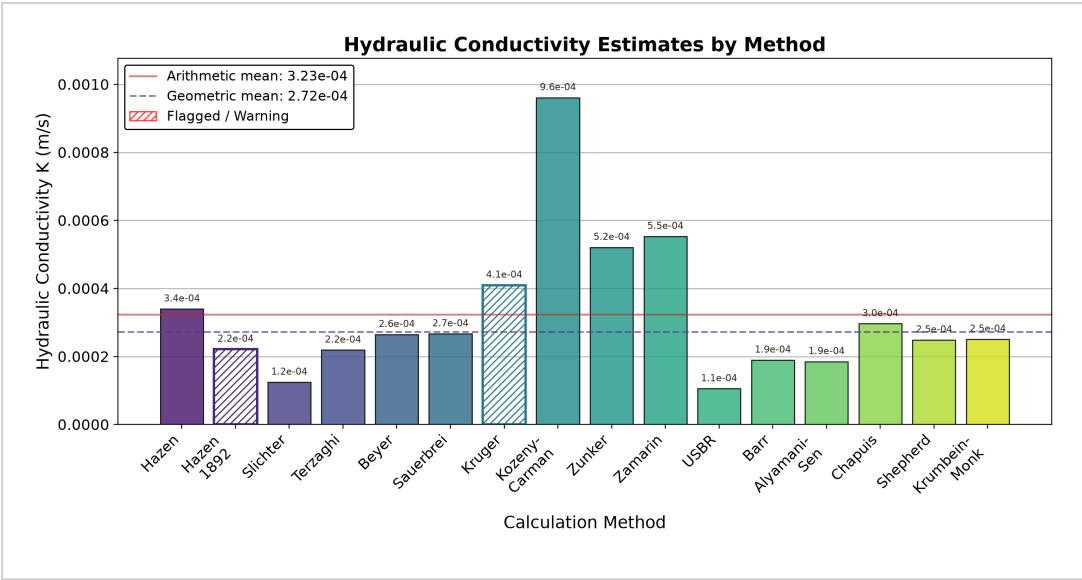
Hydraulic Conductivity Distribution



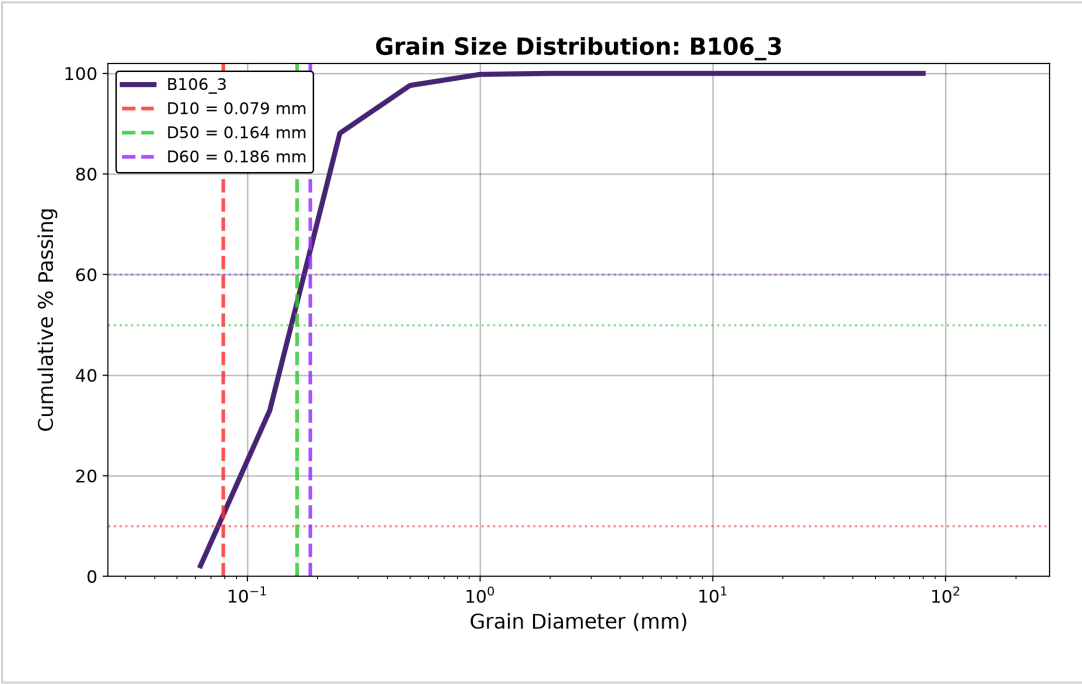


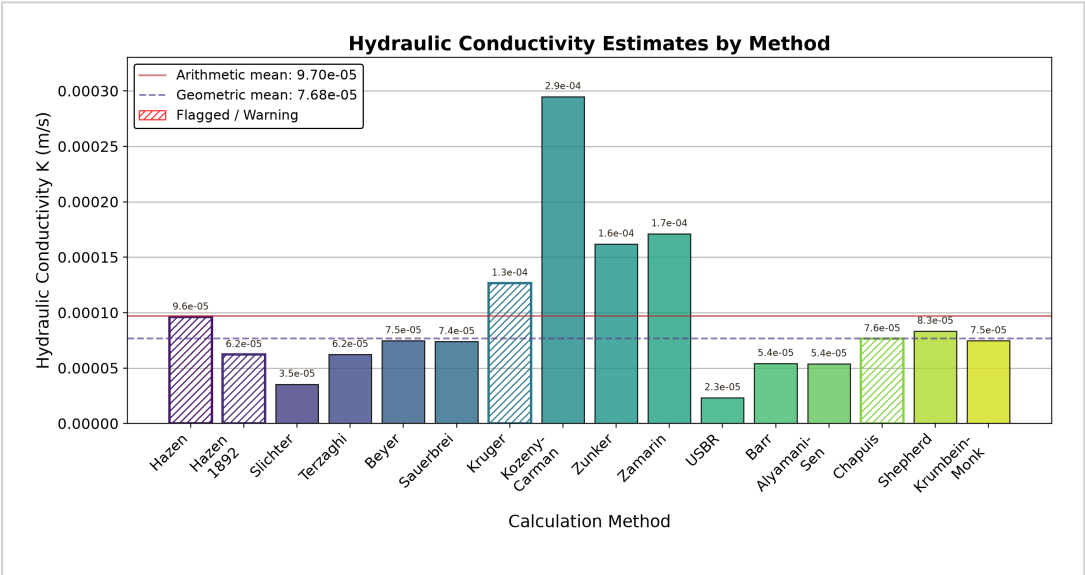
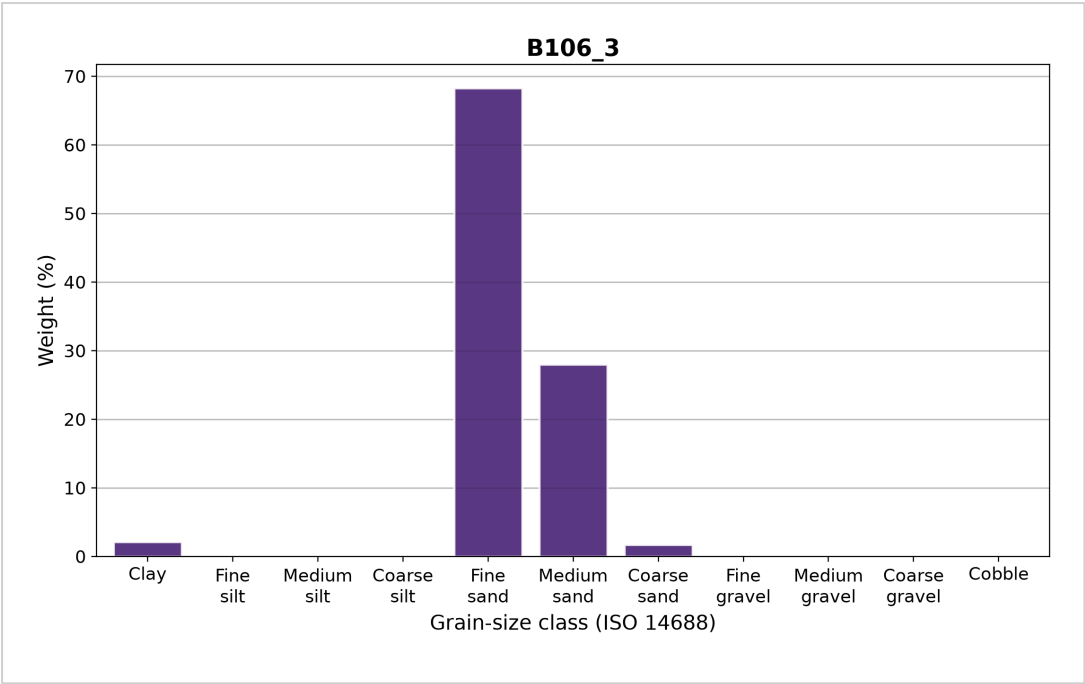
B106_2



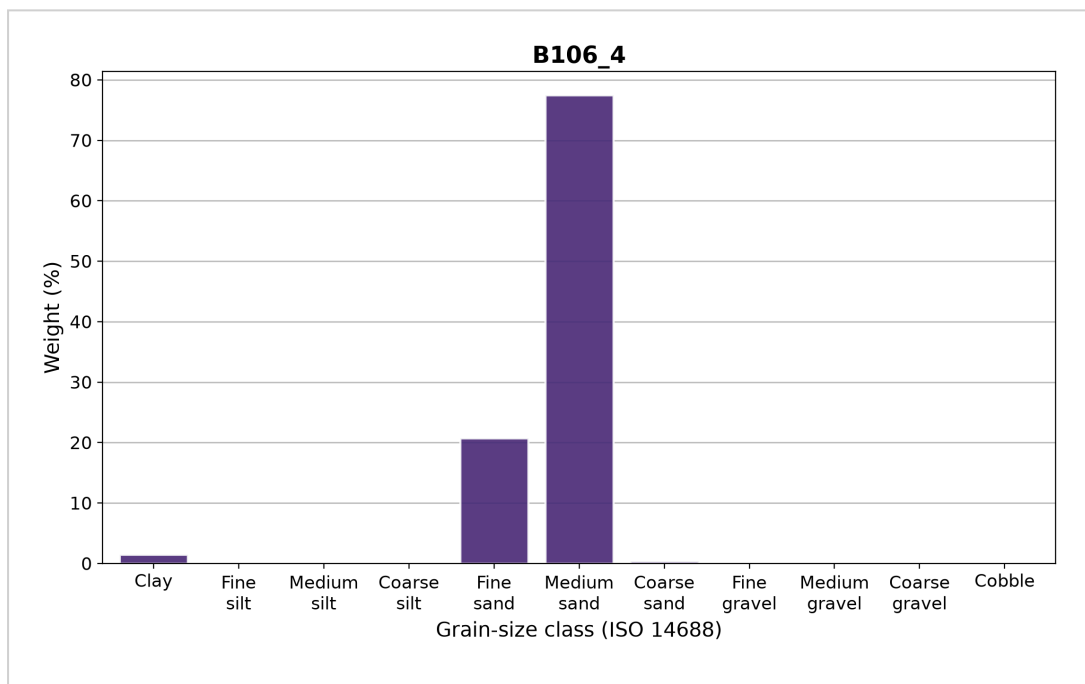
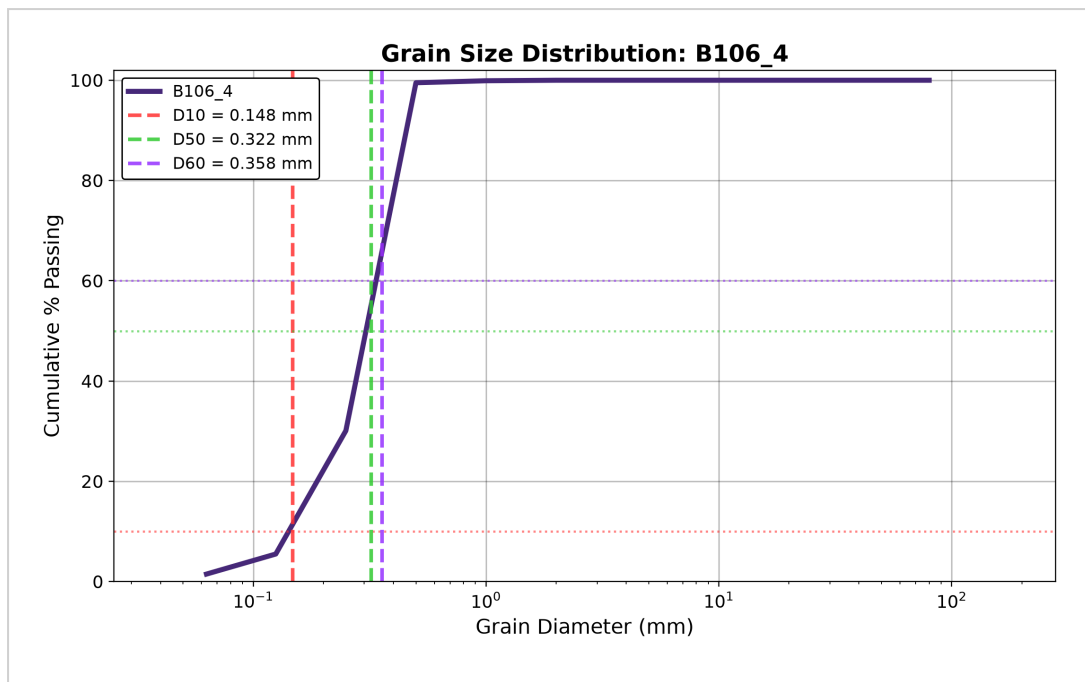


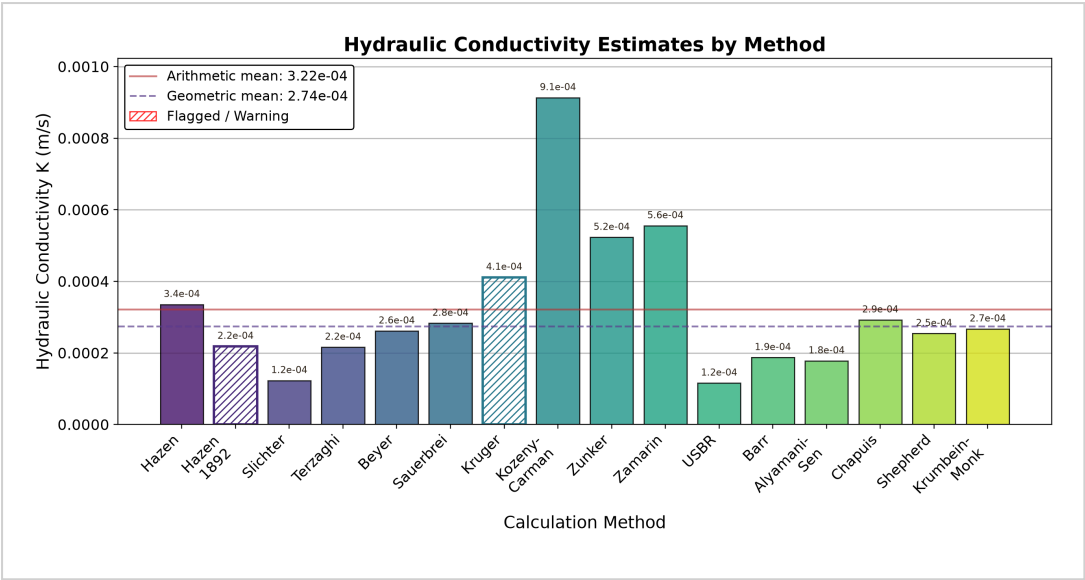
B106_3



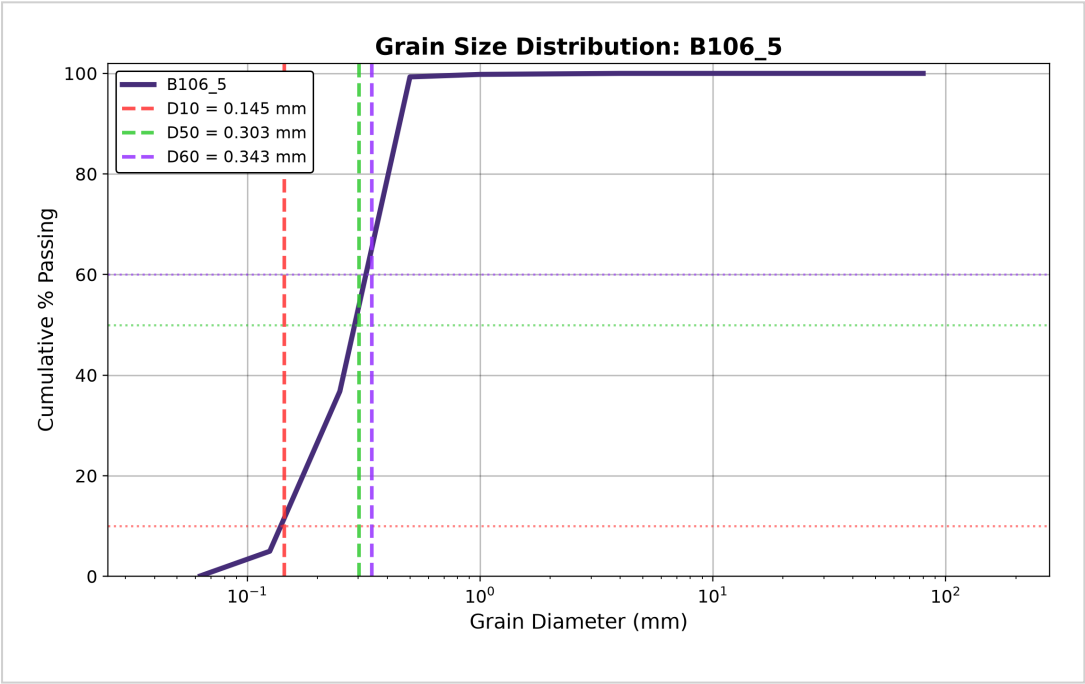


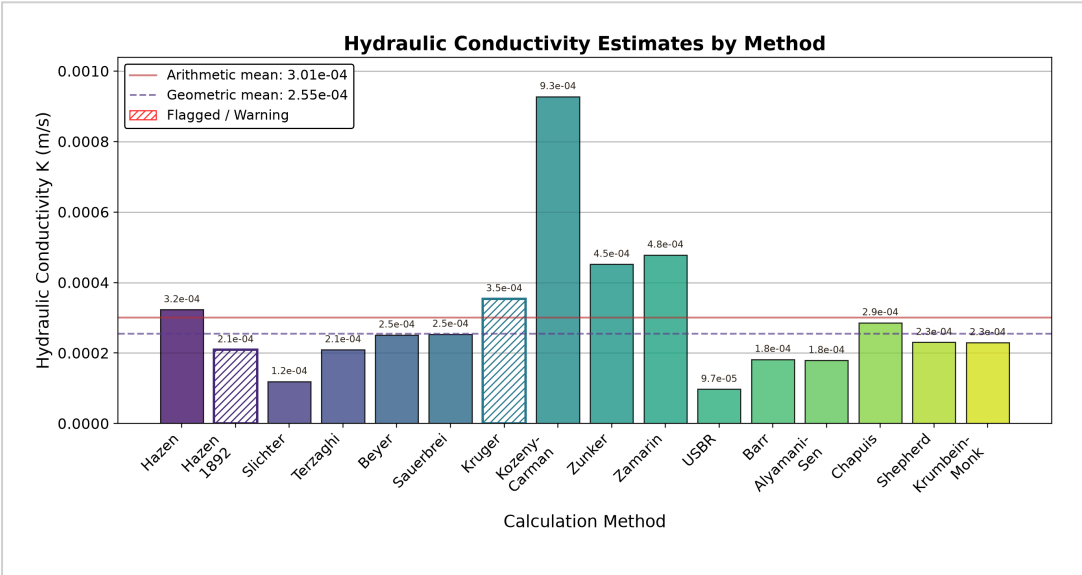
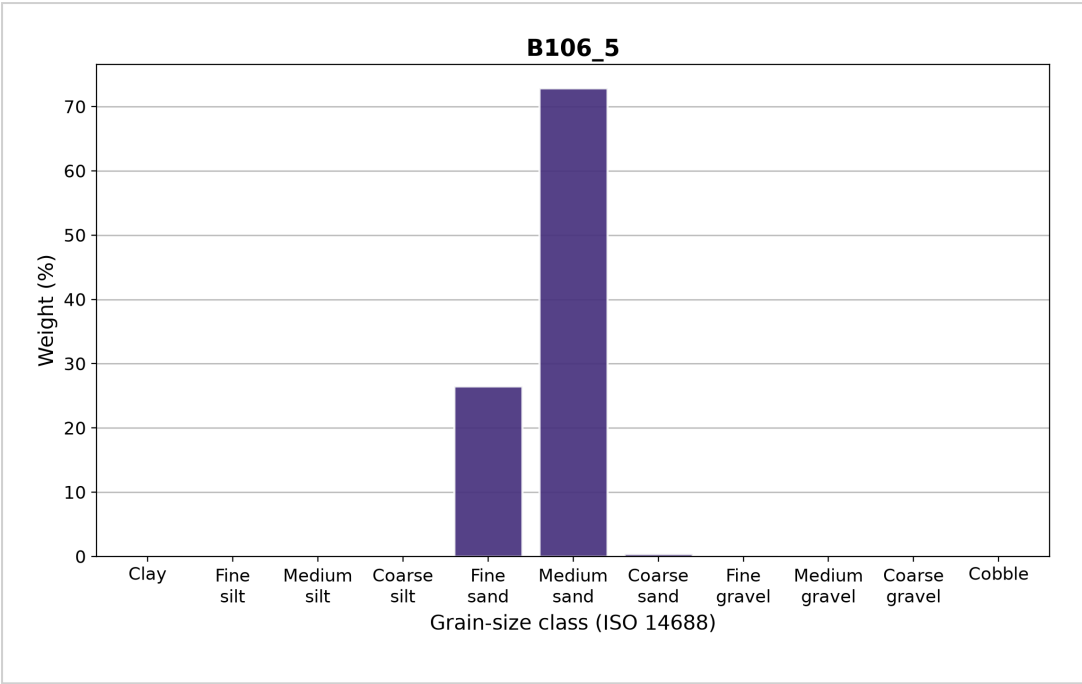
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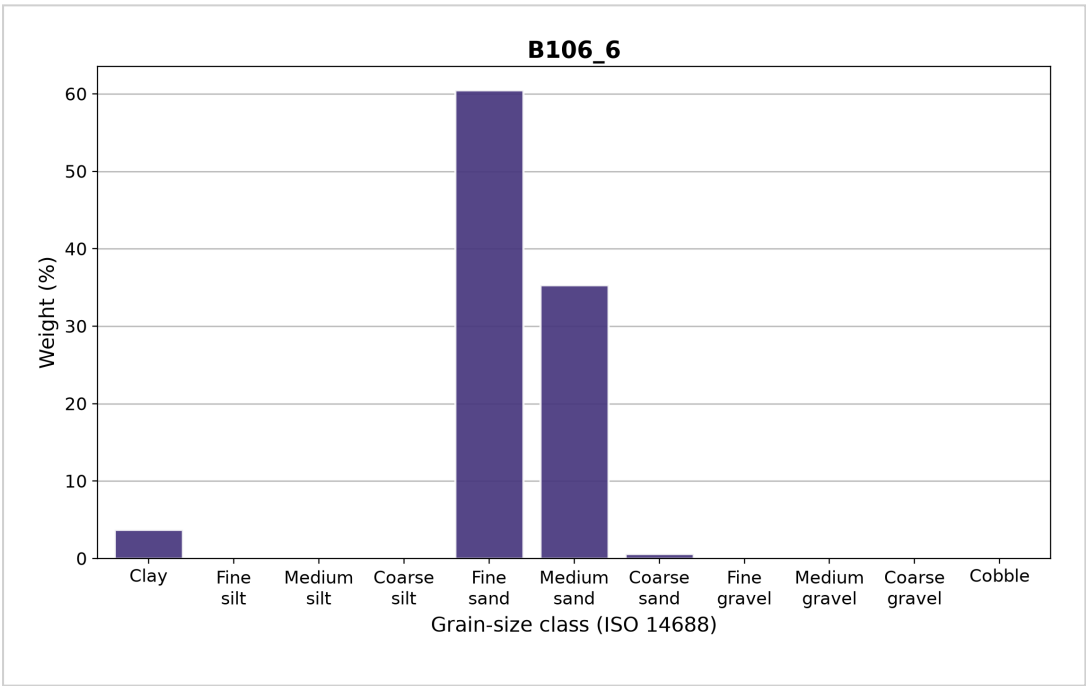
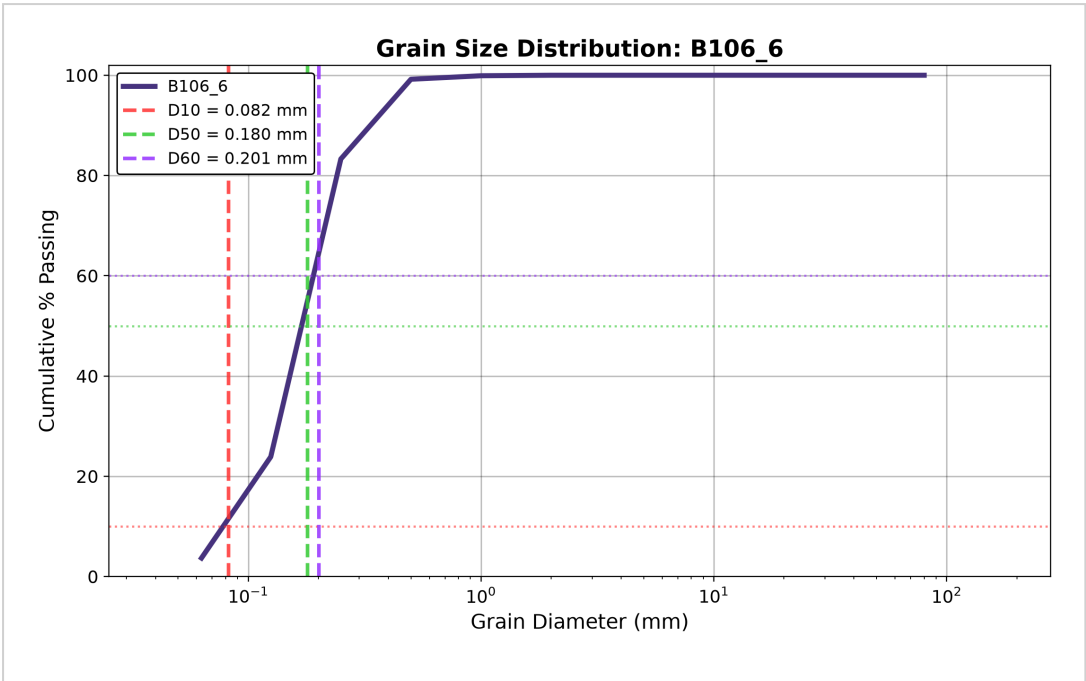


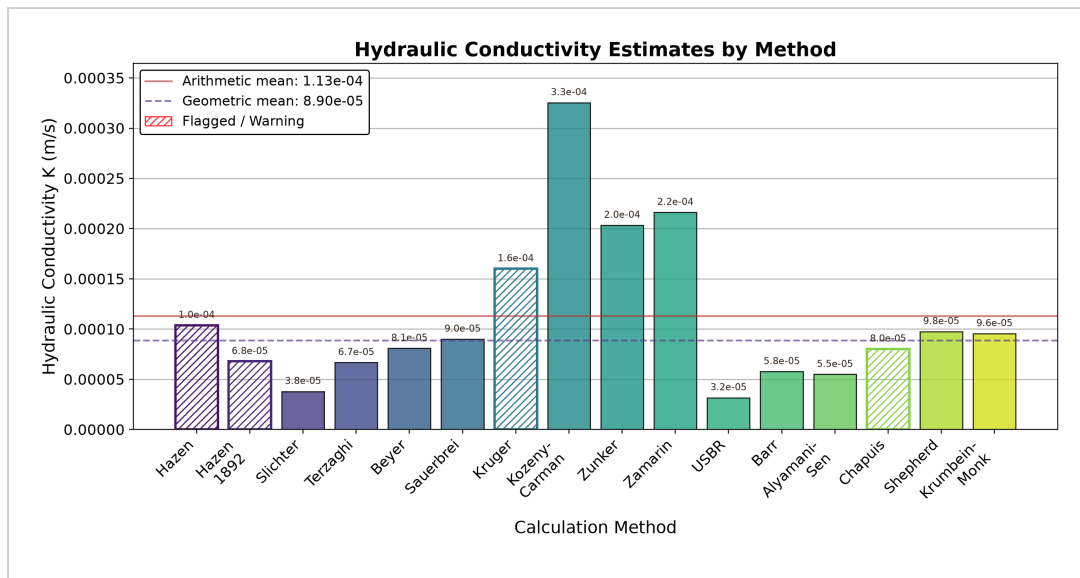
B106_5



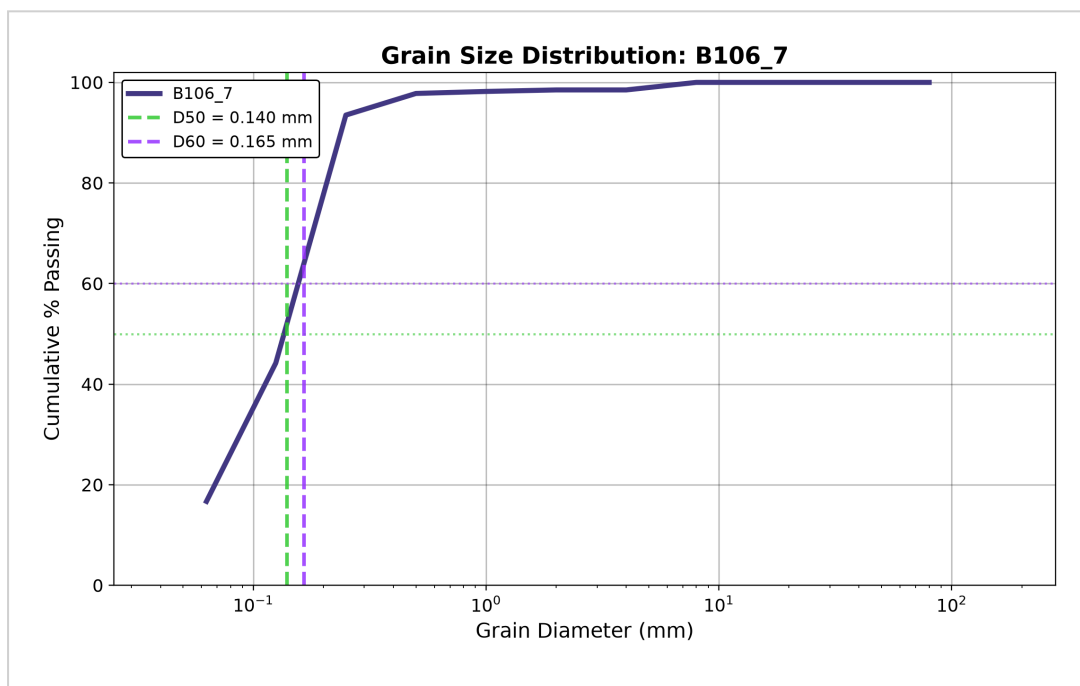


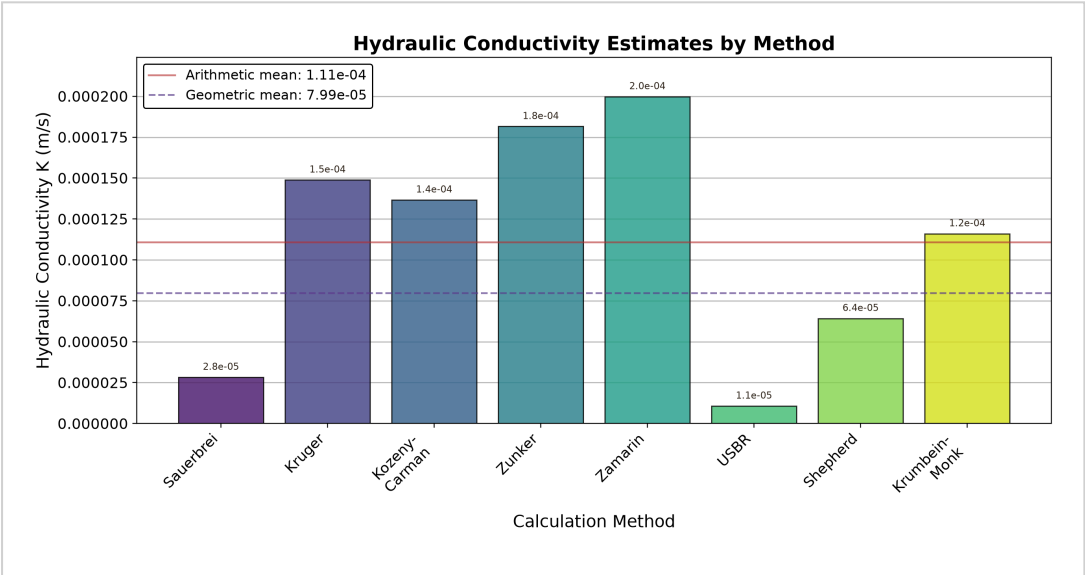
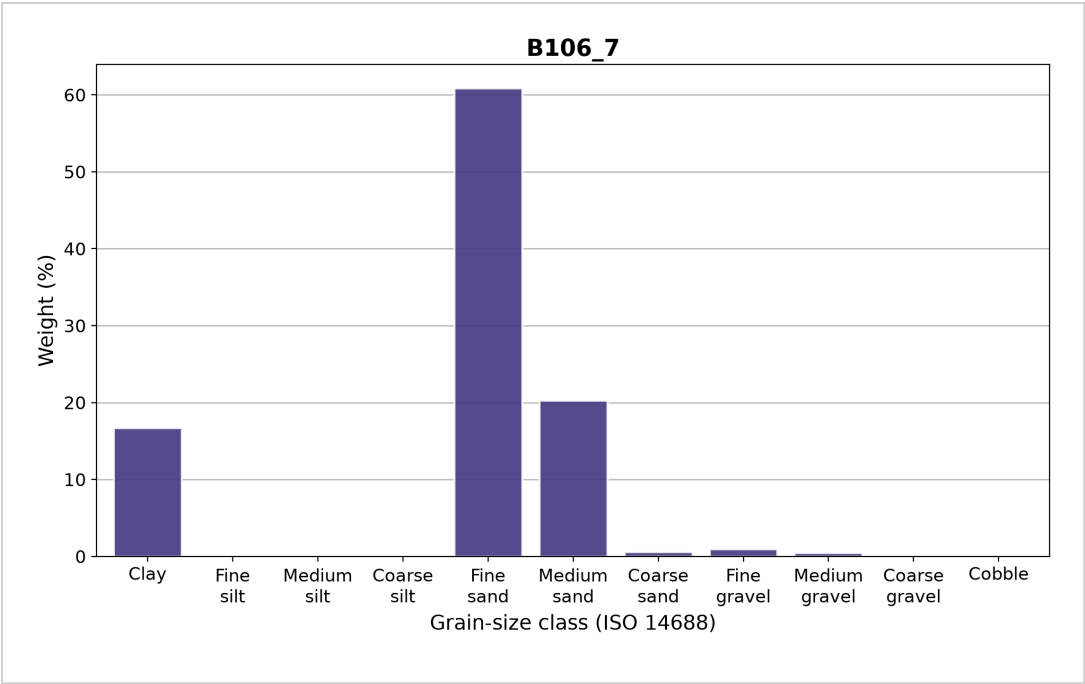
B106_6



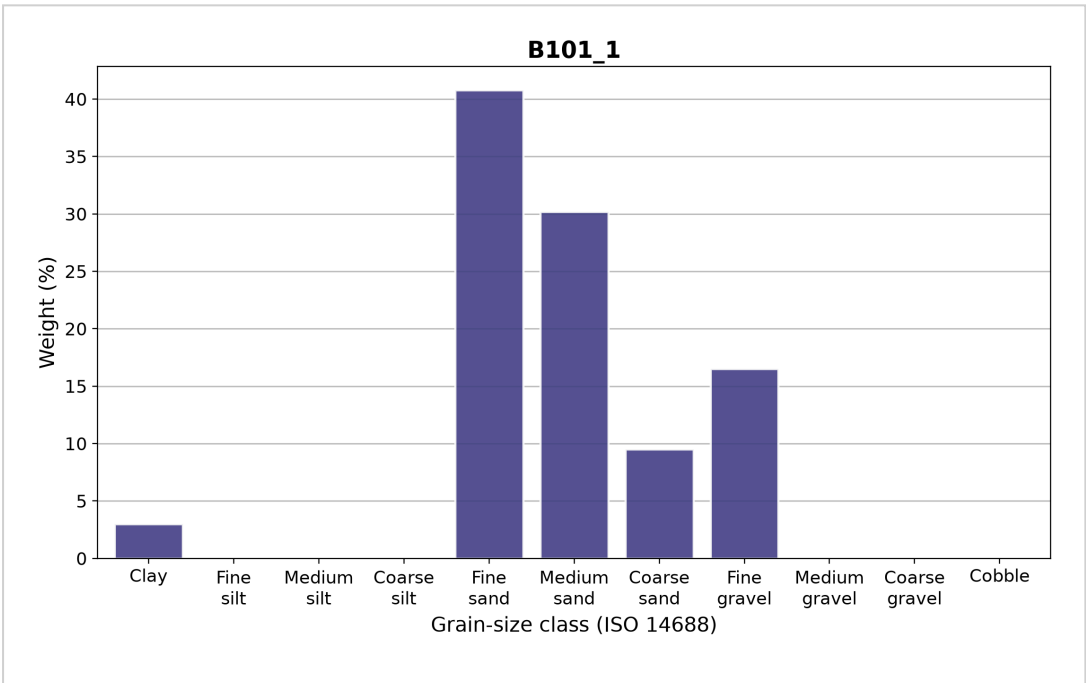
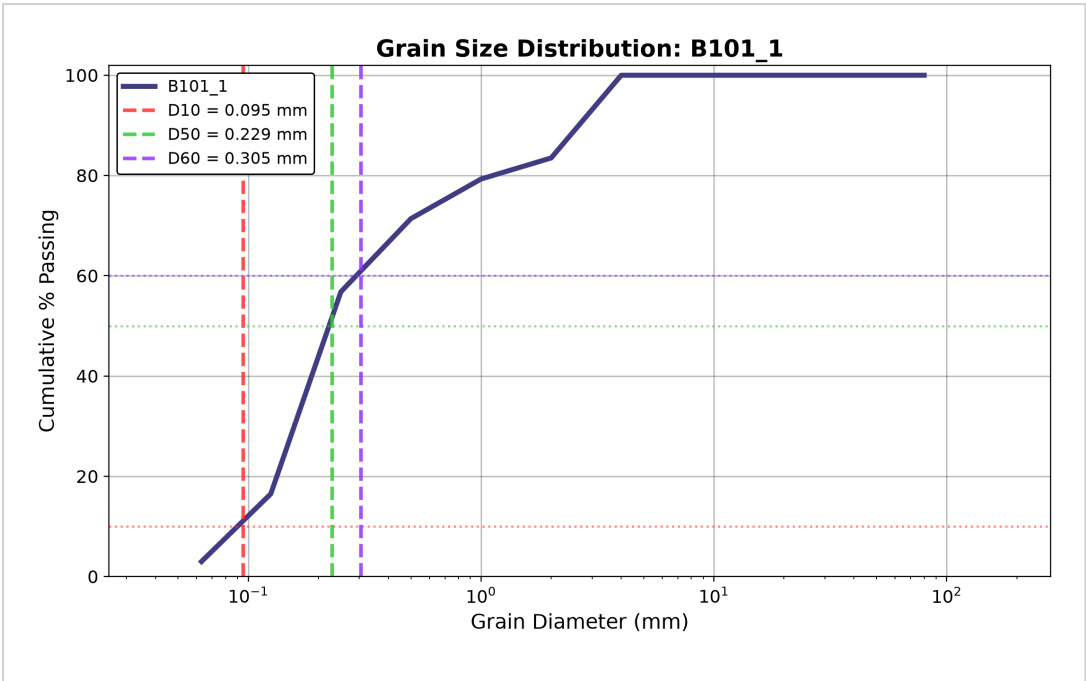


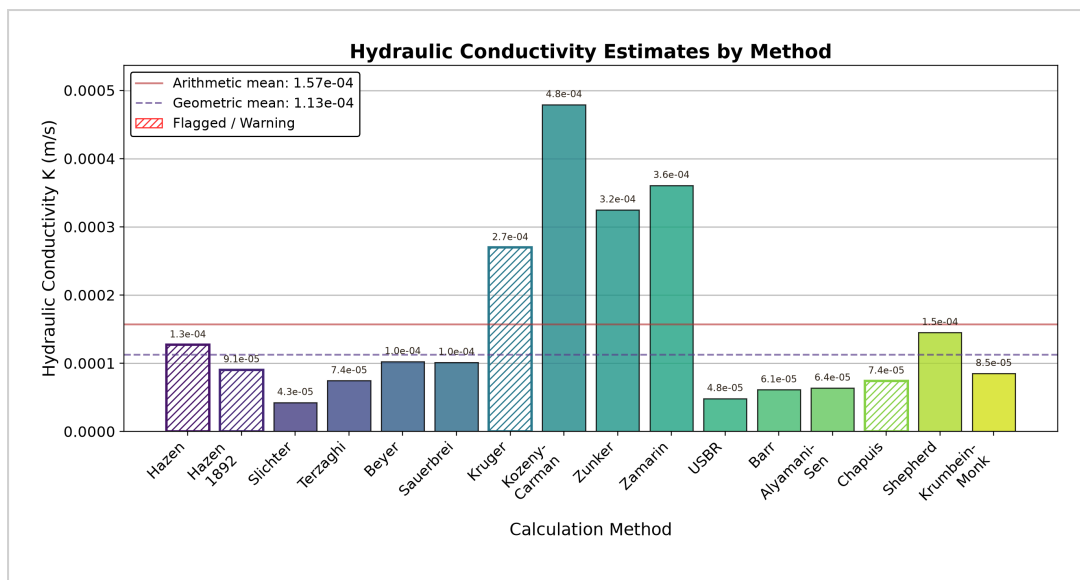
B106_7



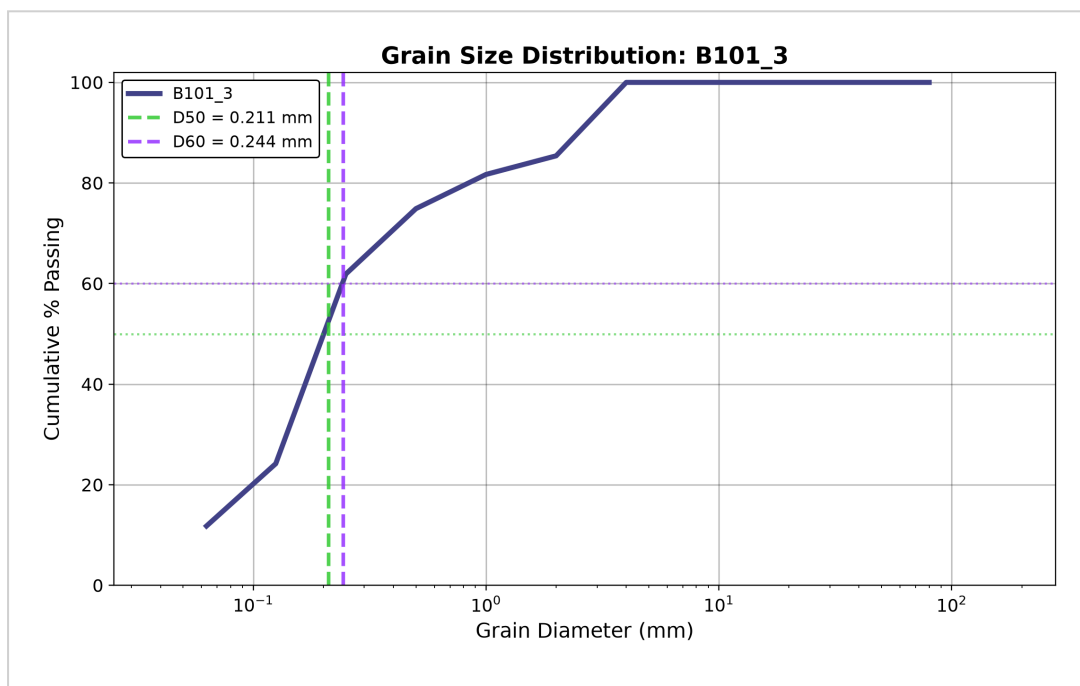


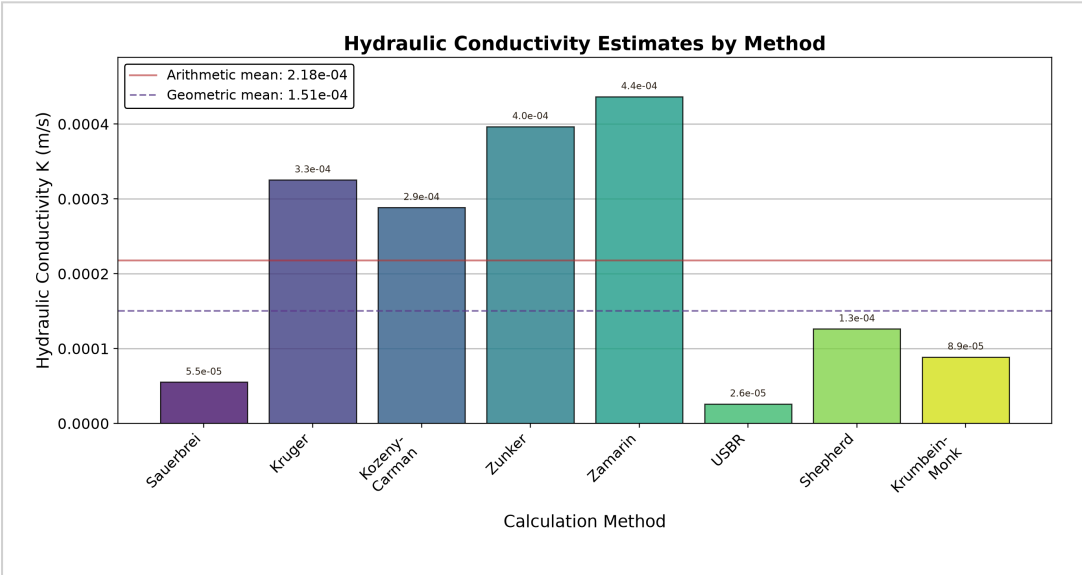
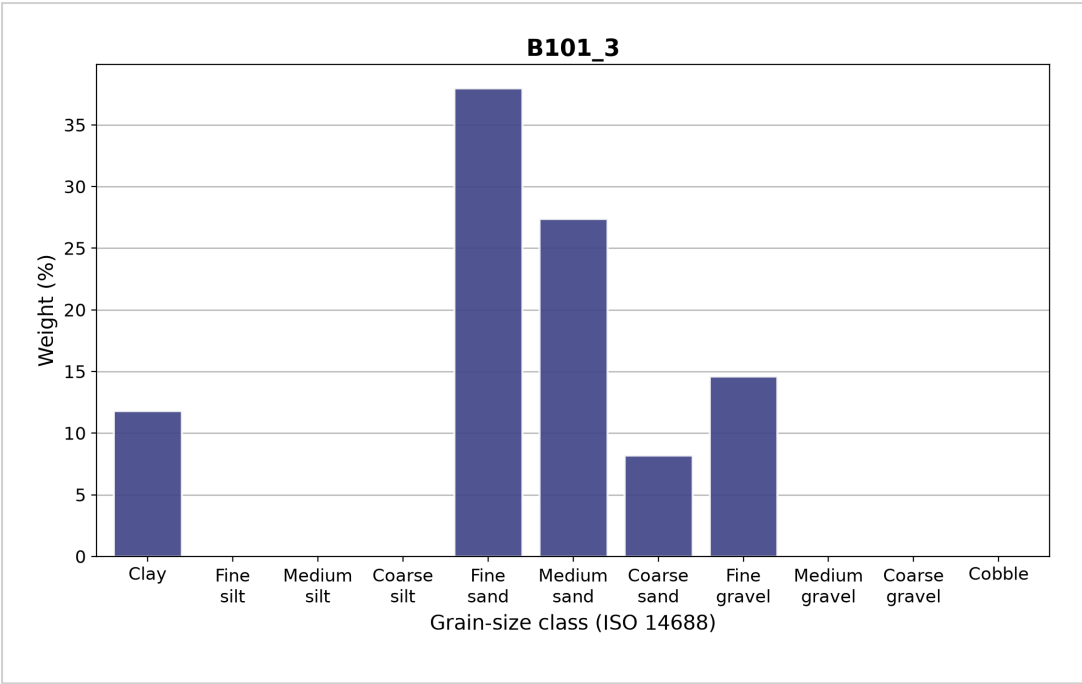
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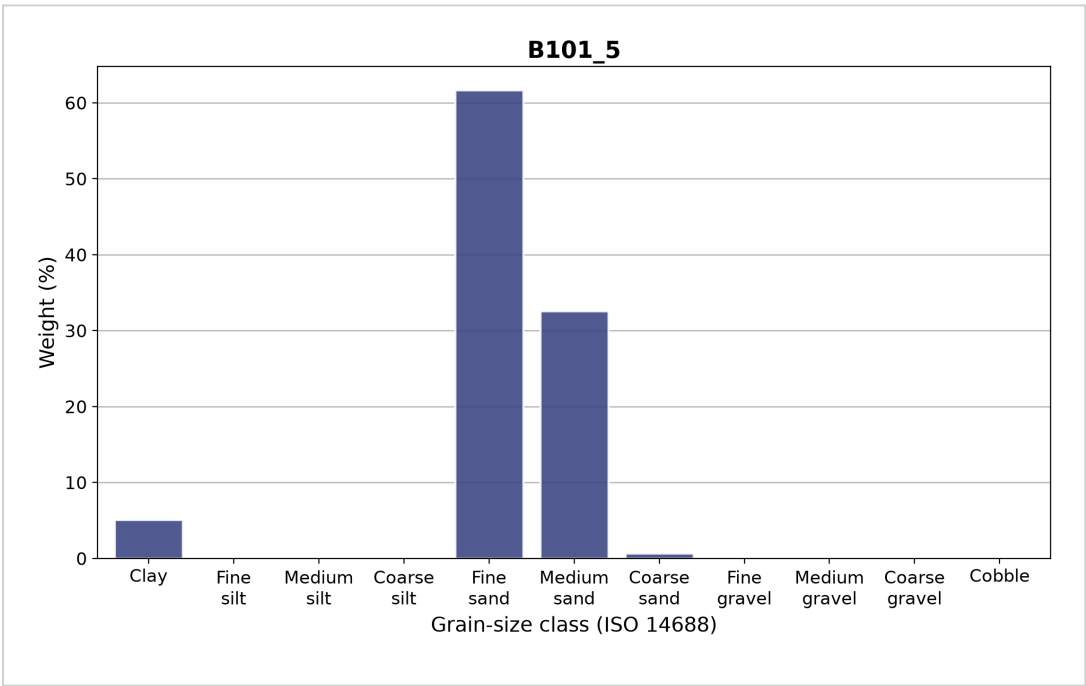
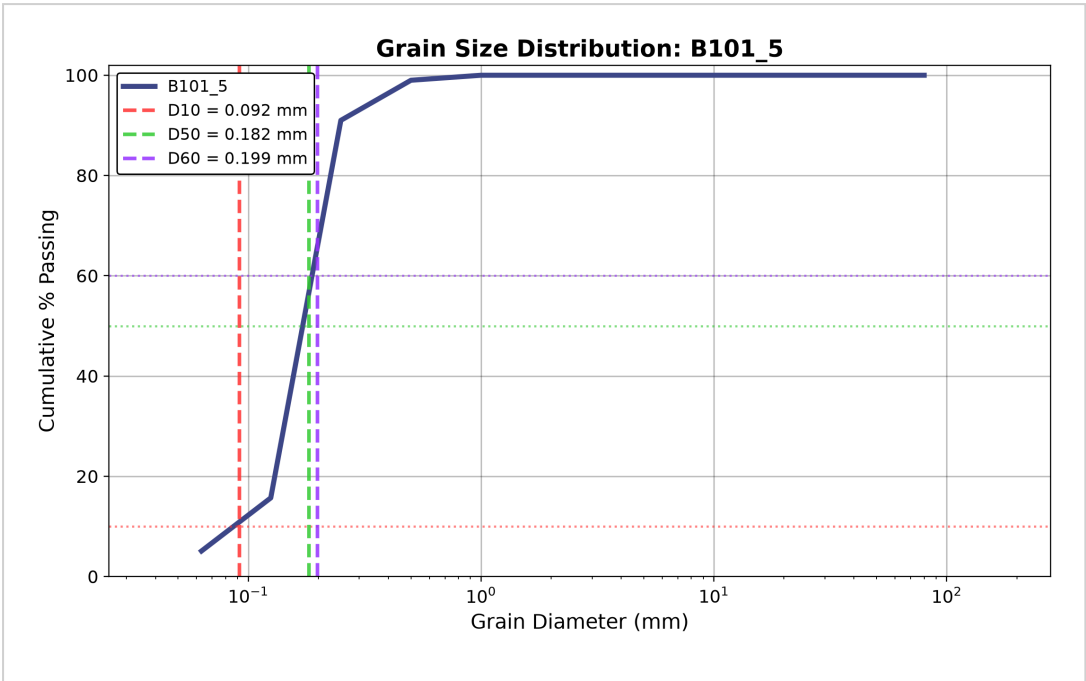


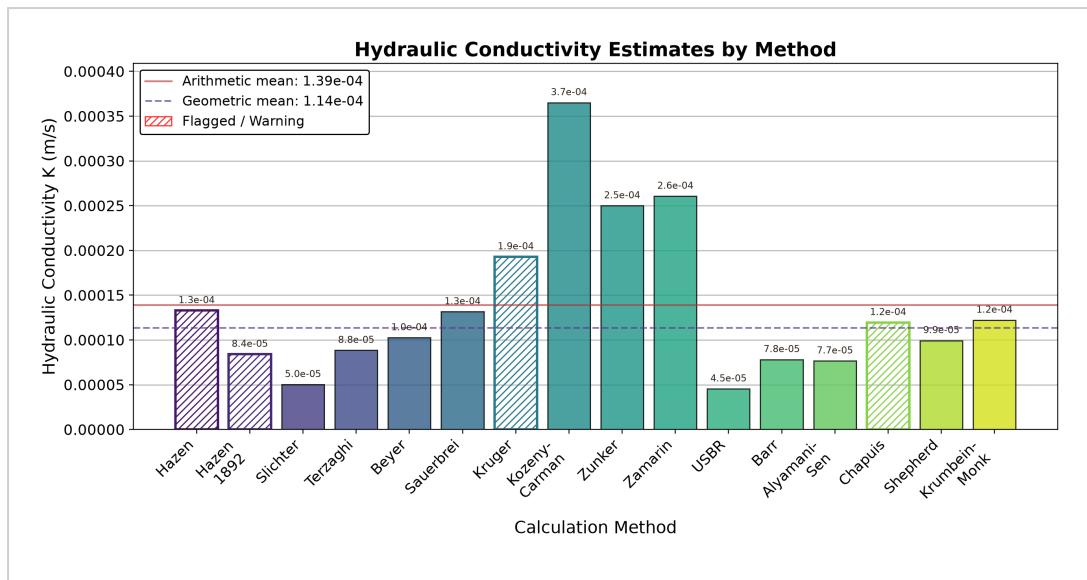
B101_3



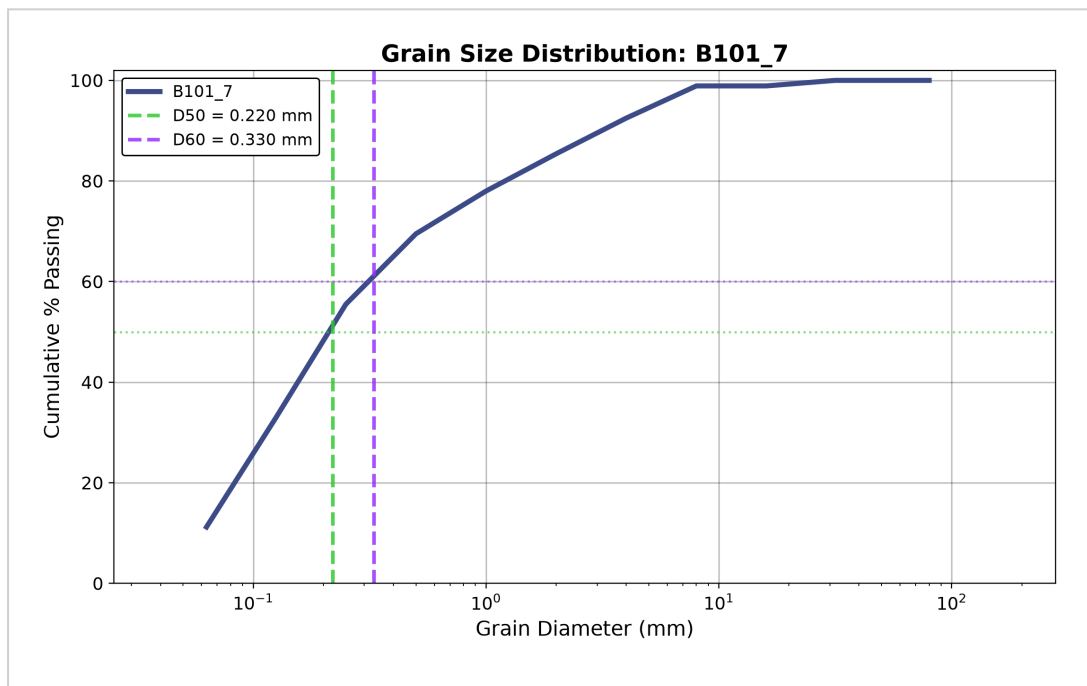


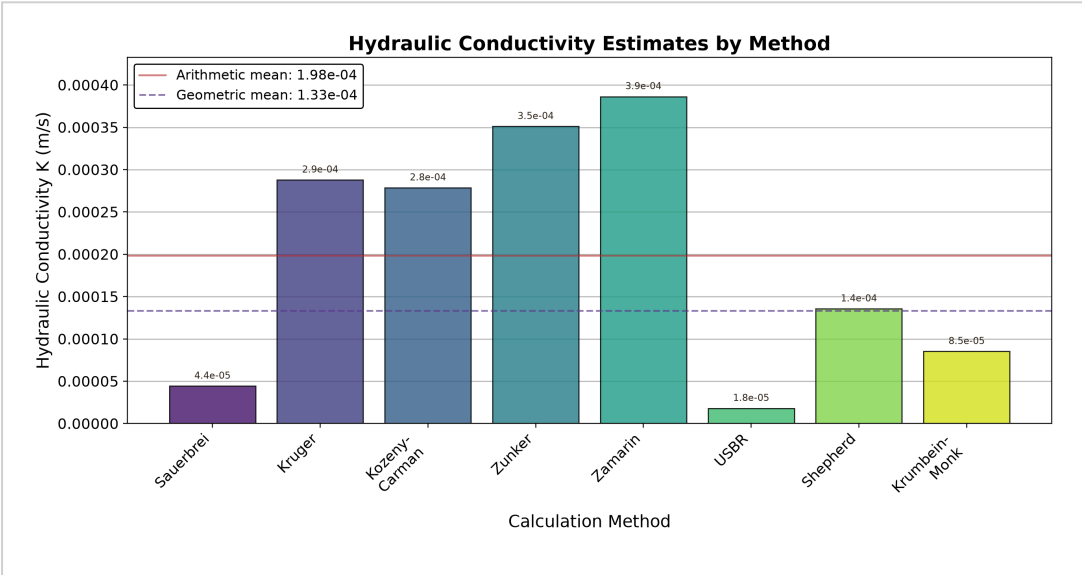
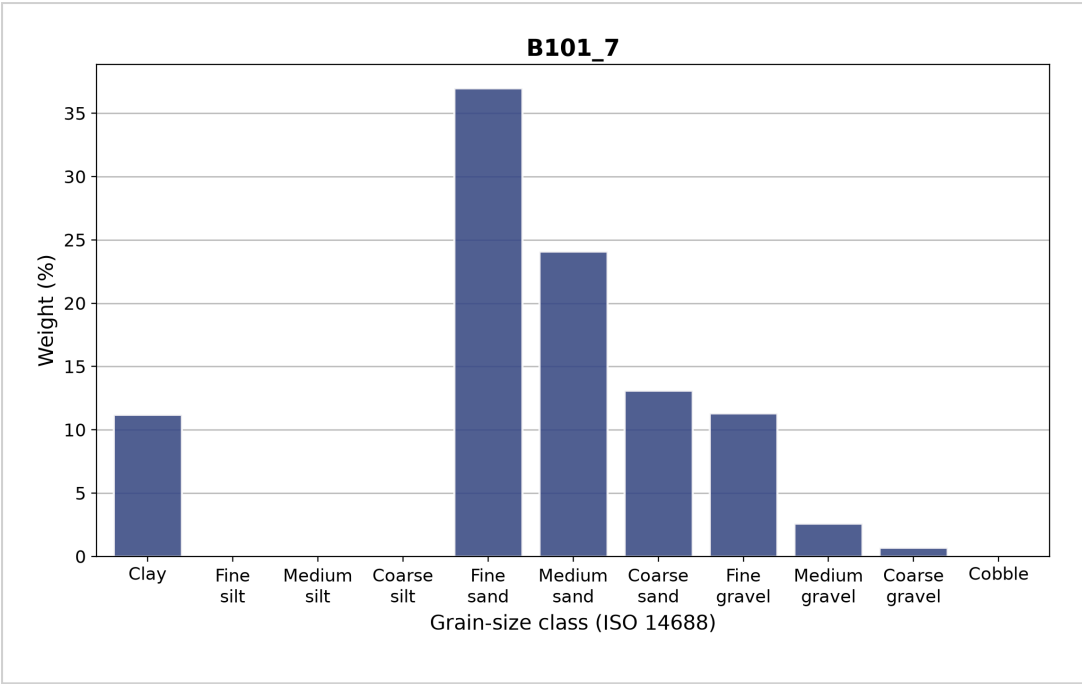
B101_5



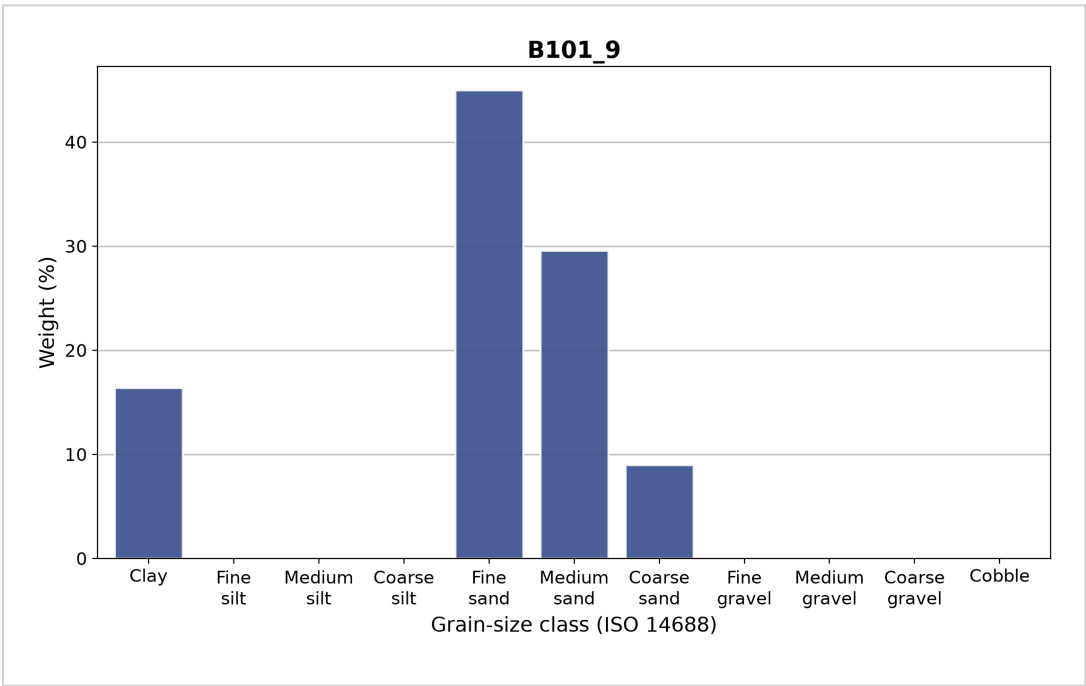
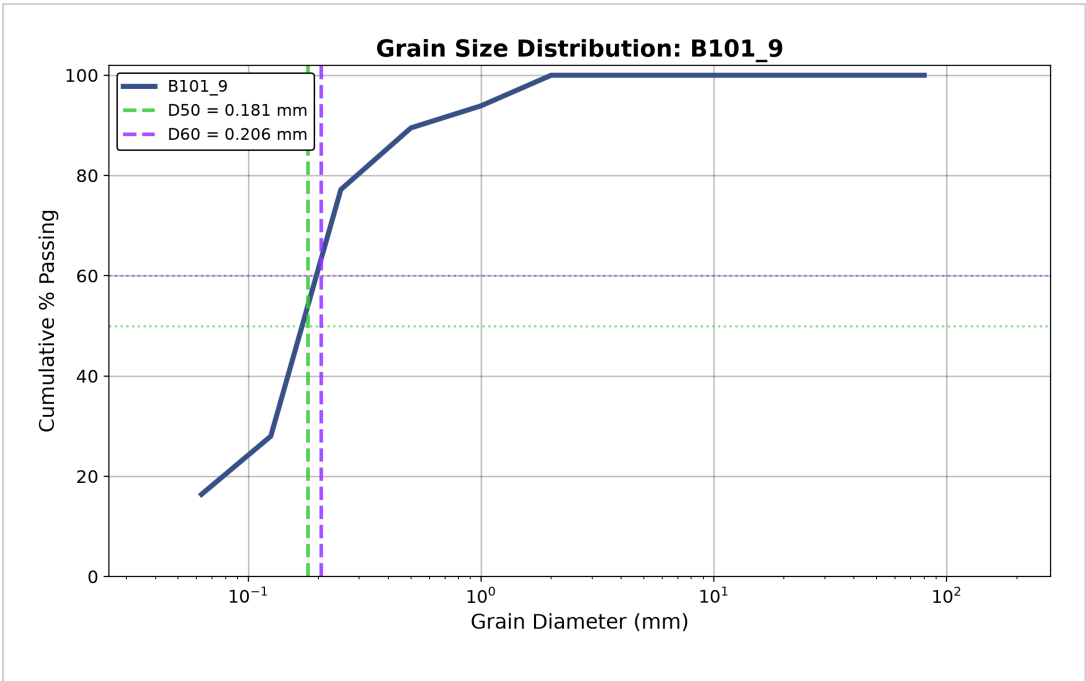


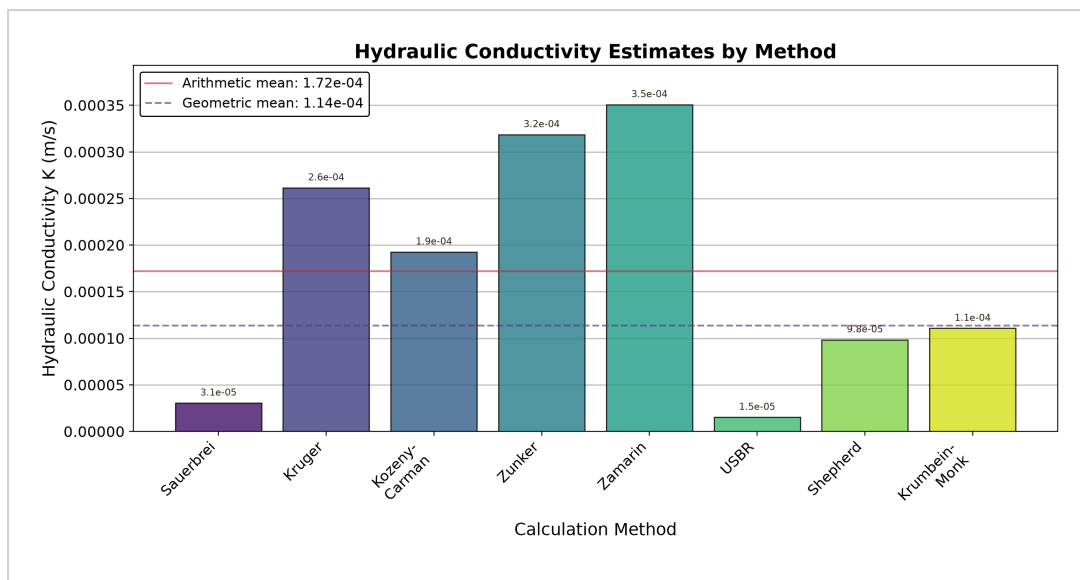
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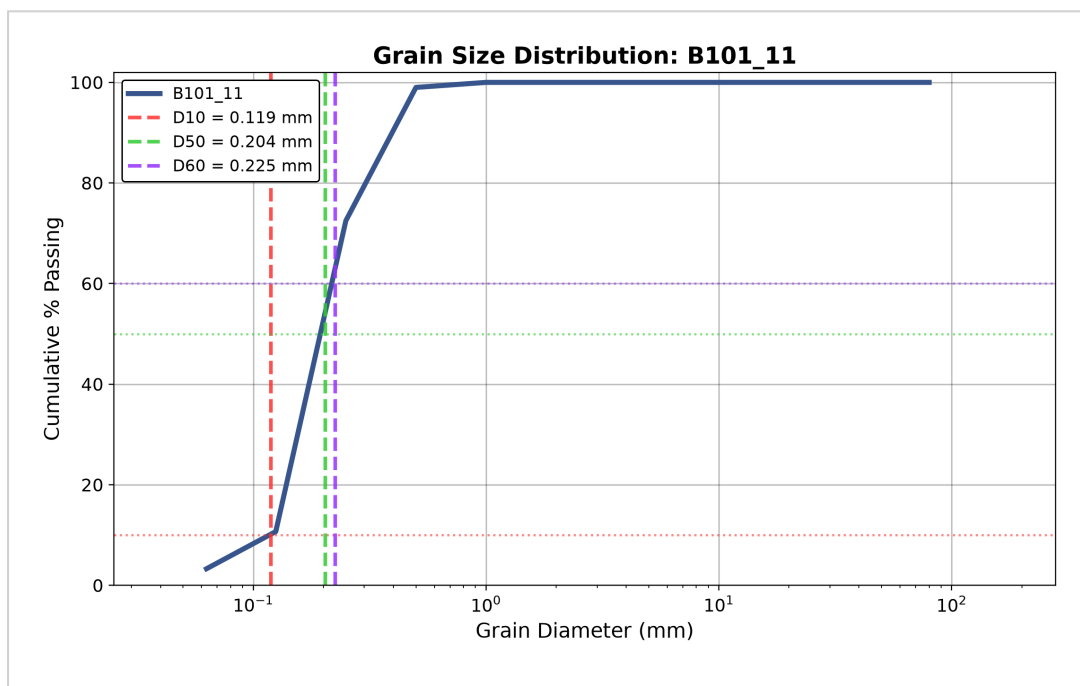


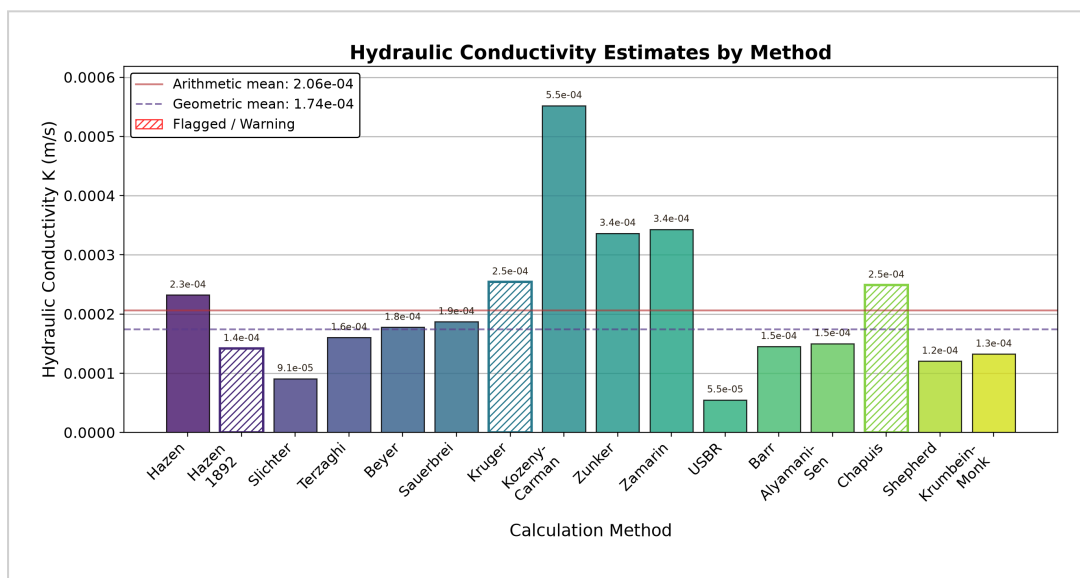
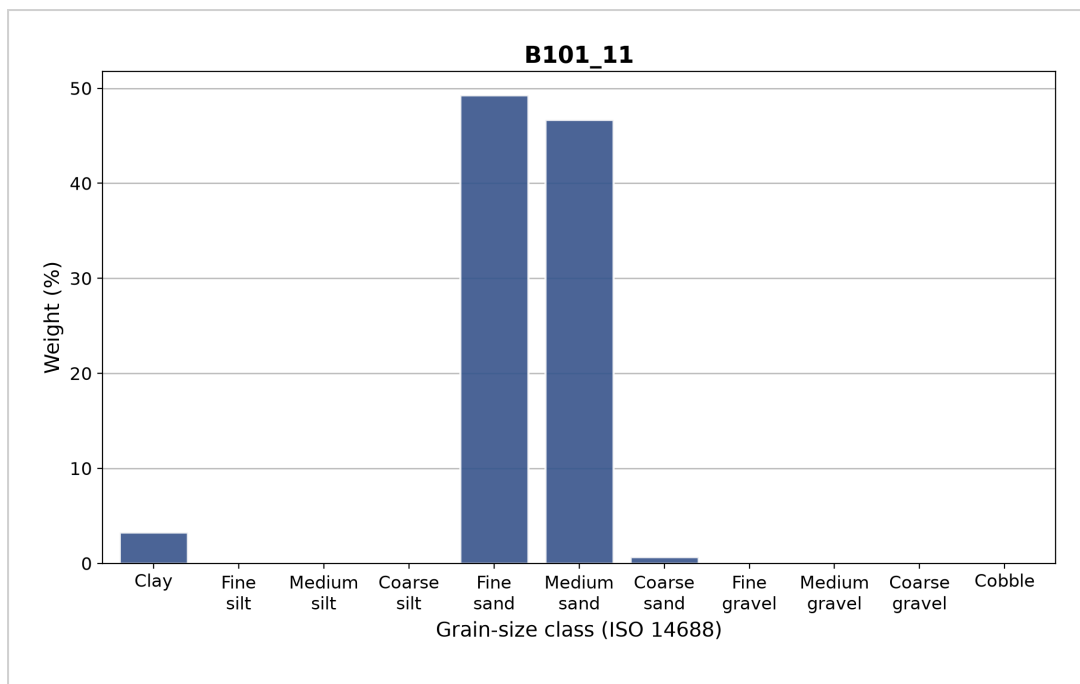
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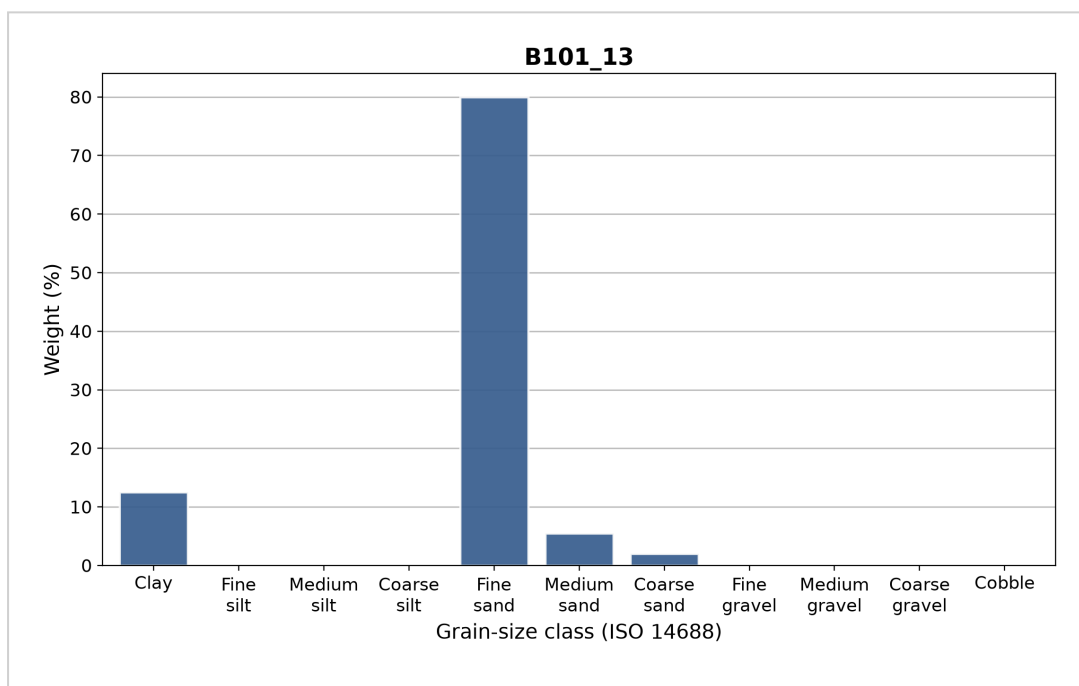
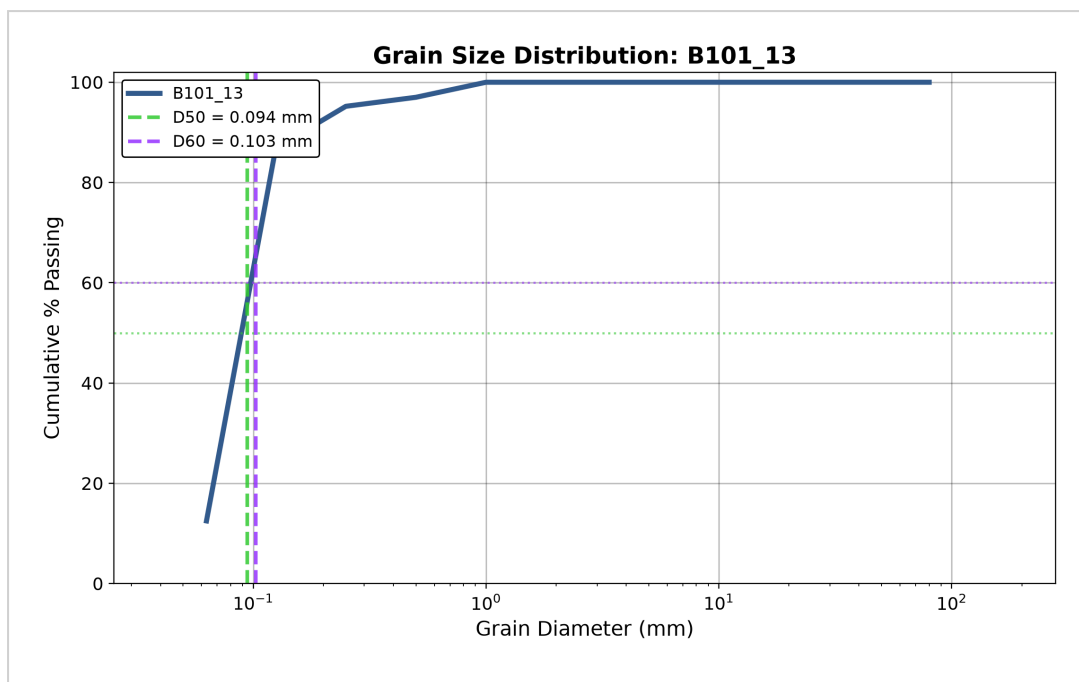


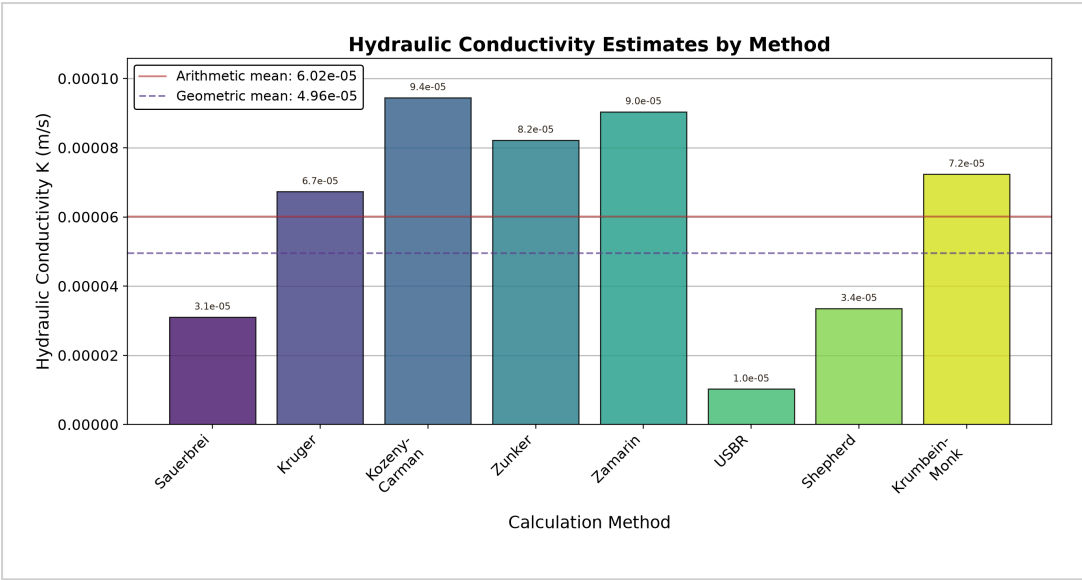
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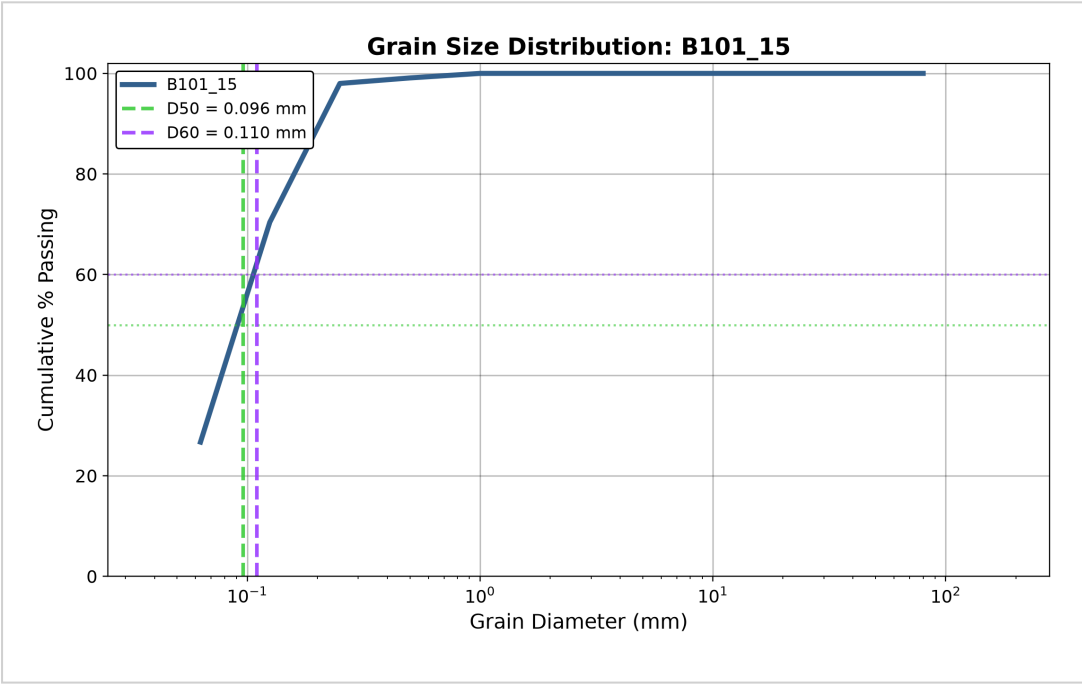


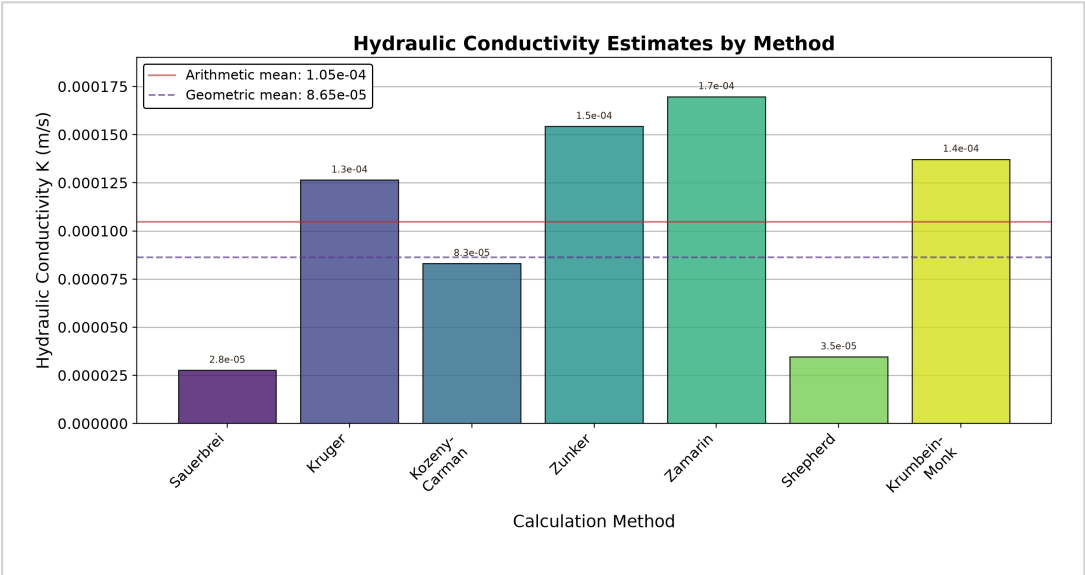
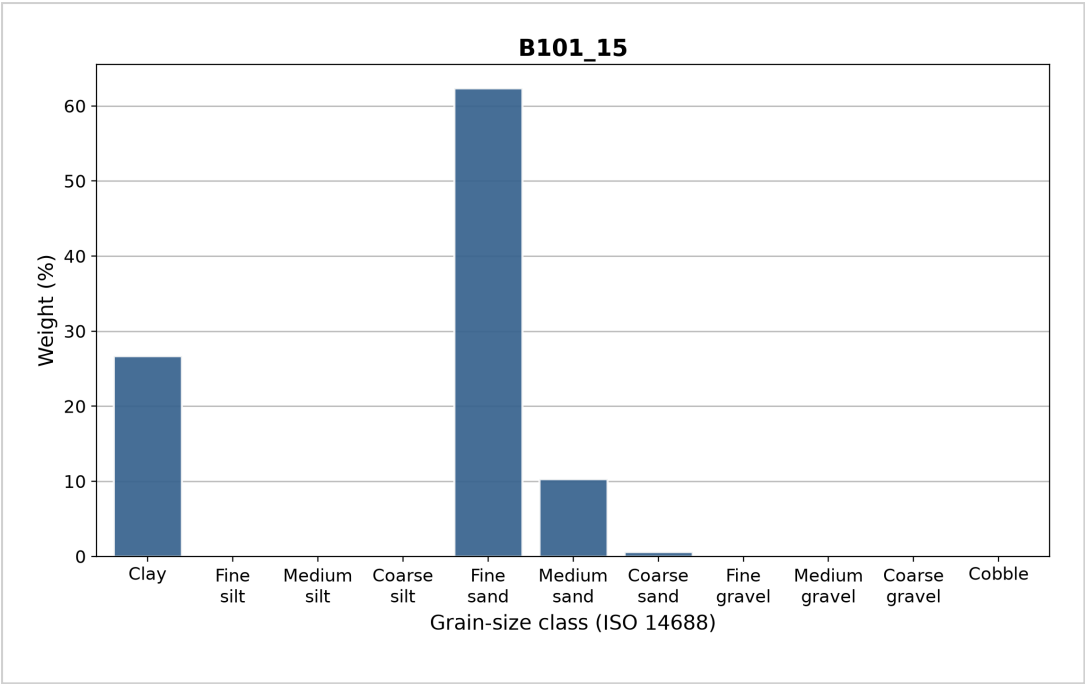
B101_13



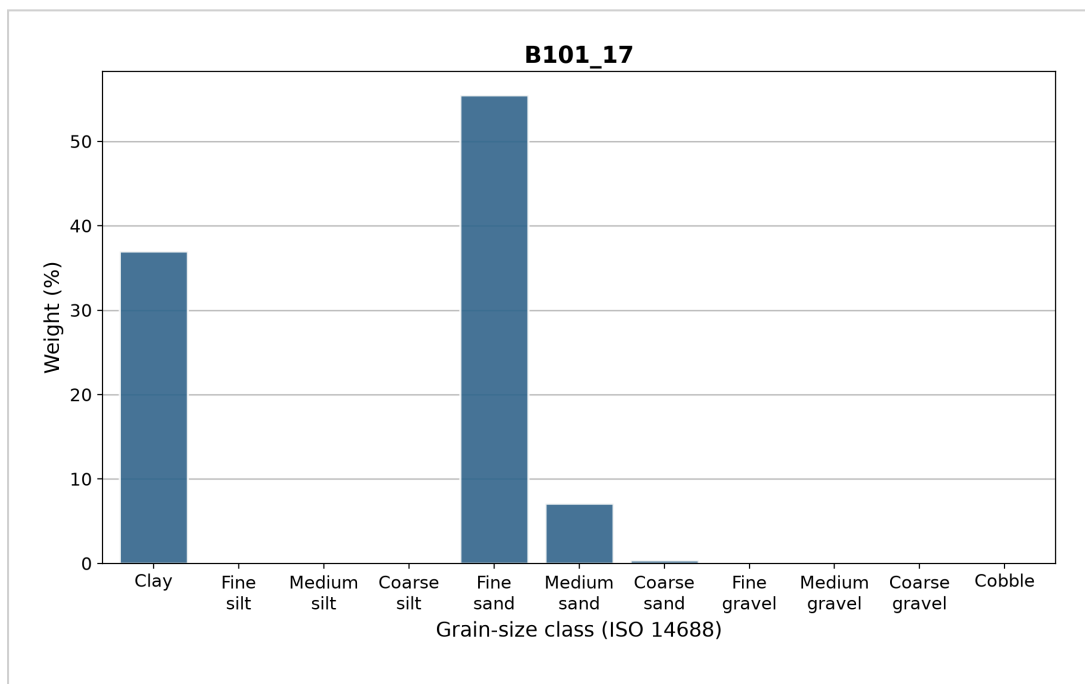
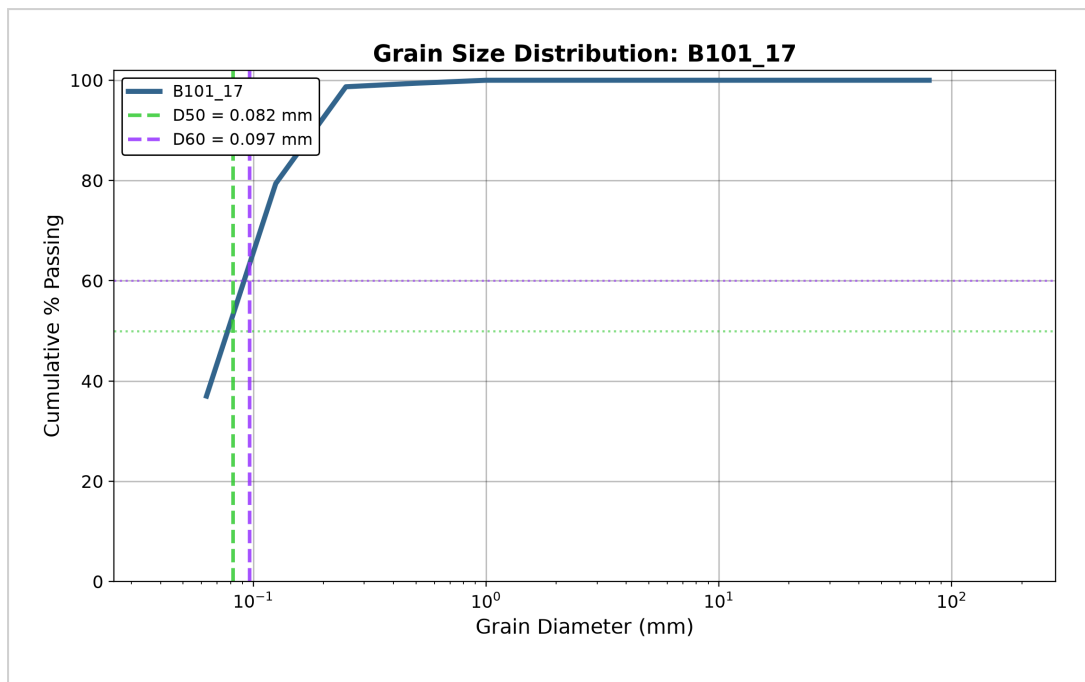


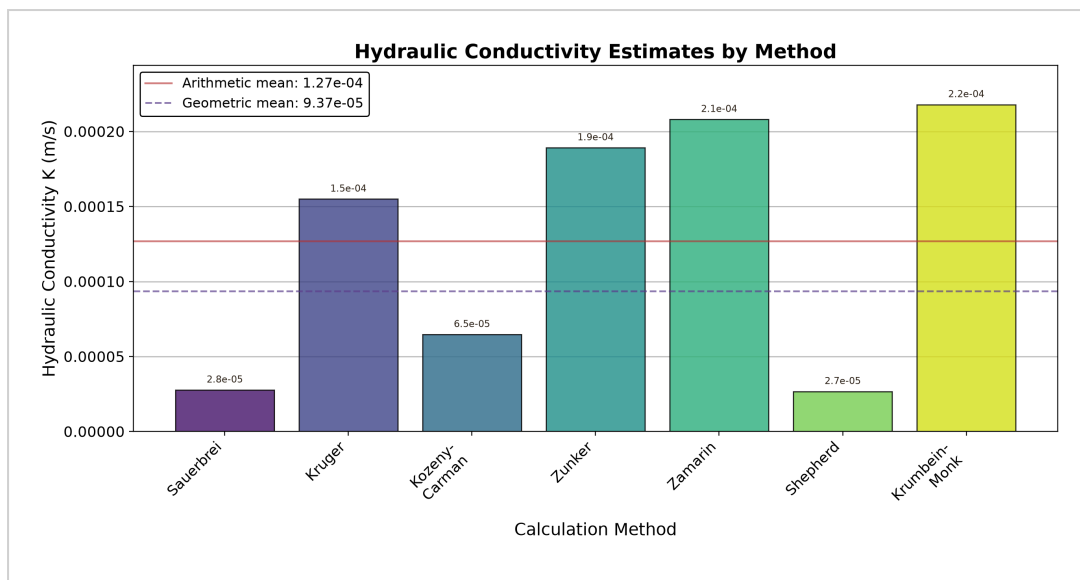
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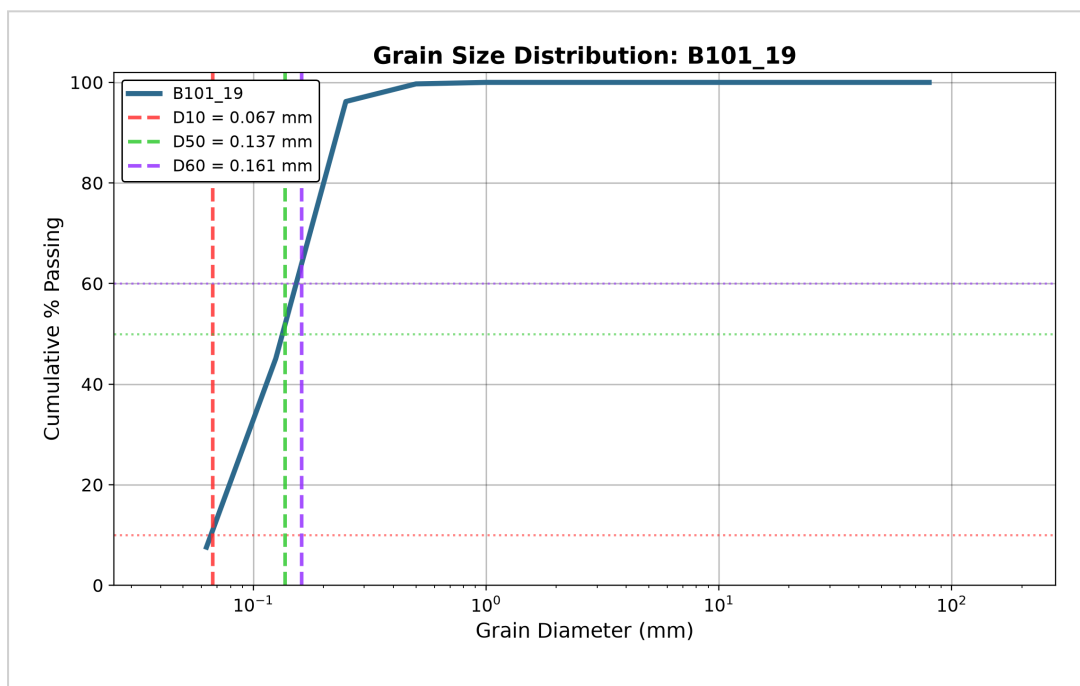


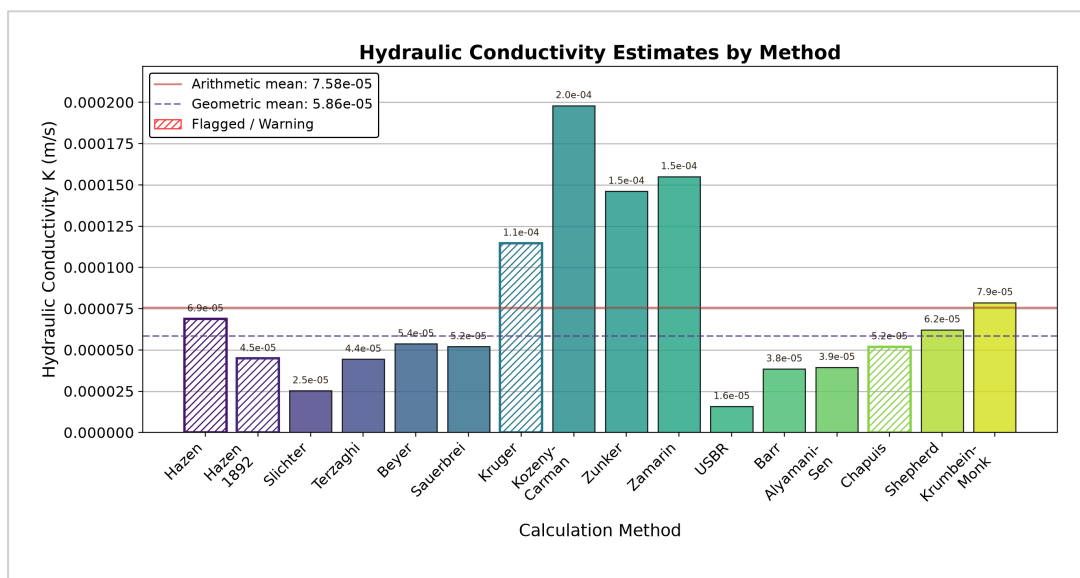
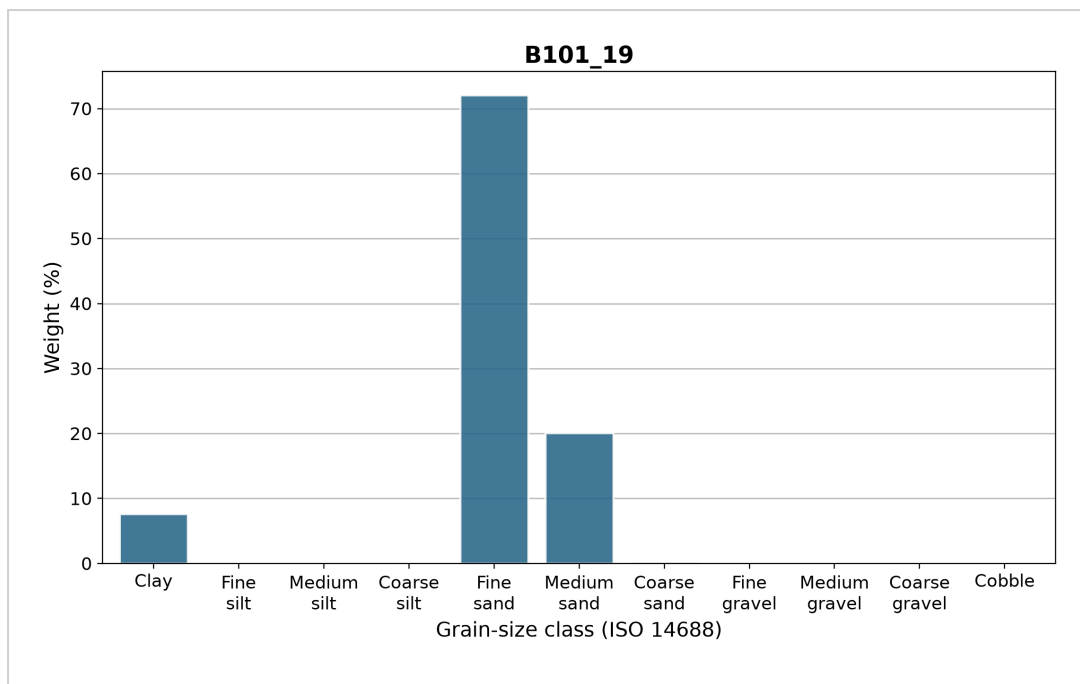
B101_17



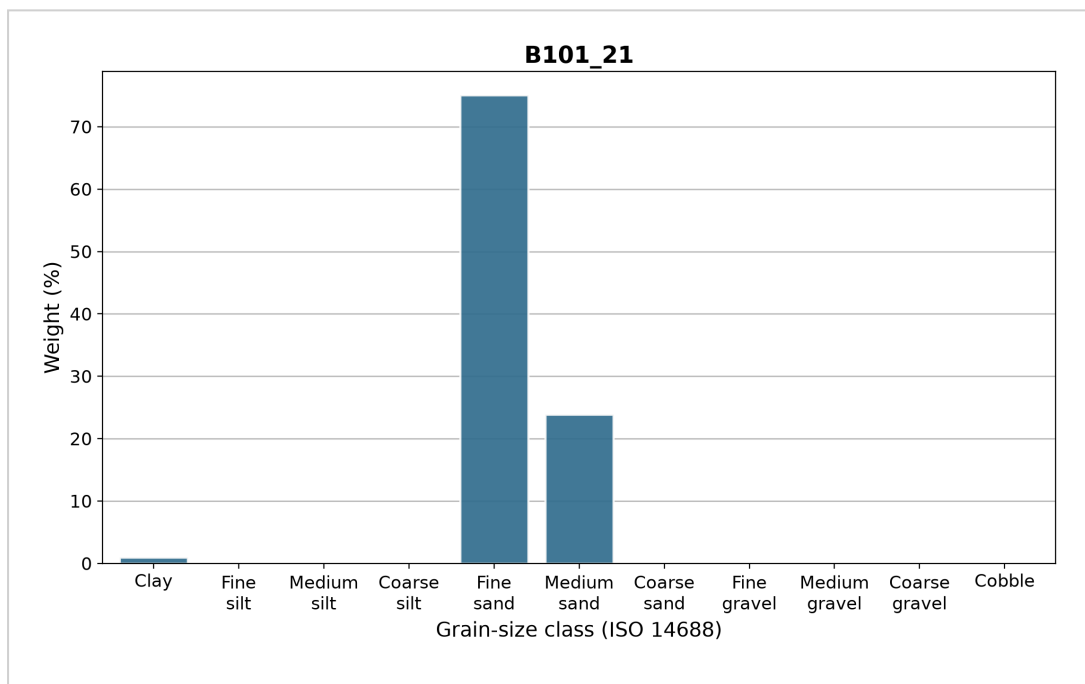
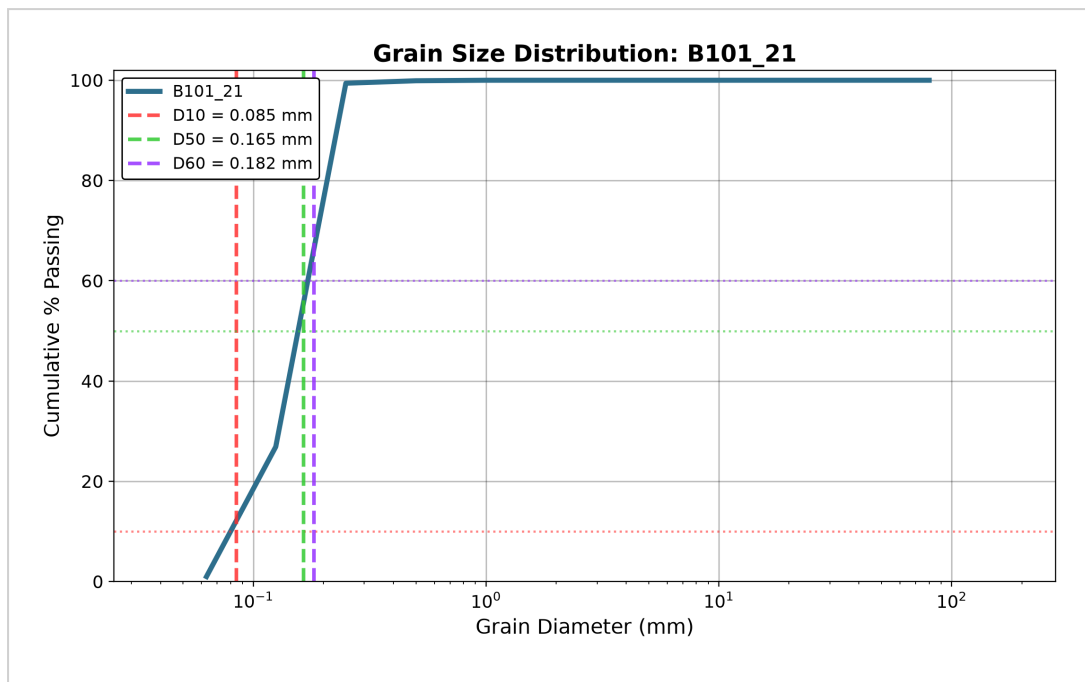


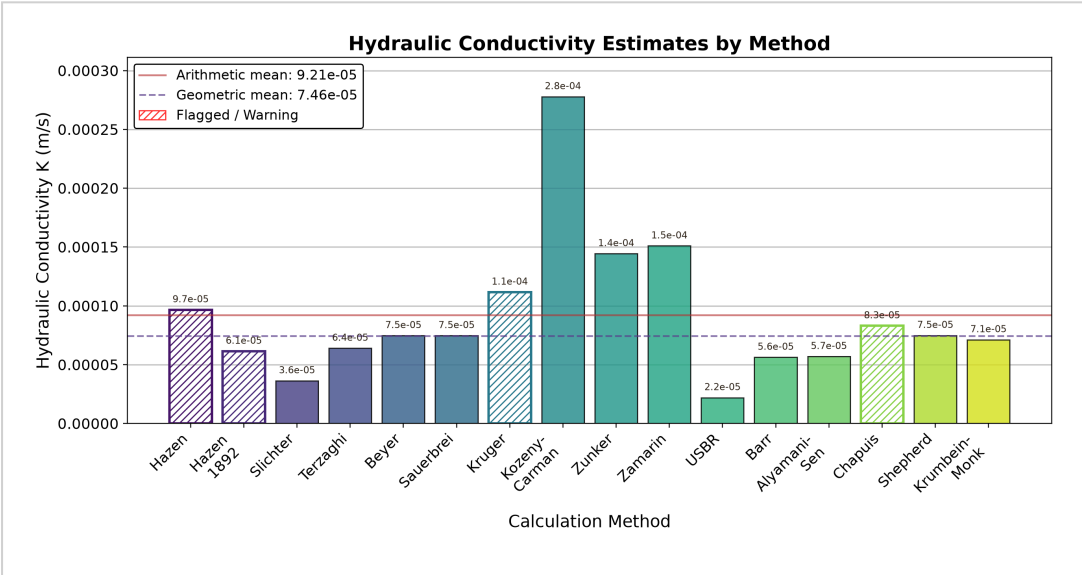
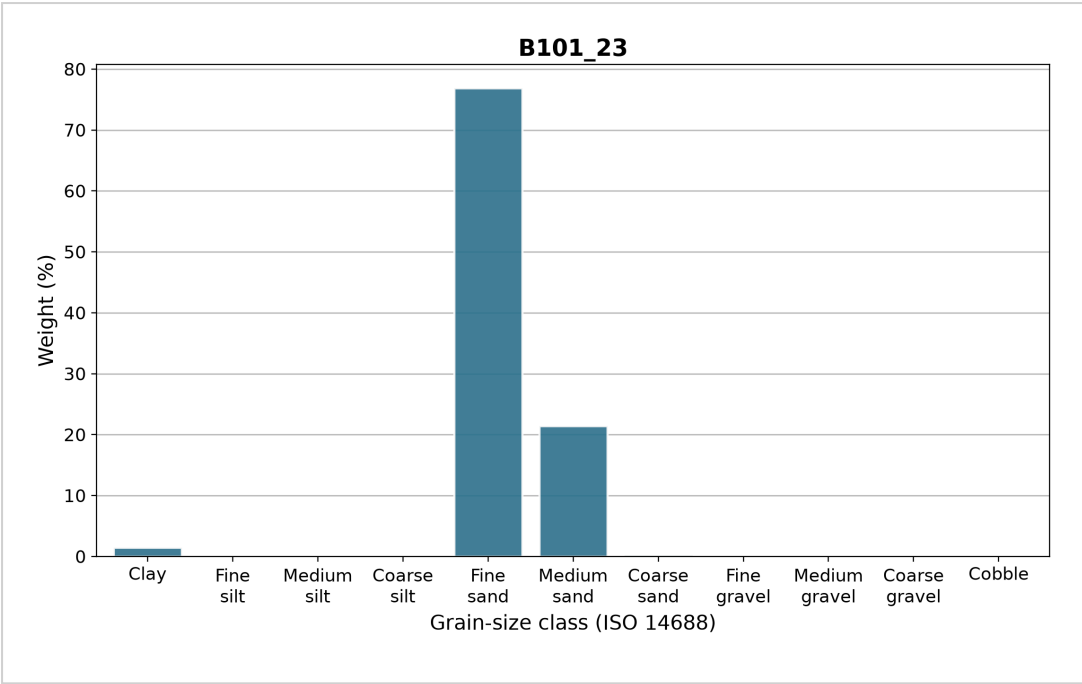
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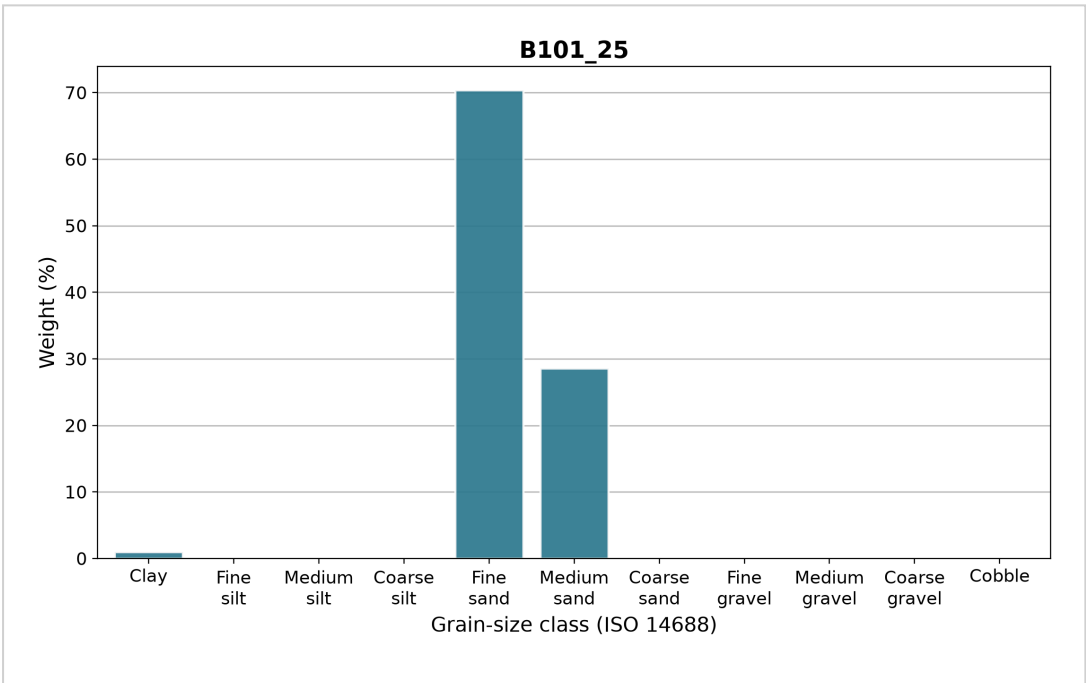
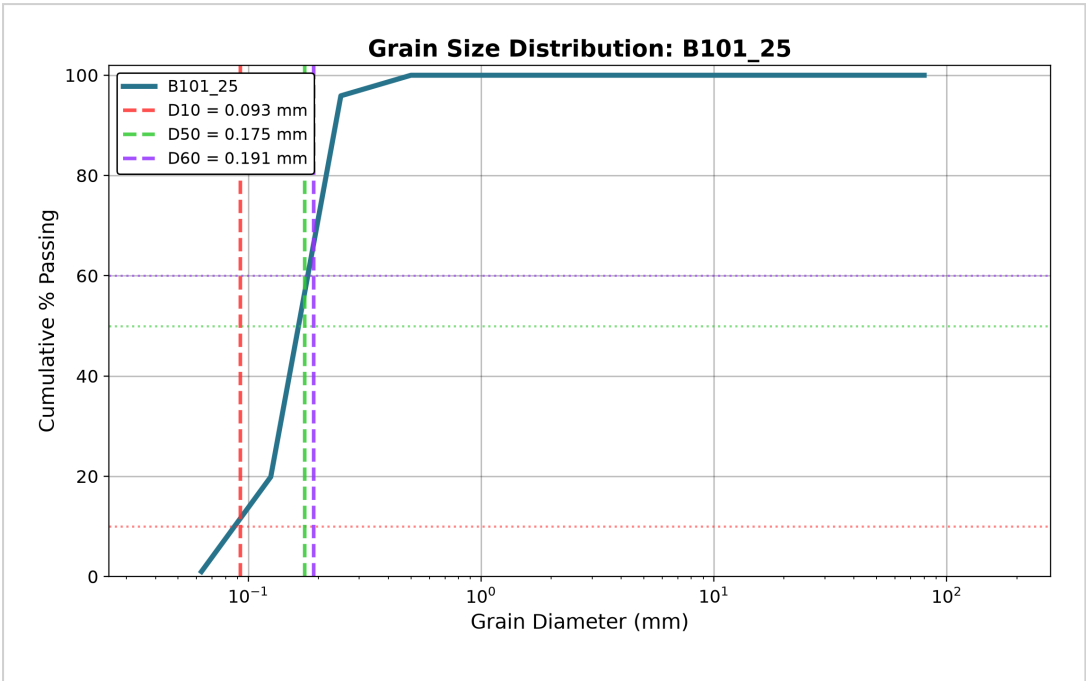


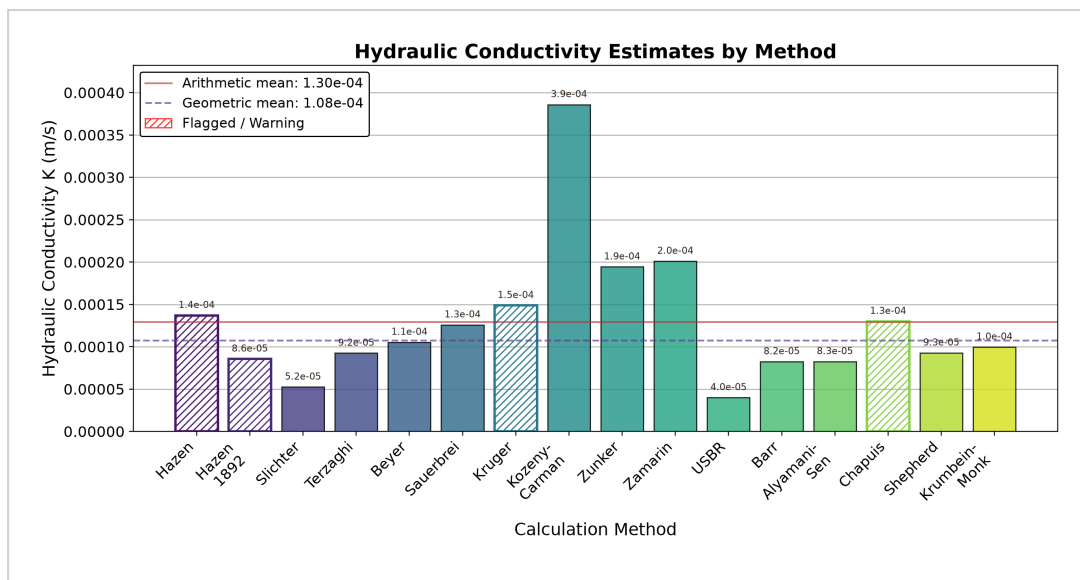
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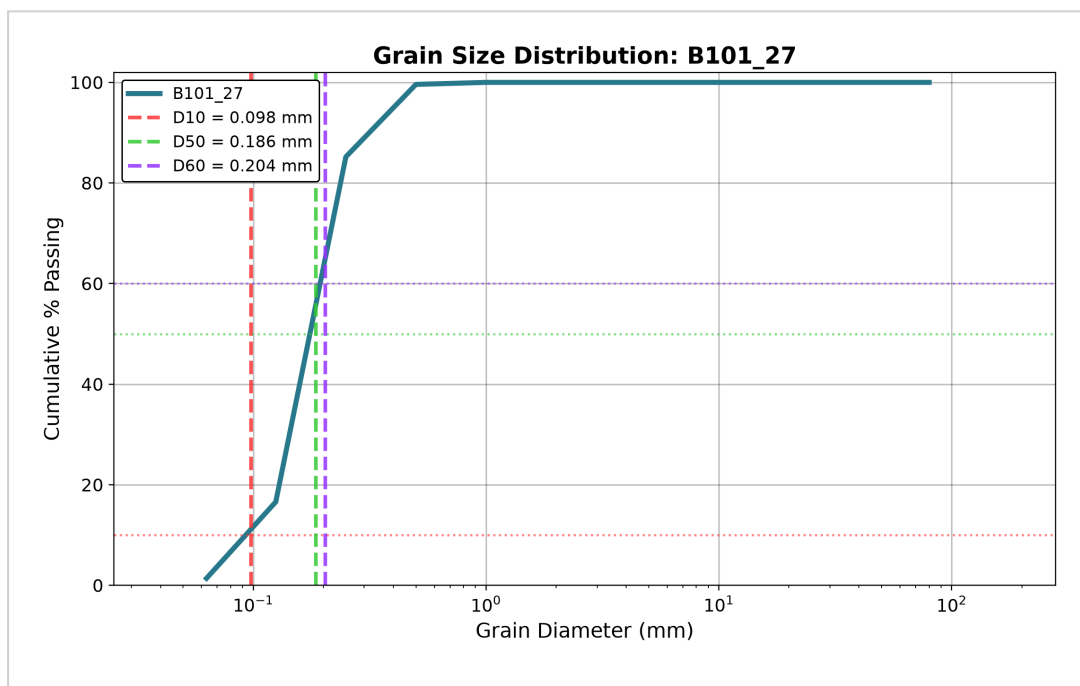


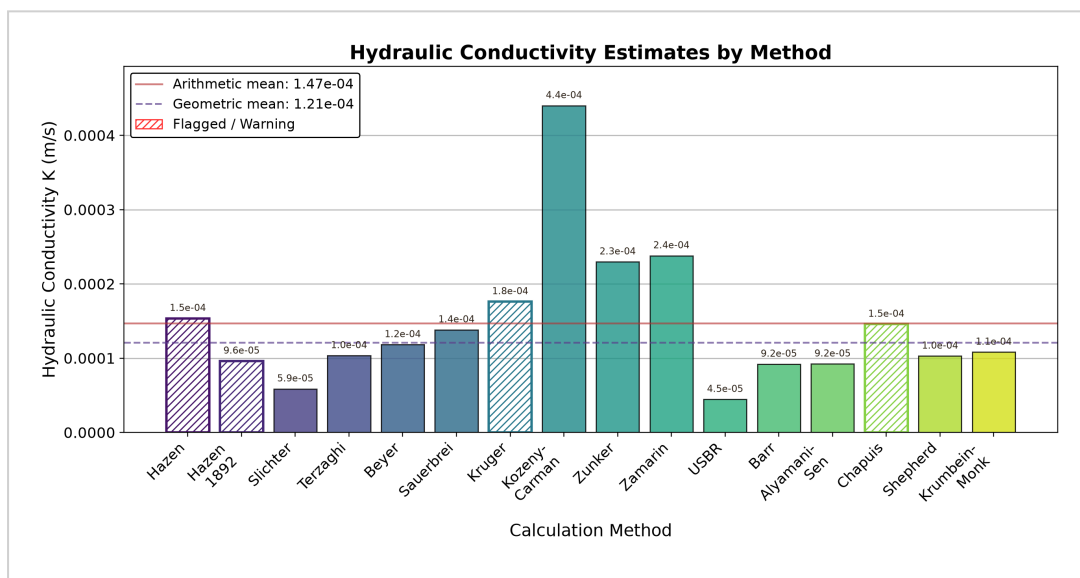
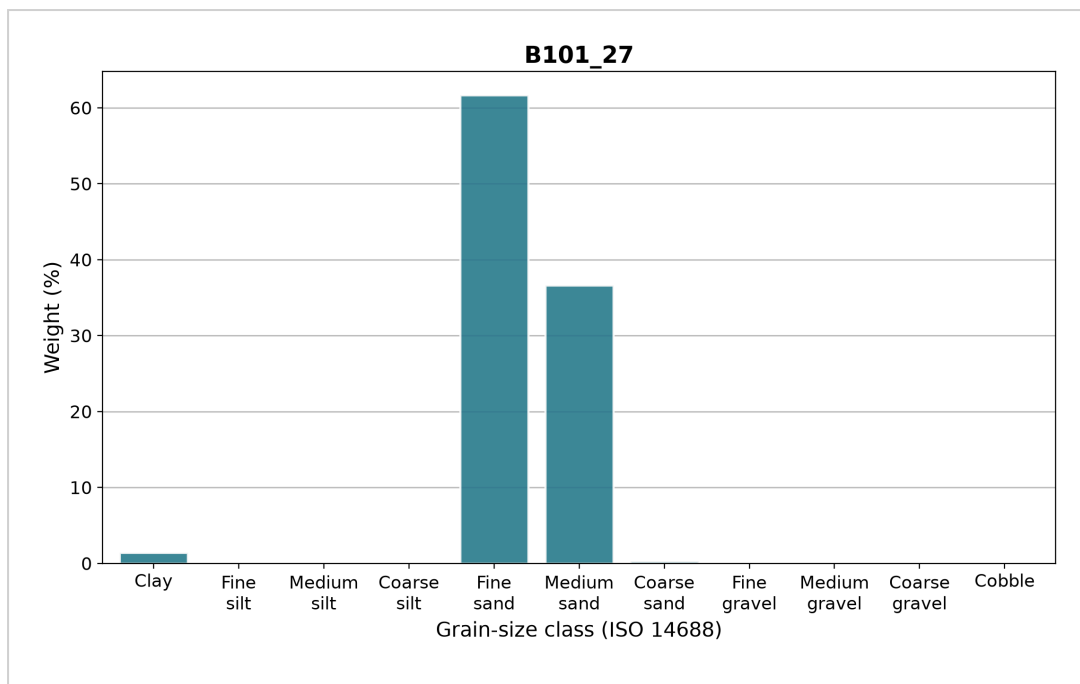
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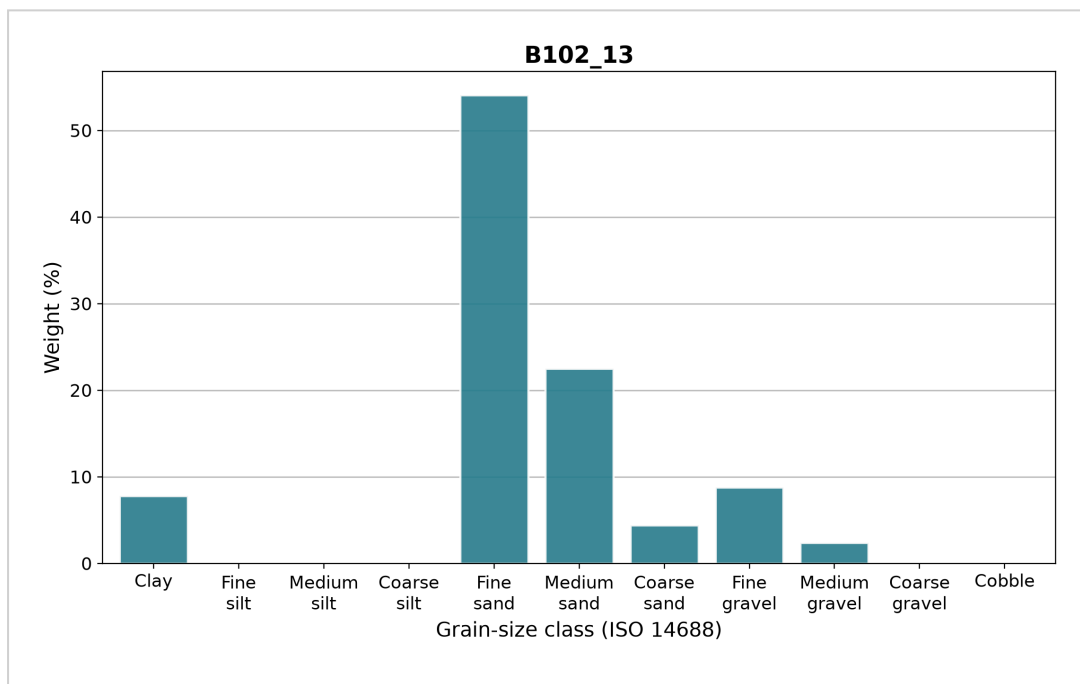
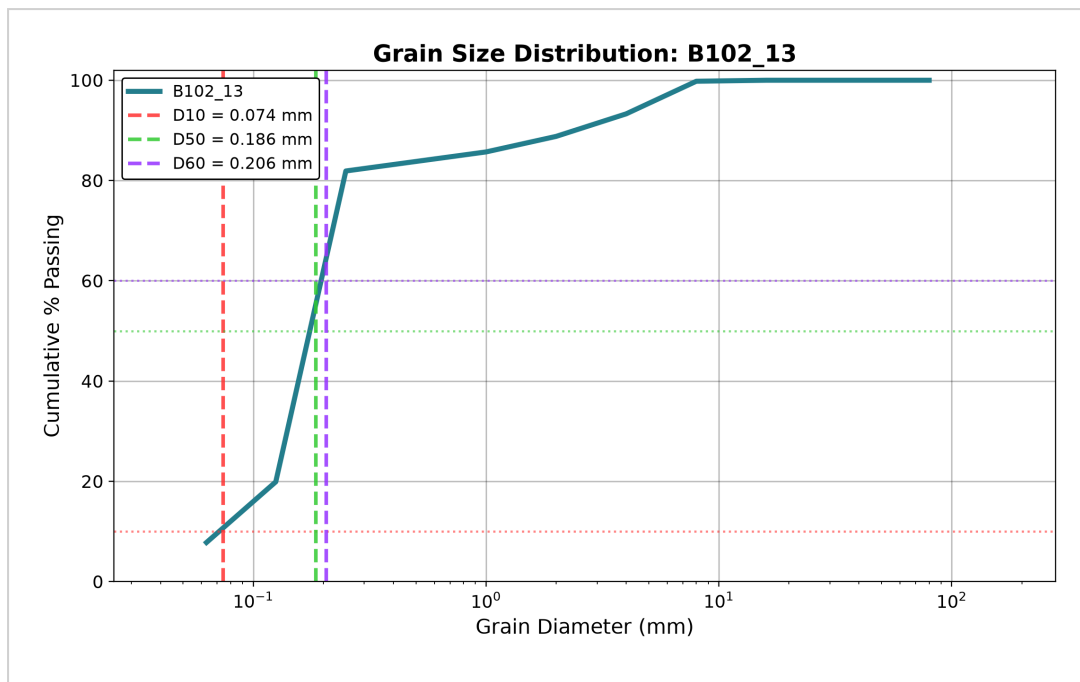


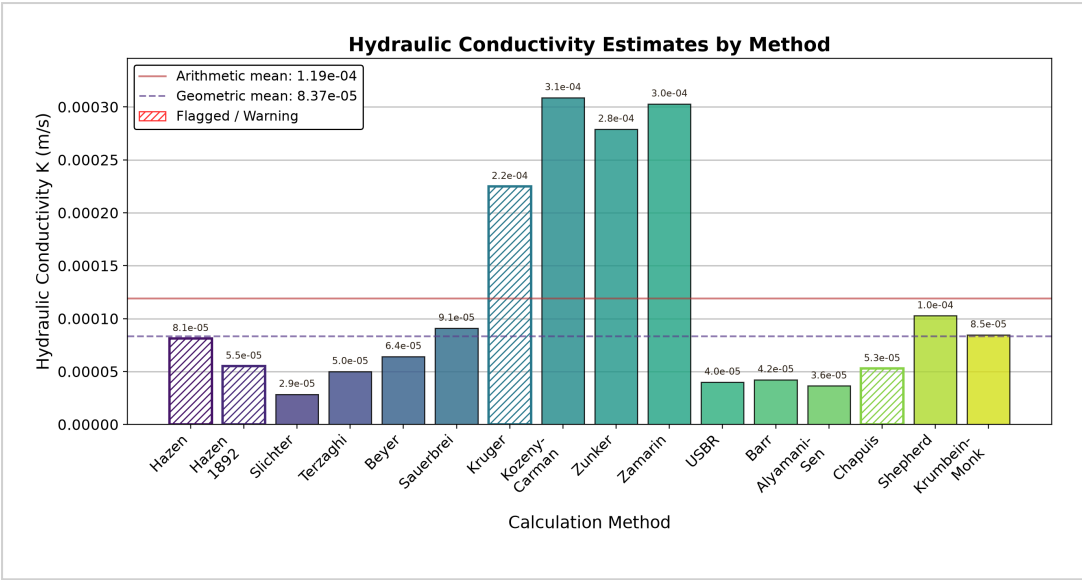
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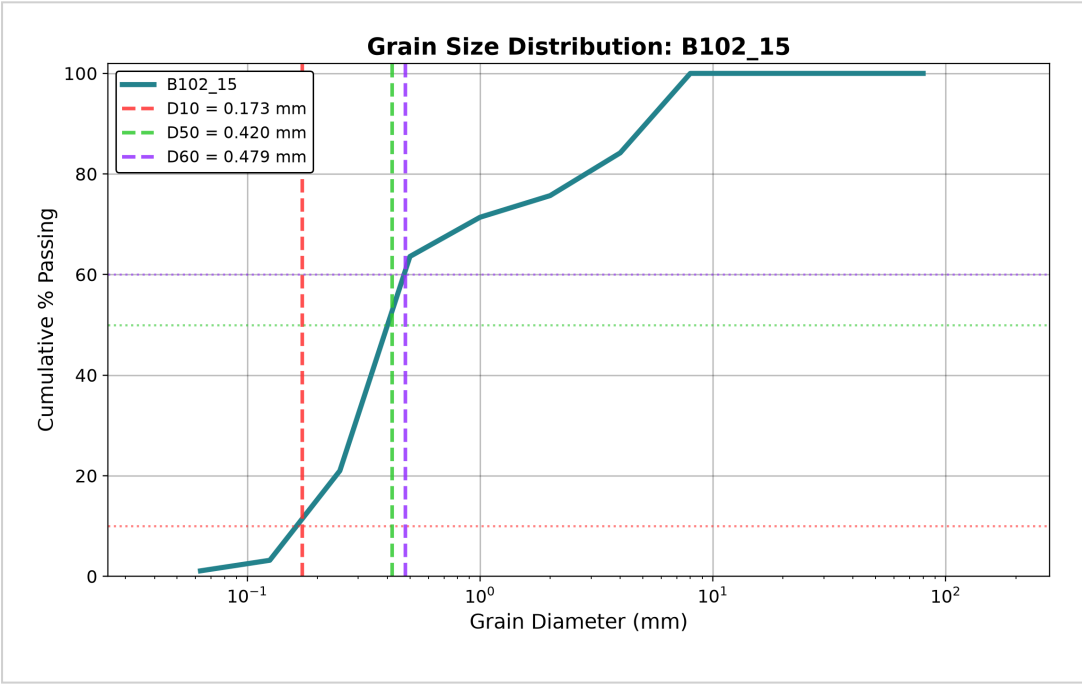


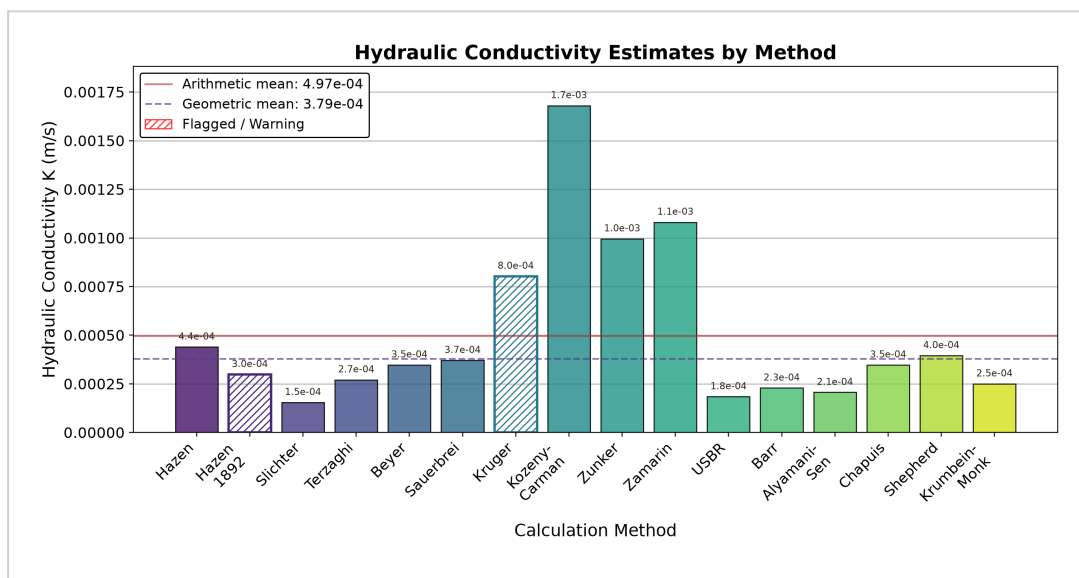
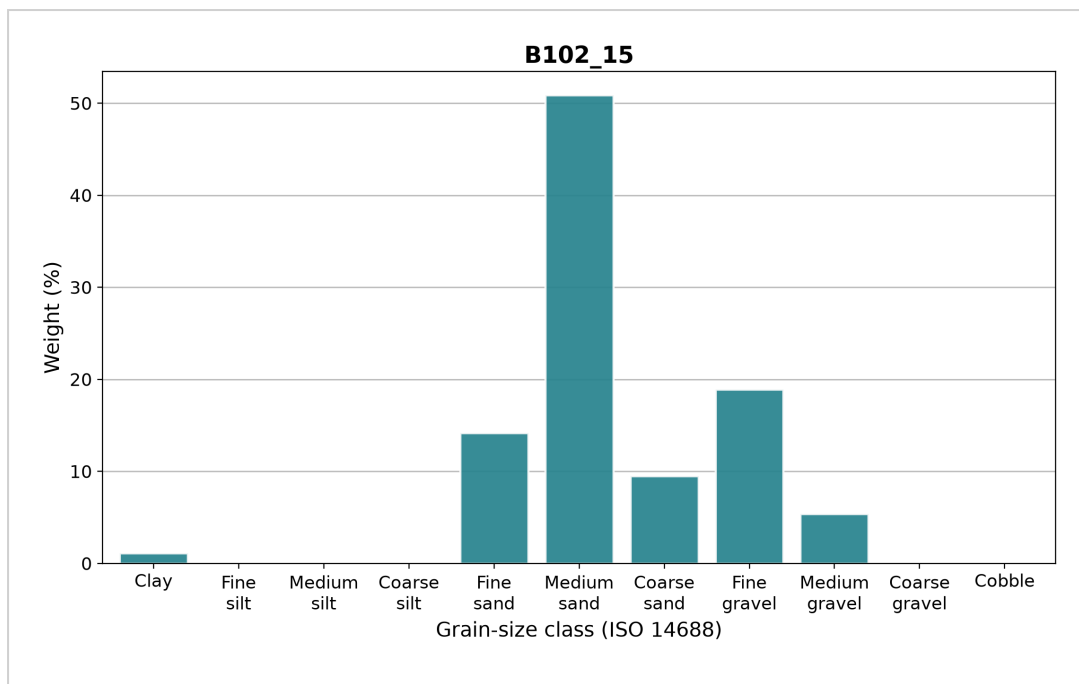
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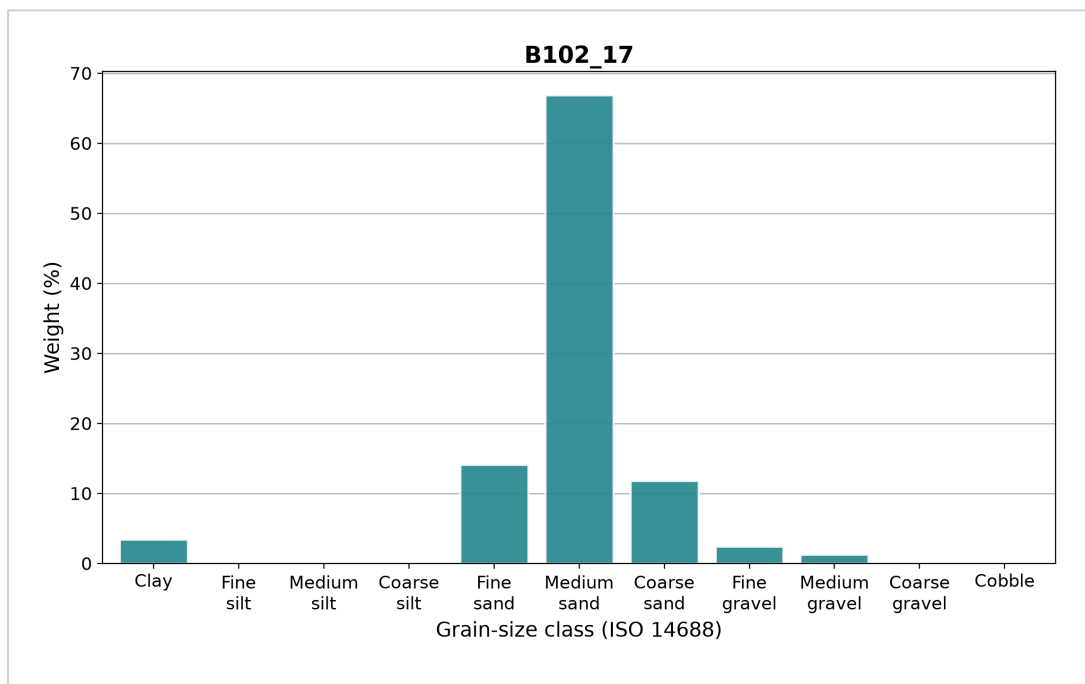
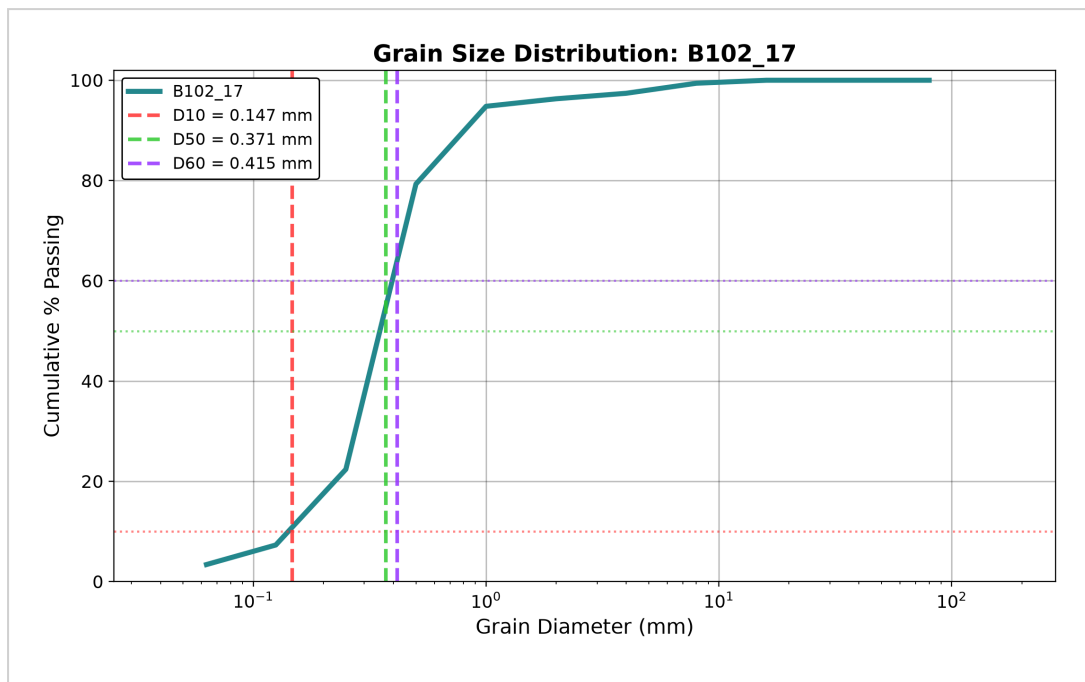


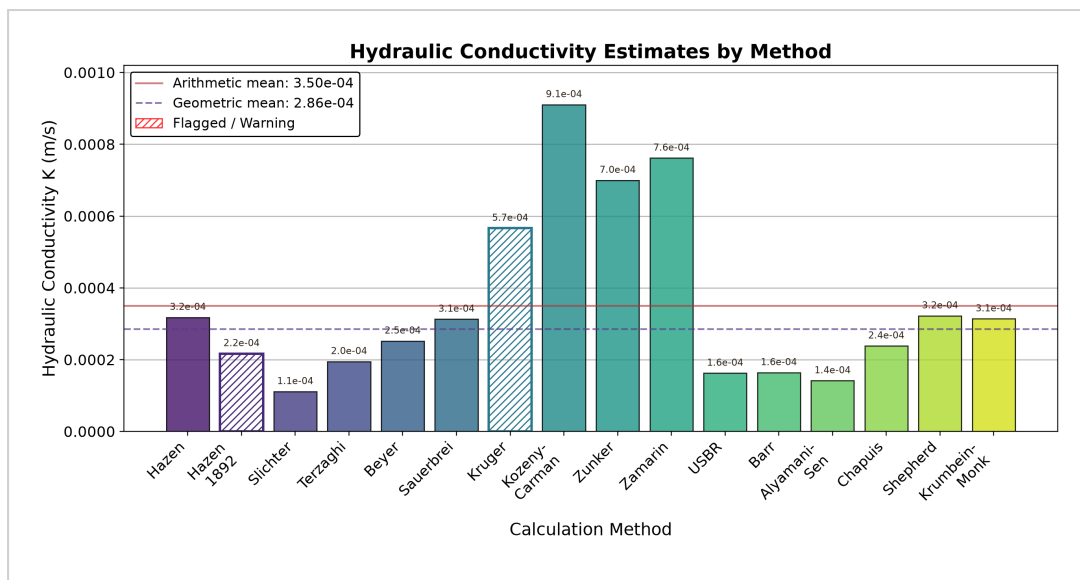
B102_15



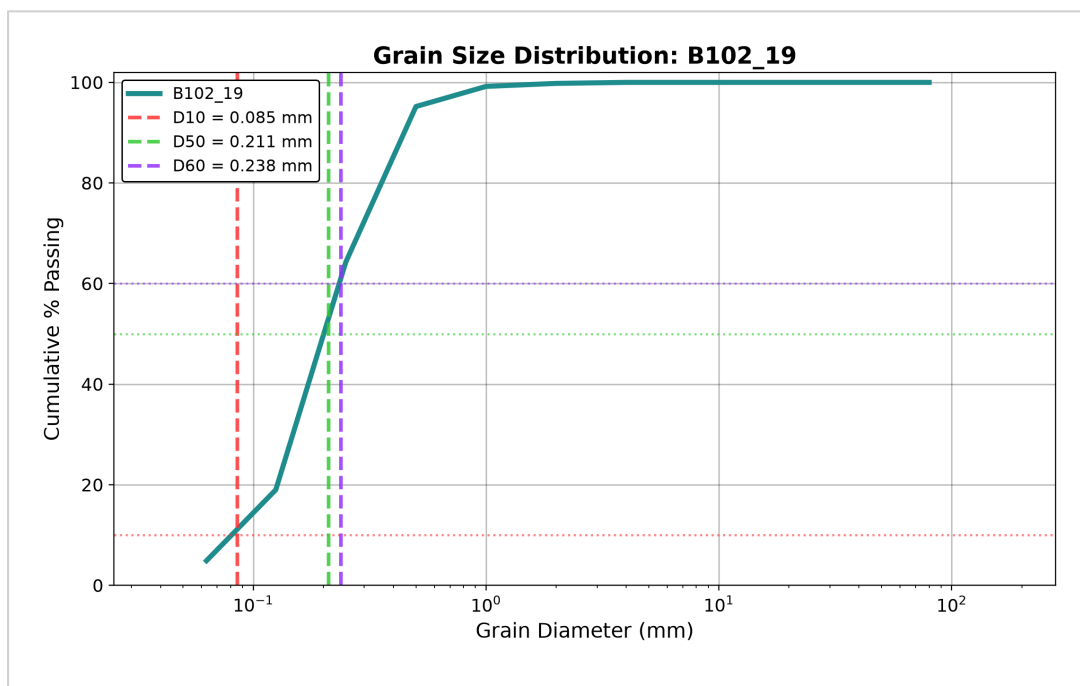


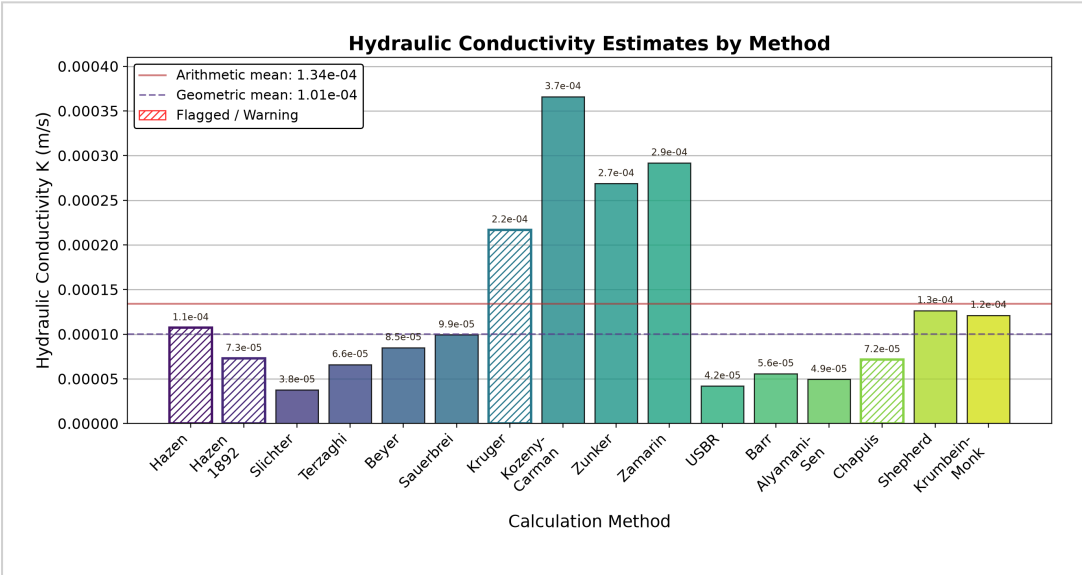
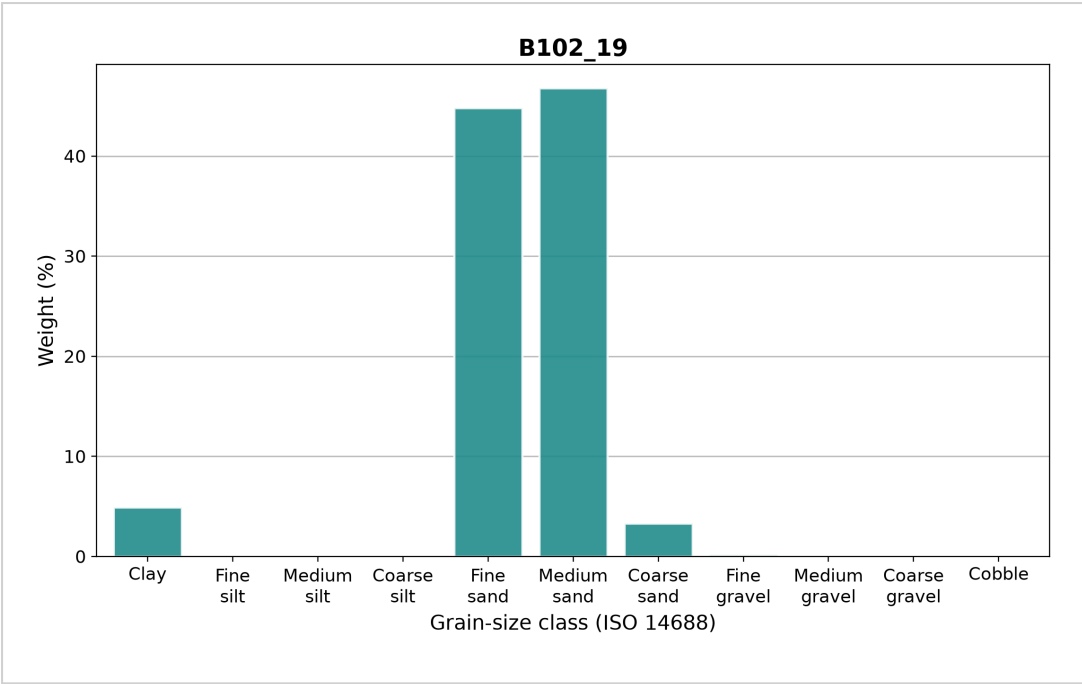
B102_17



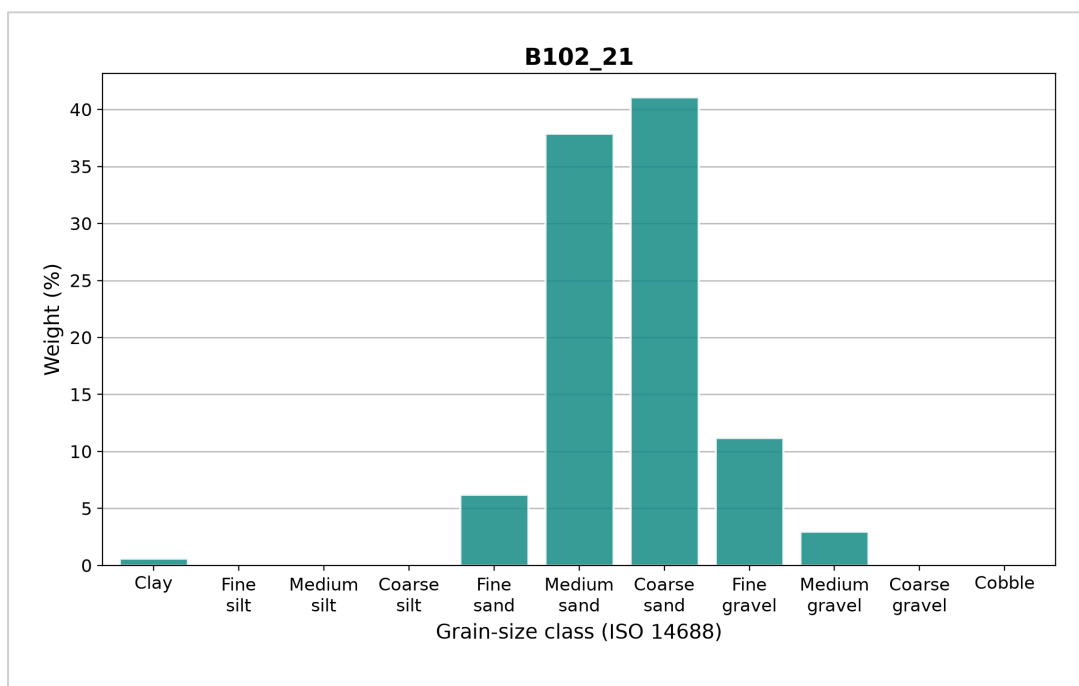
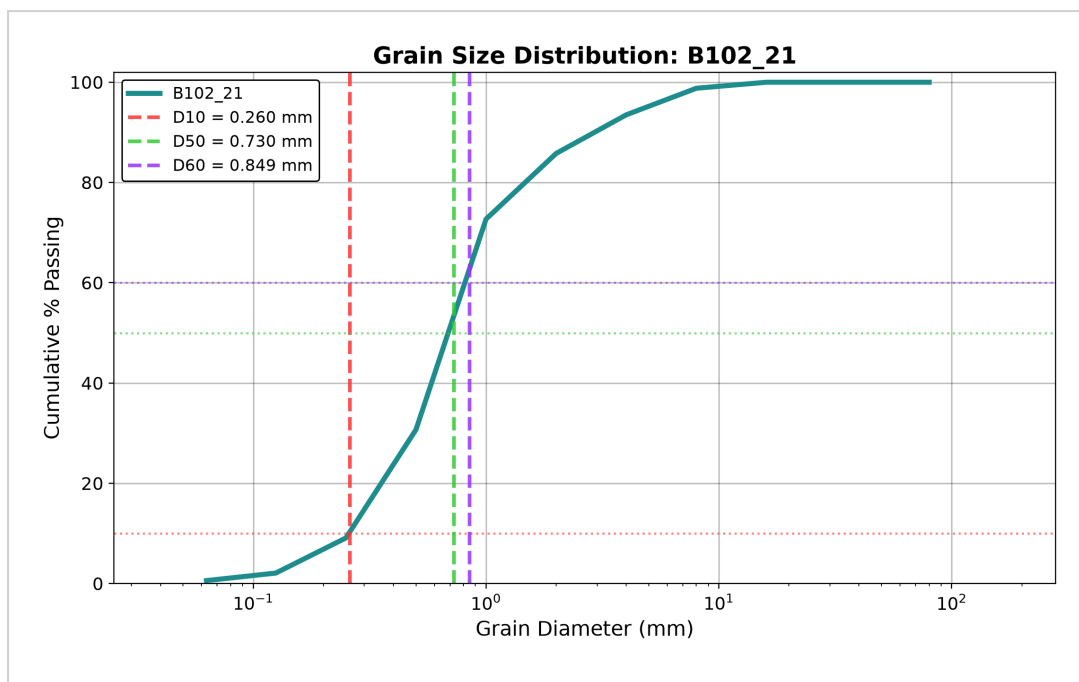


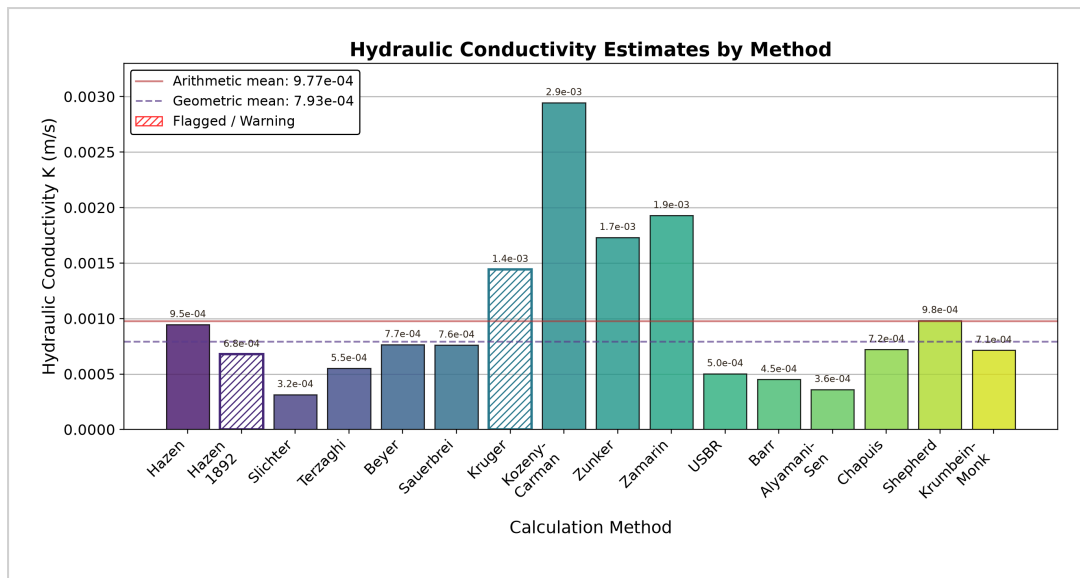
B102_19



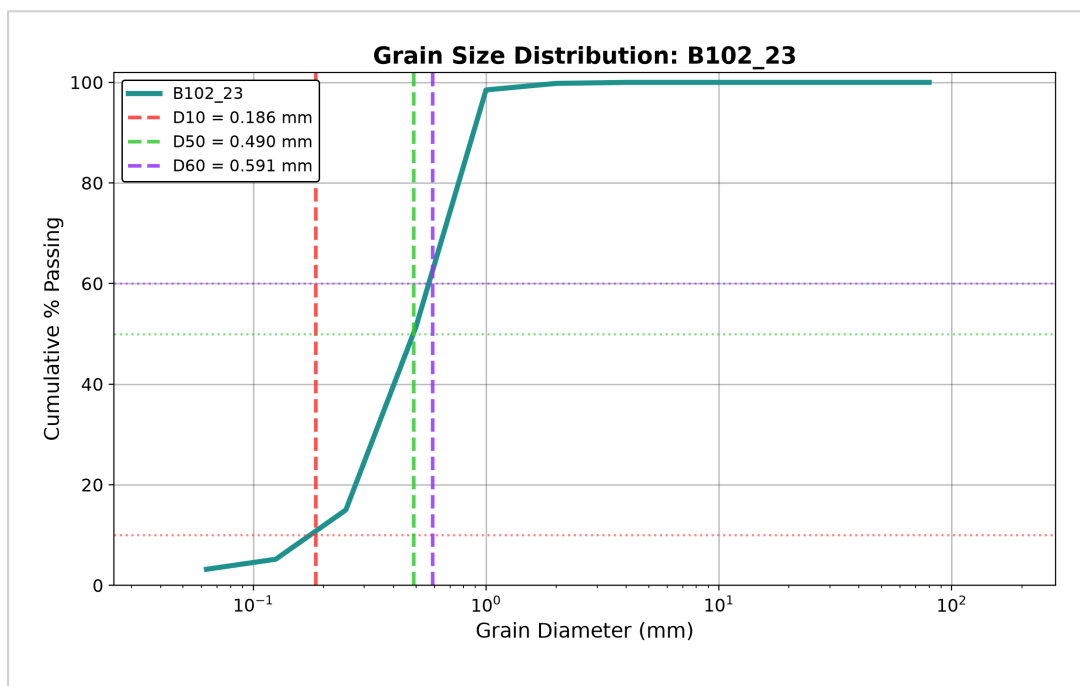


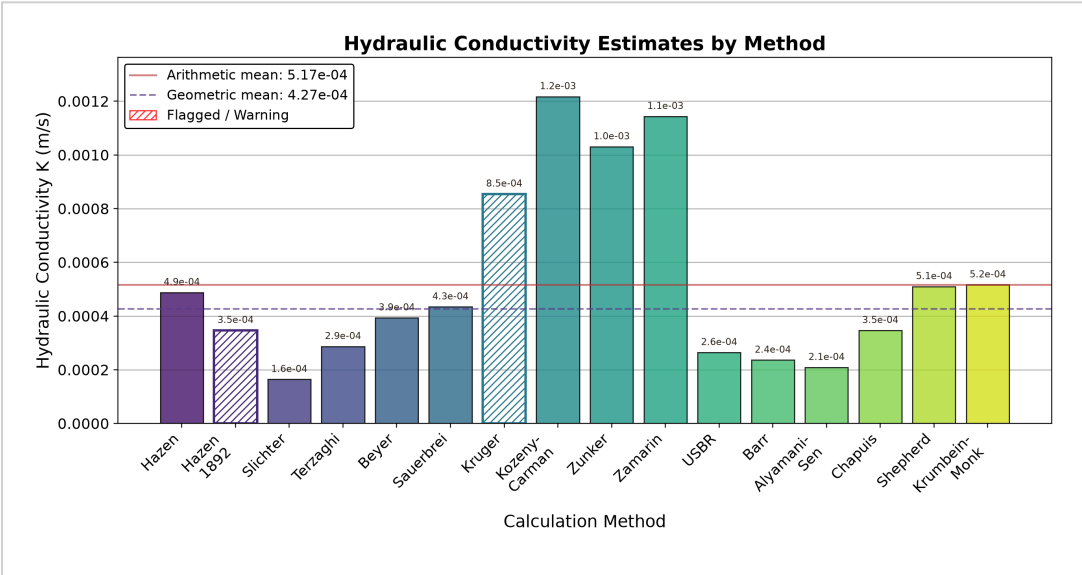
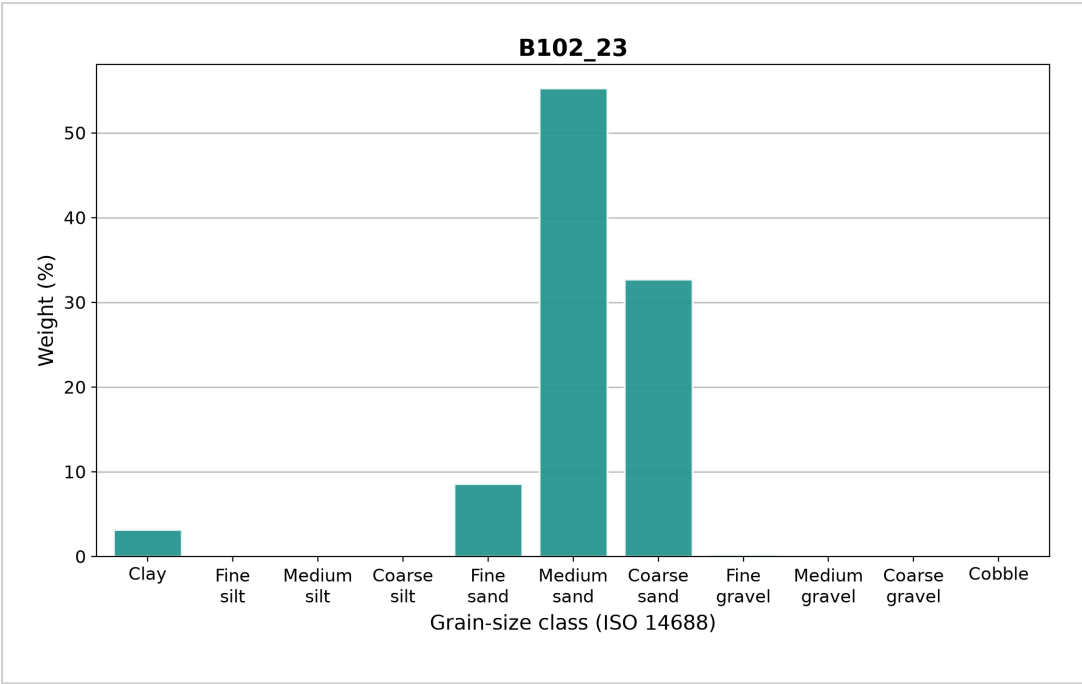
B102_21



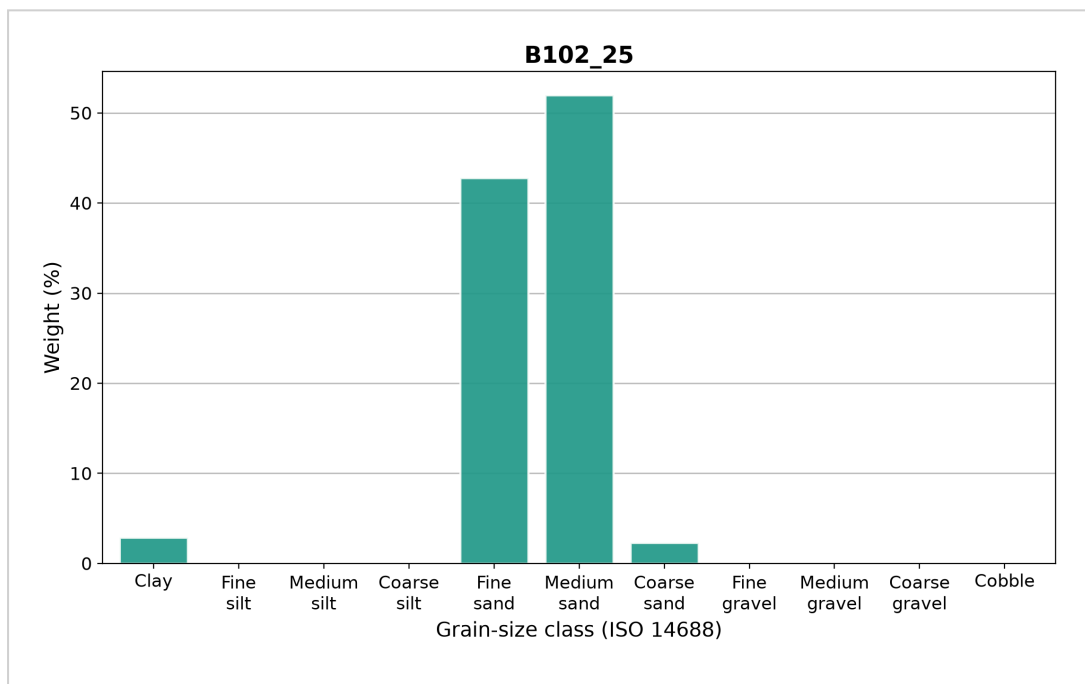
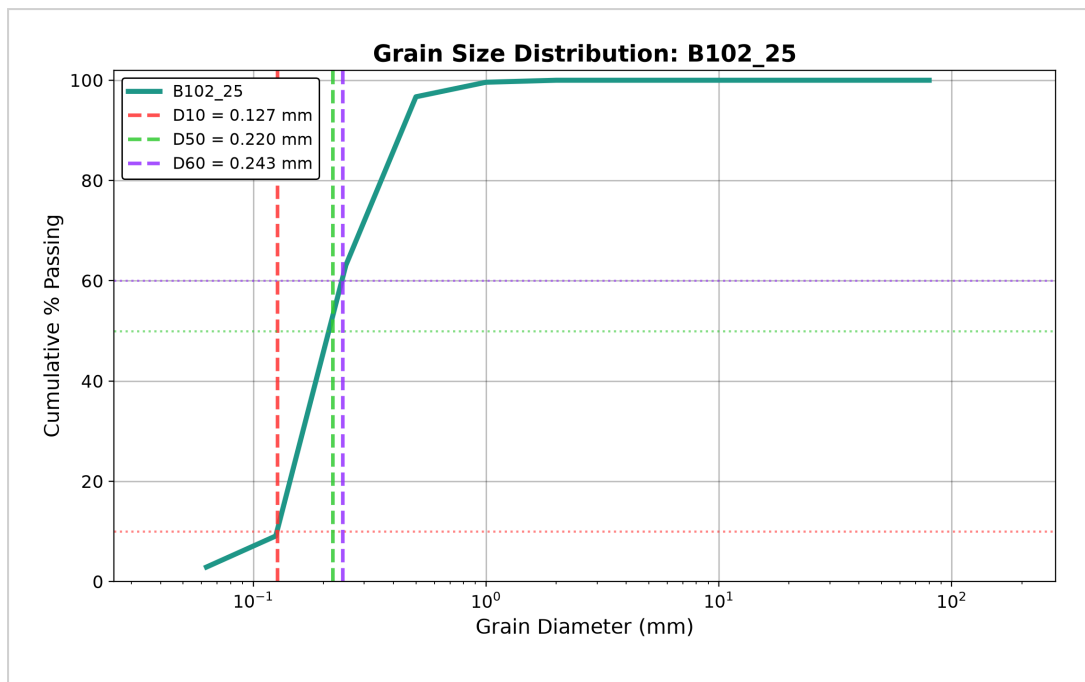


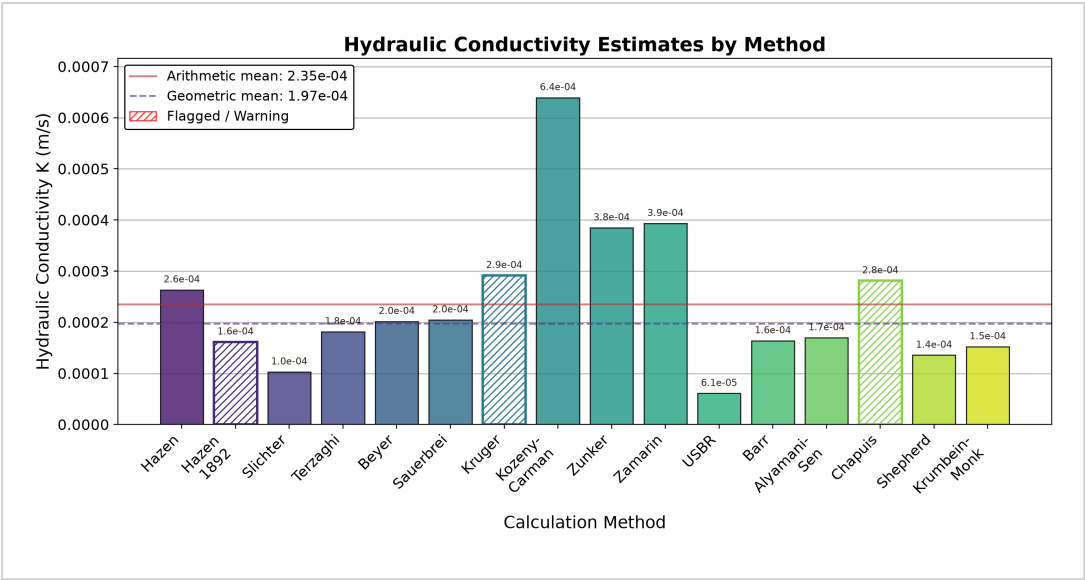
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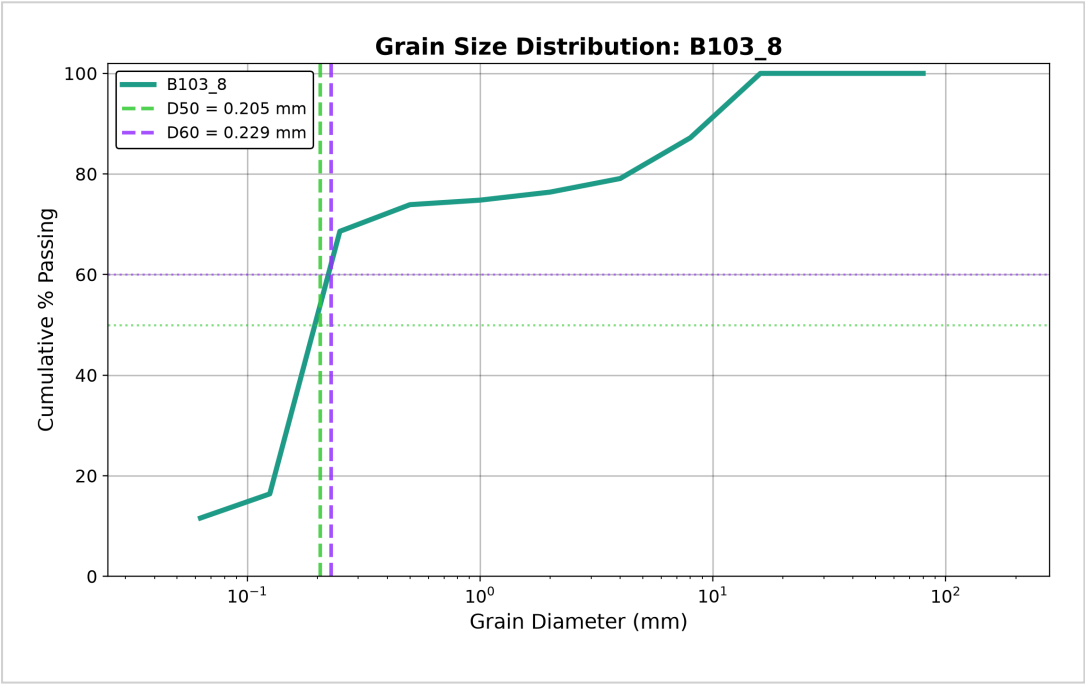


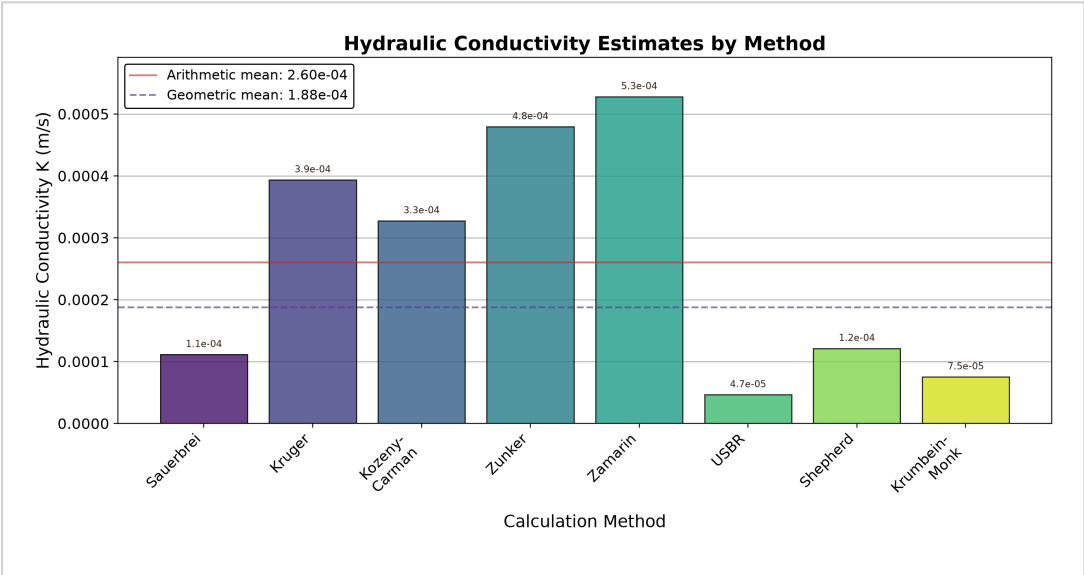
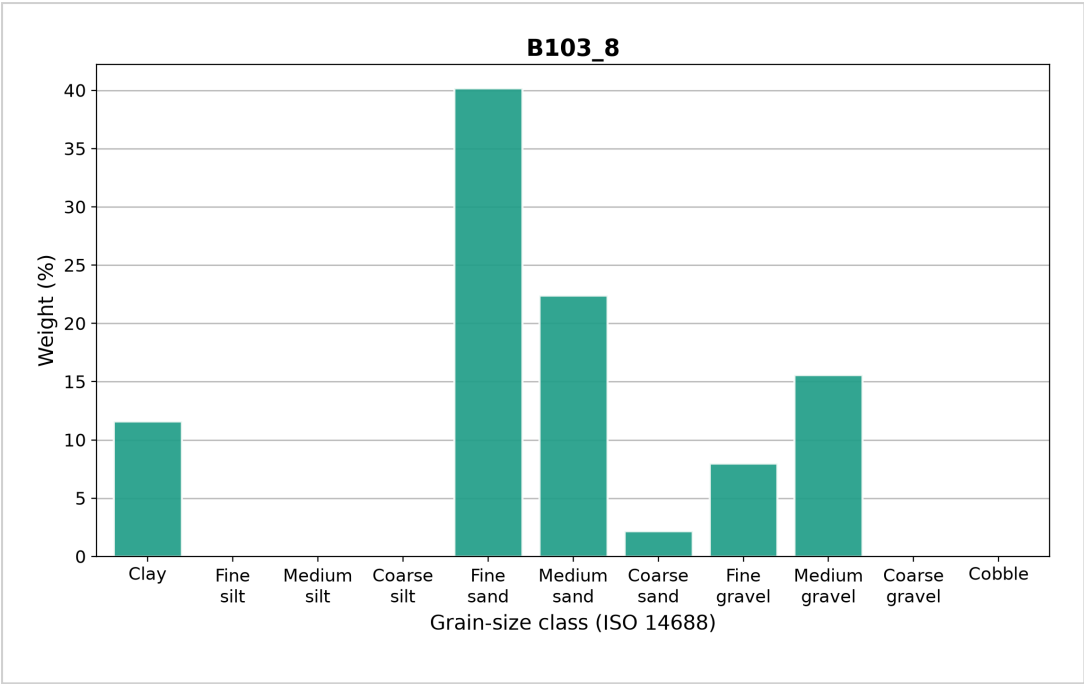
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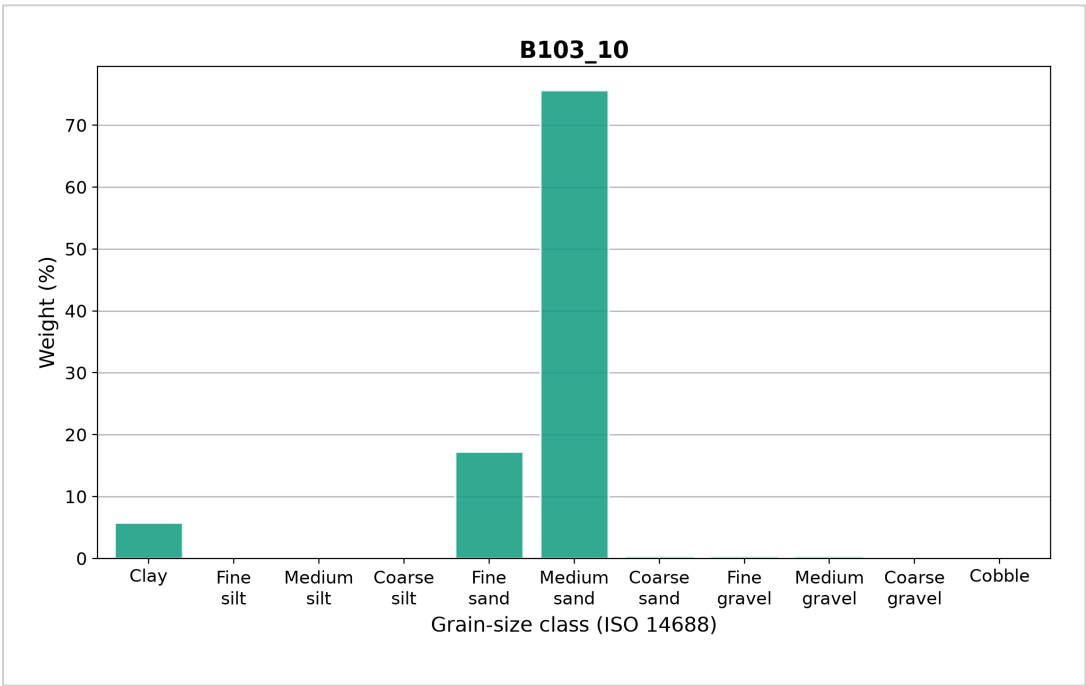
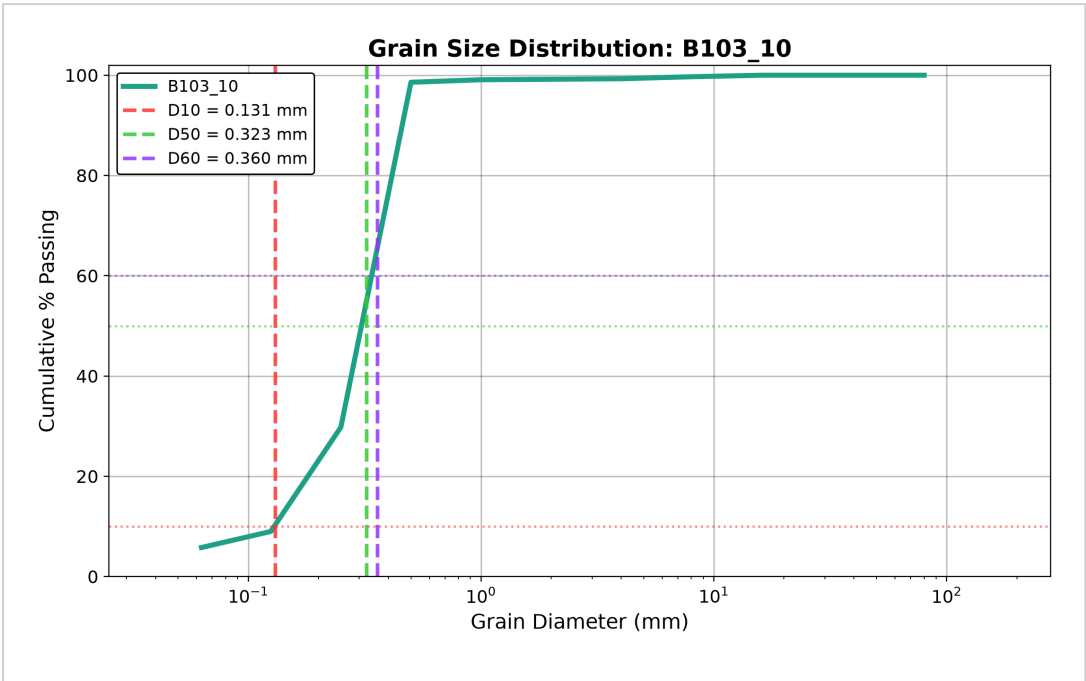


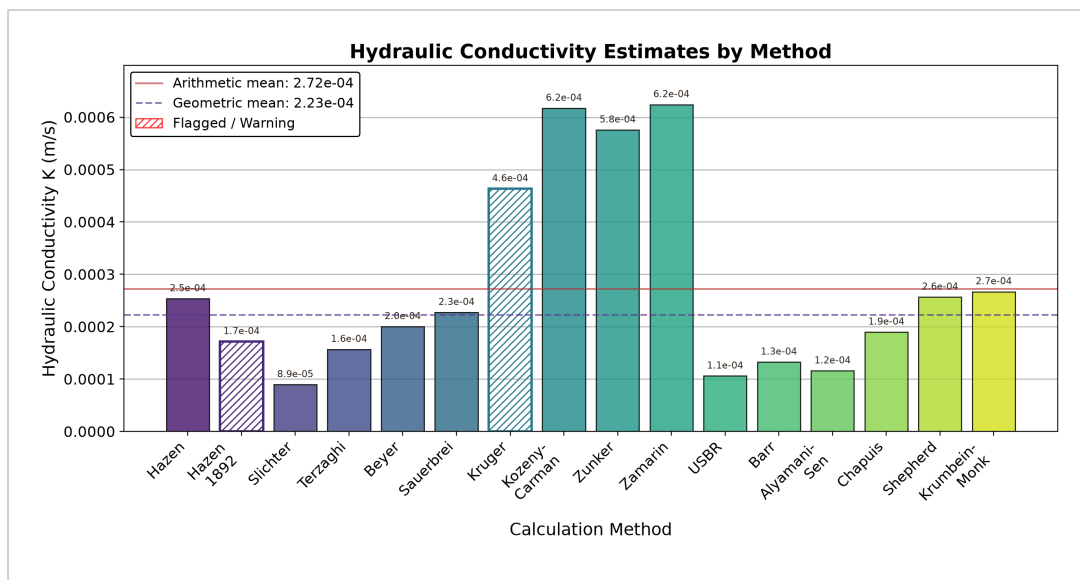
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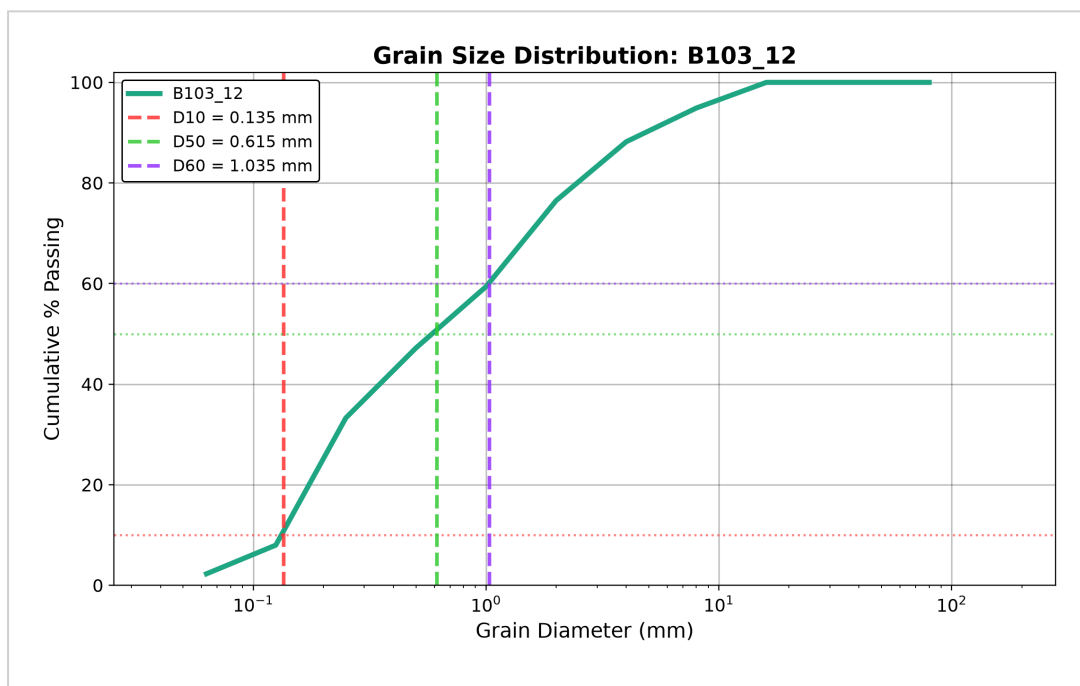


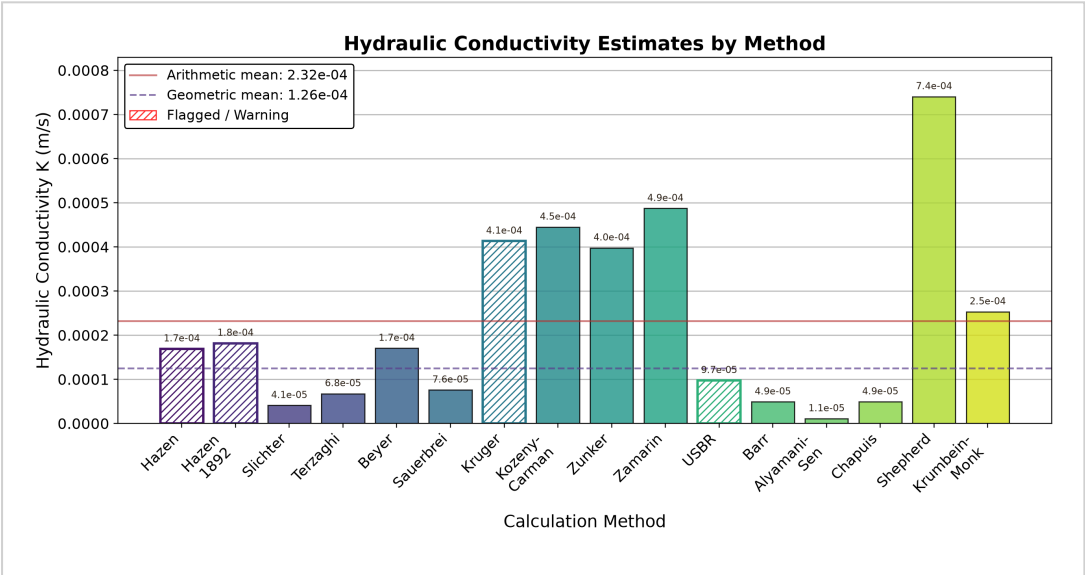
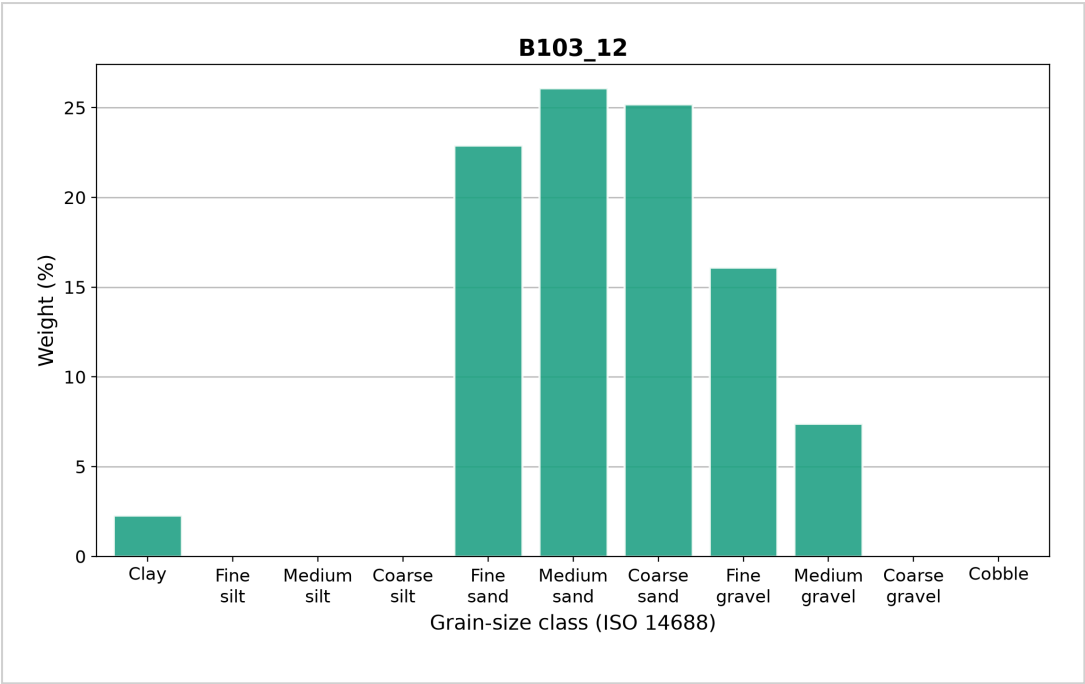
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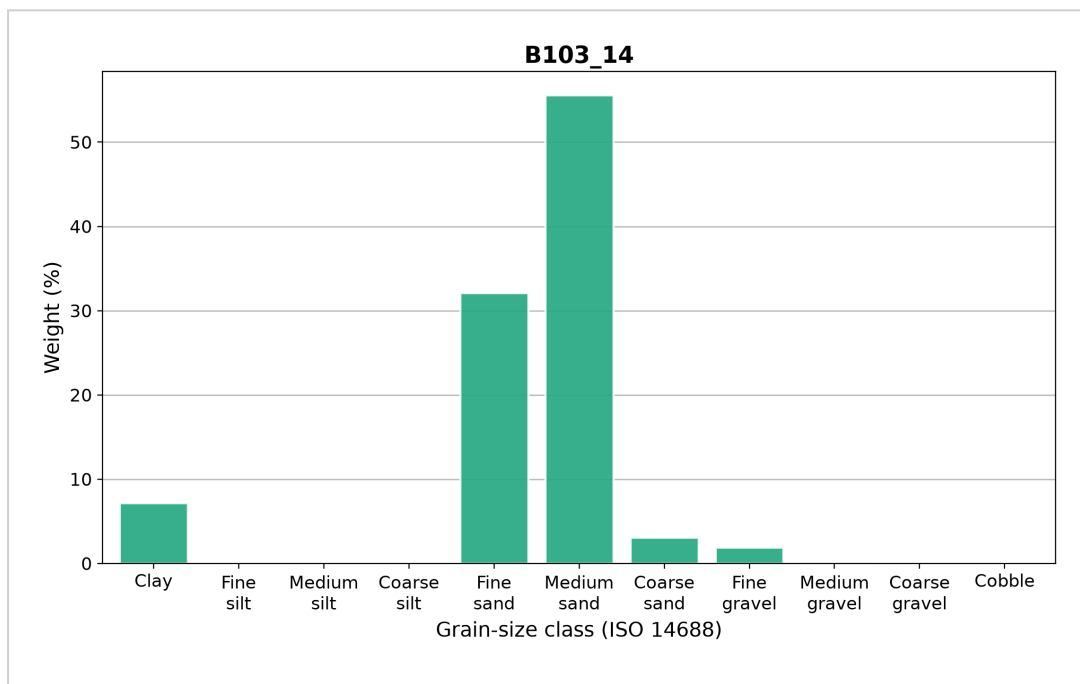
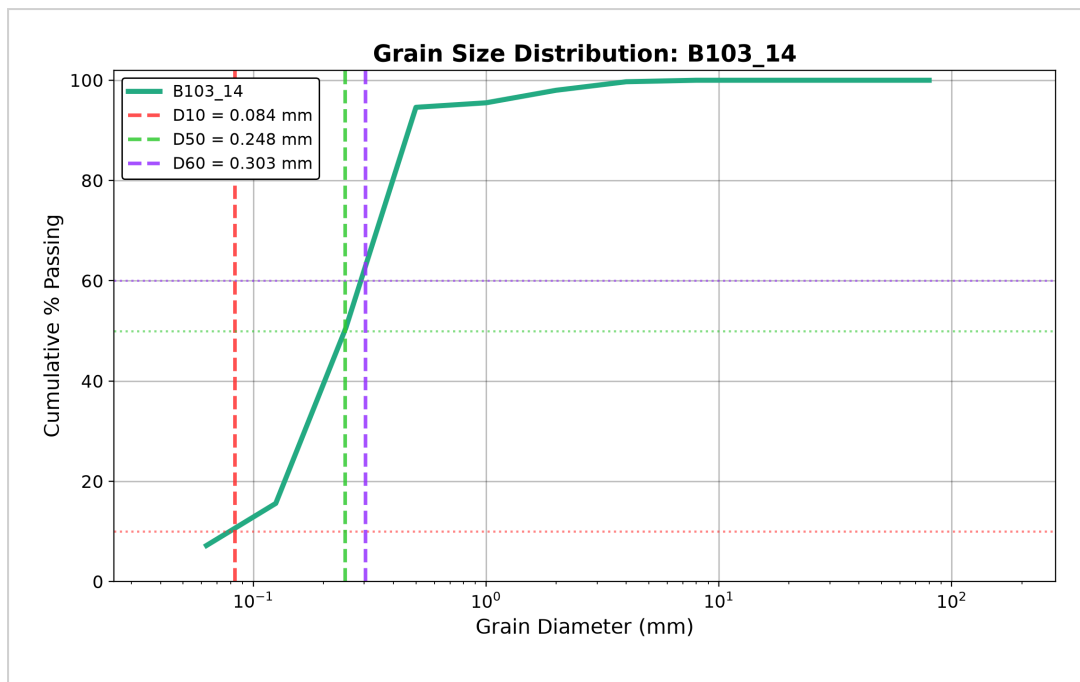


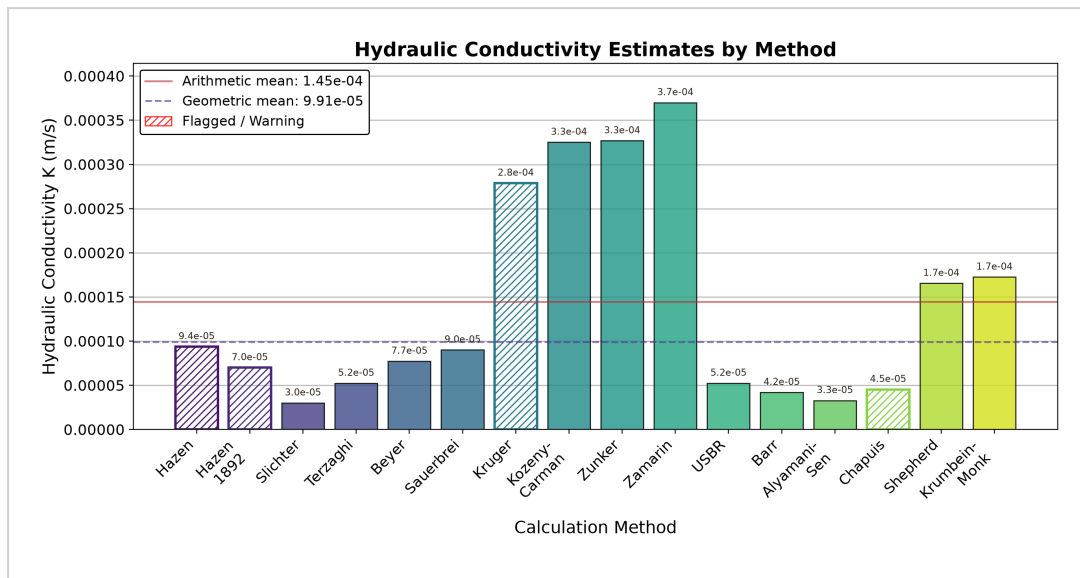
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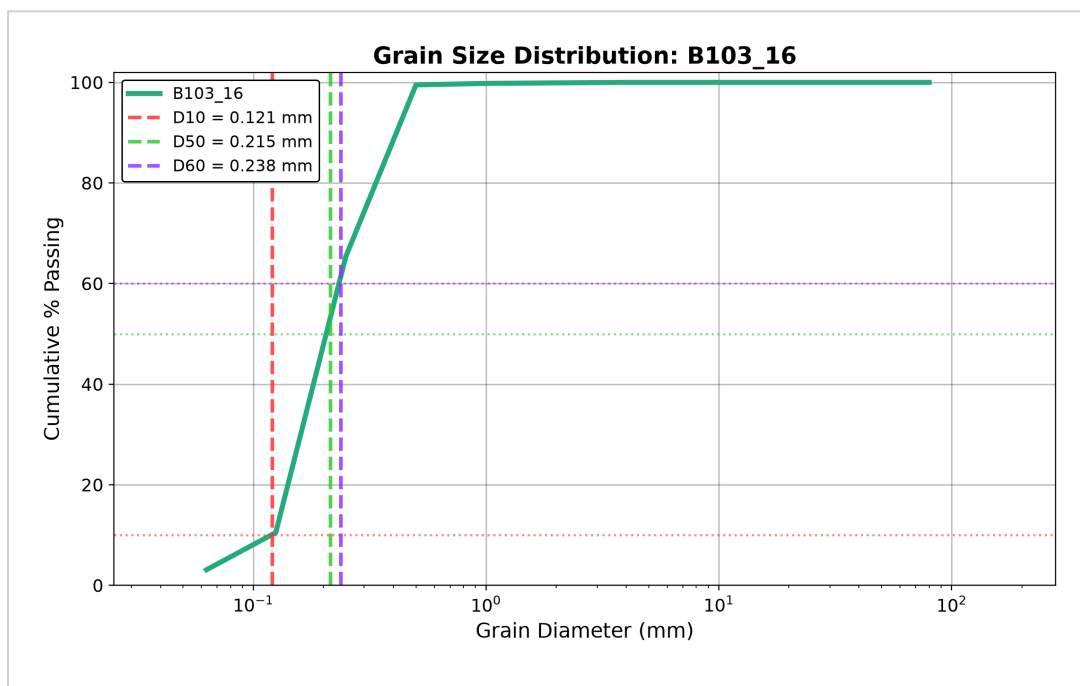


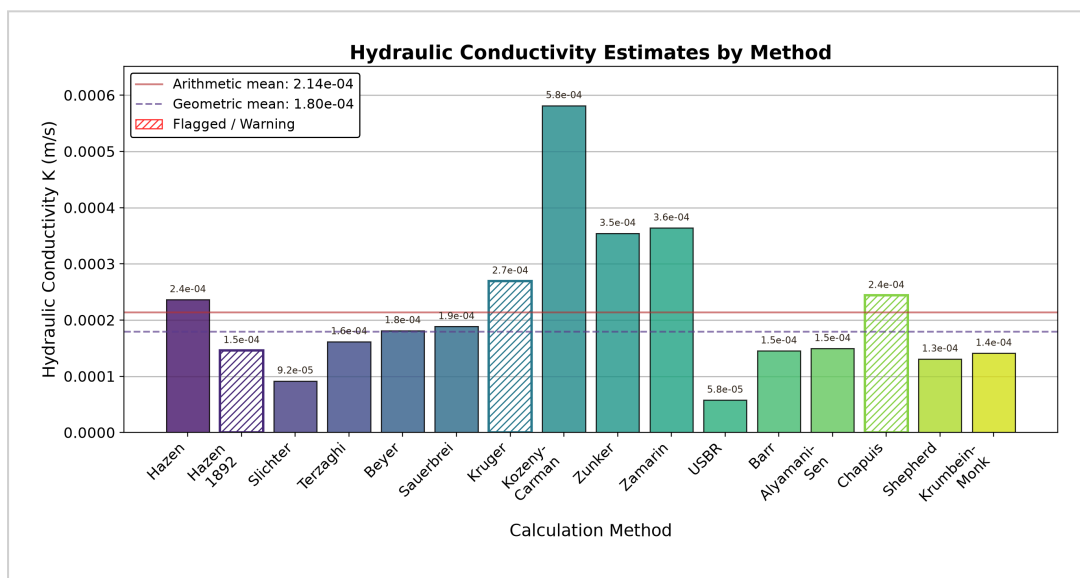
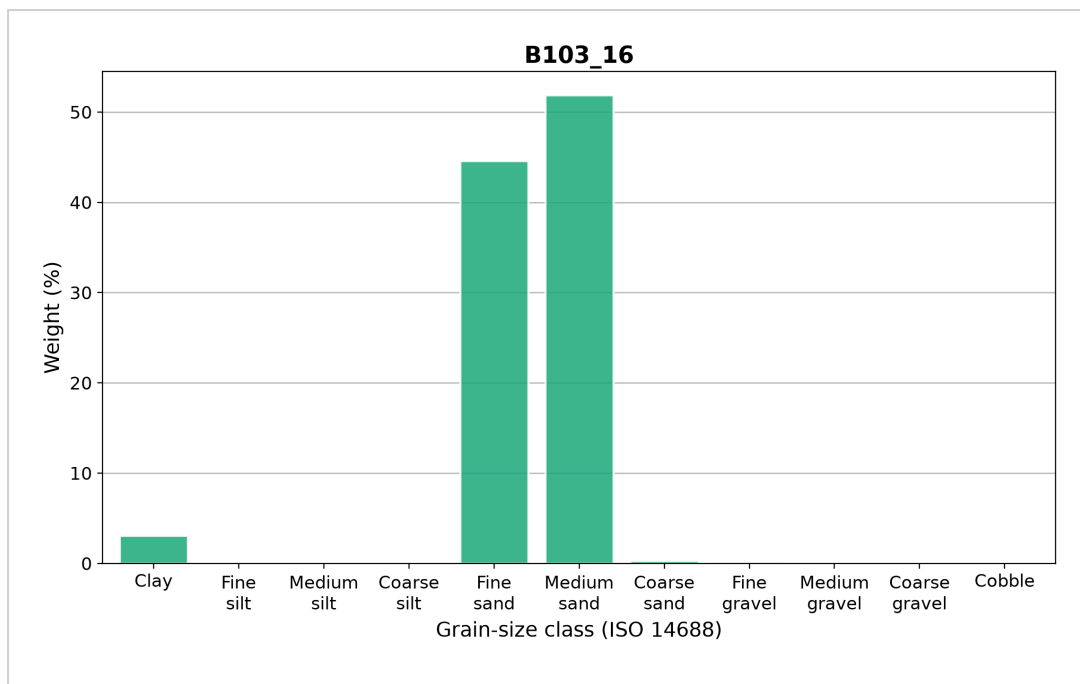
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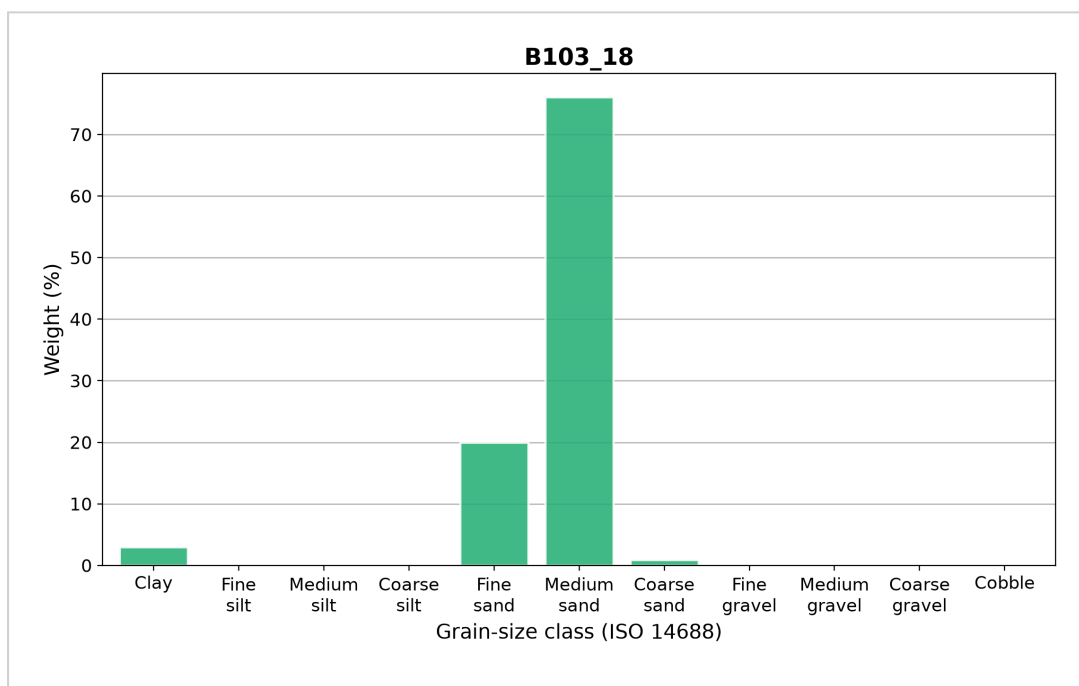
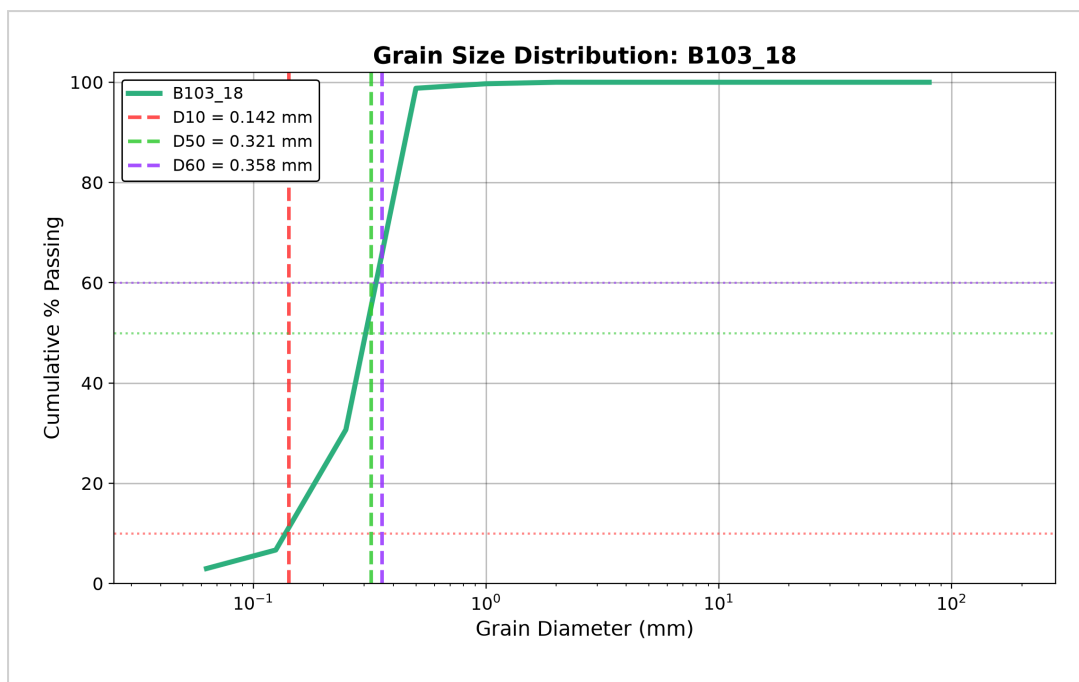


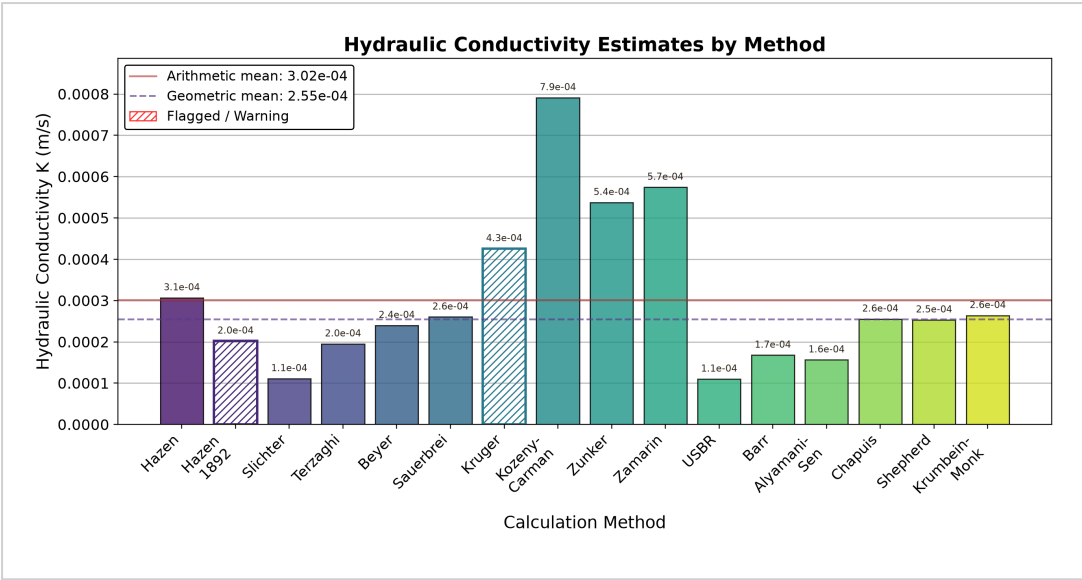
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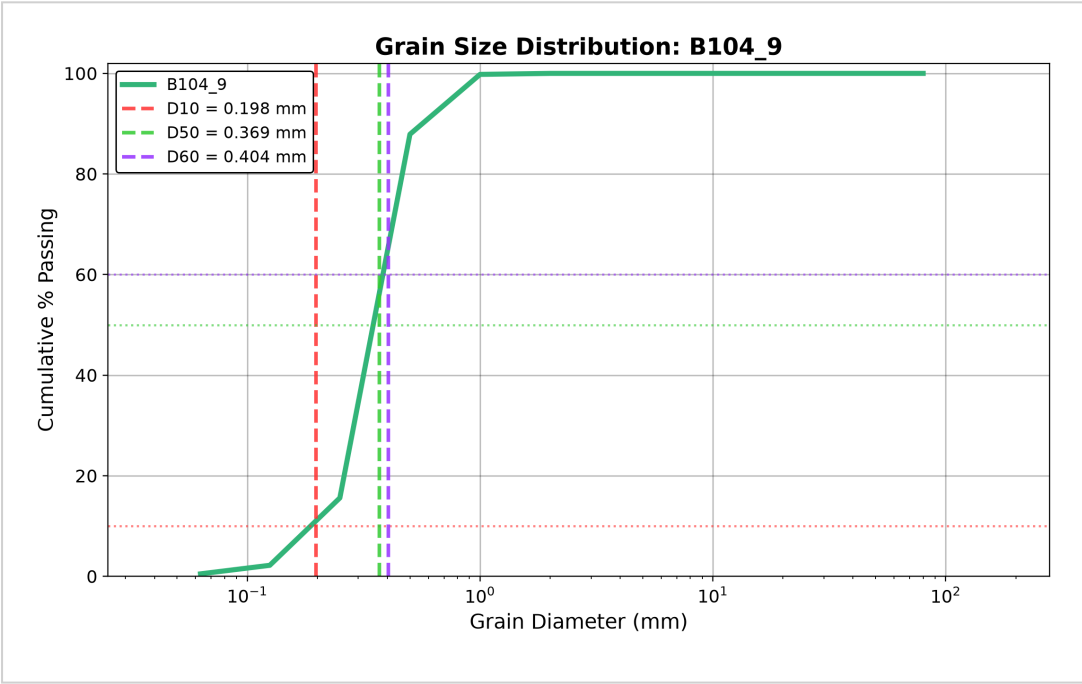


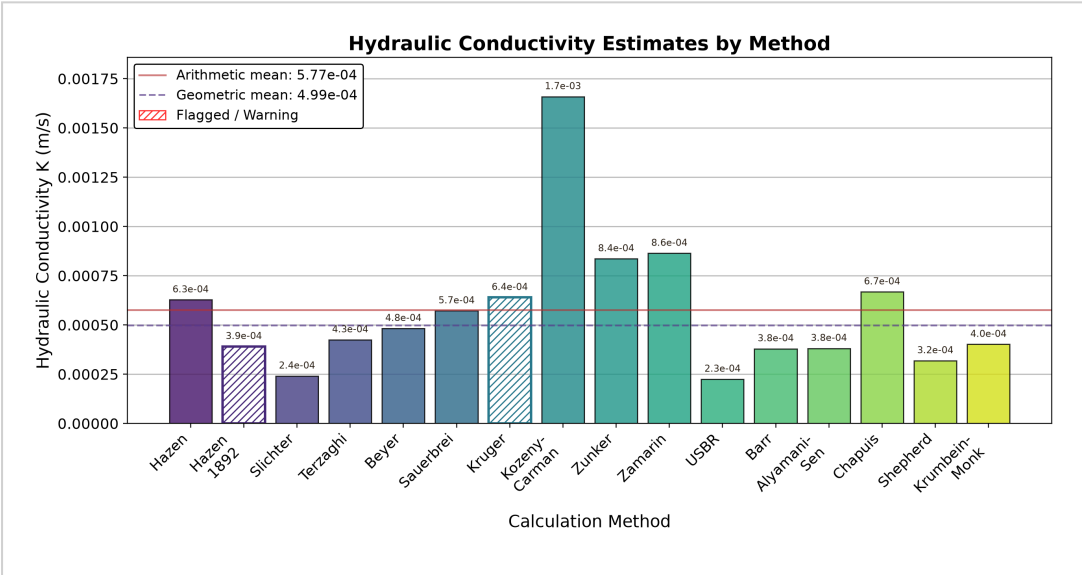
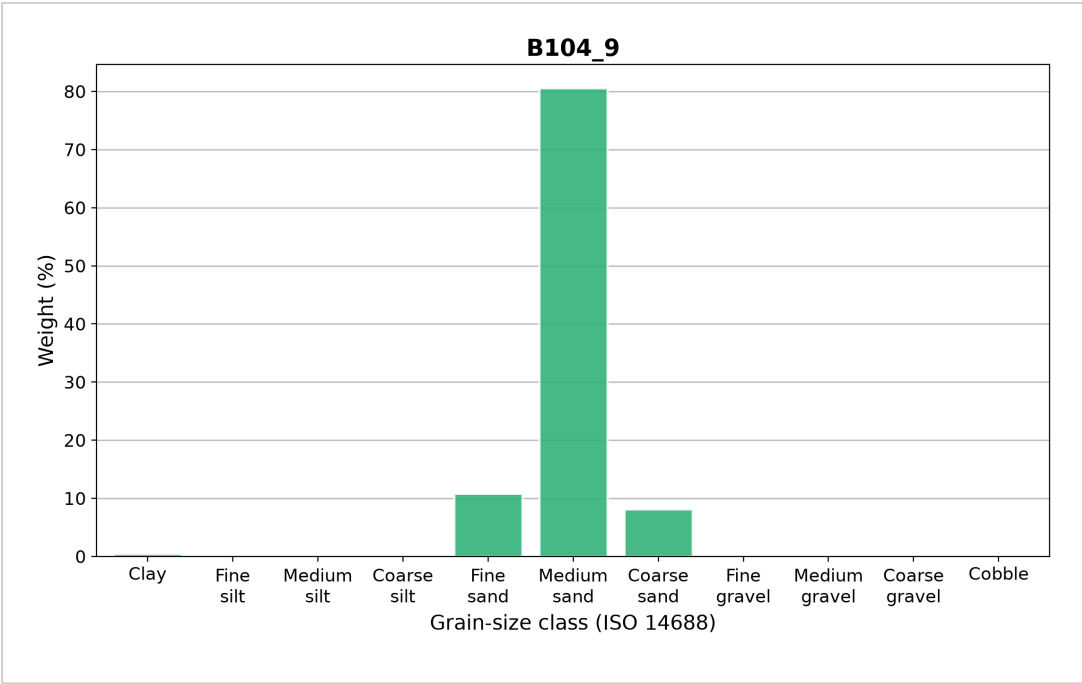
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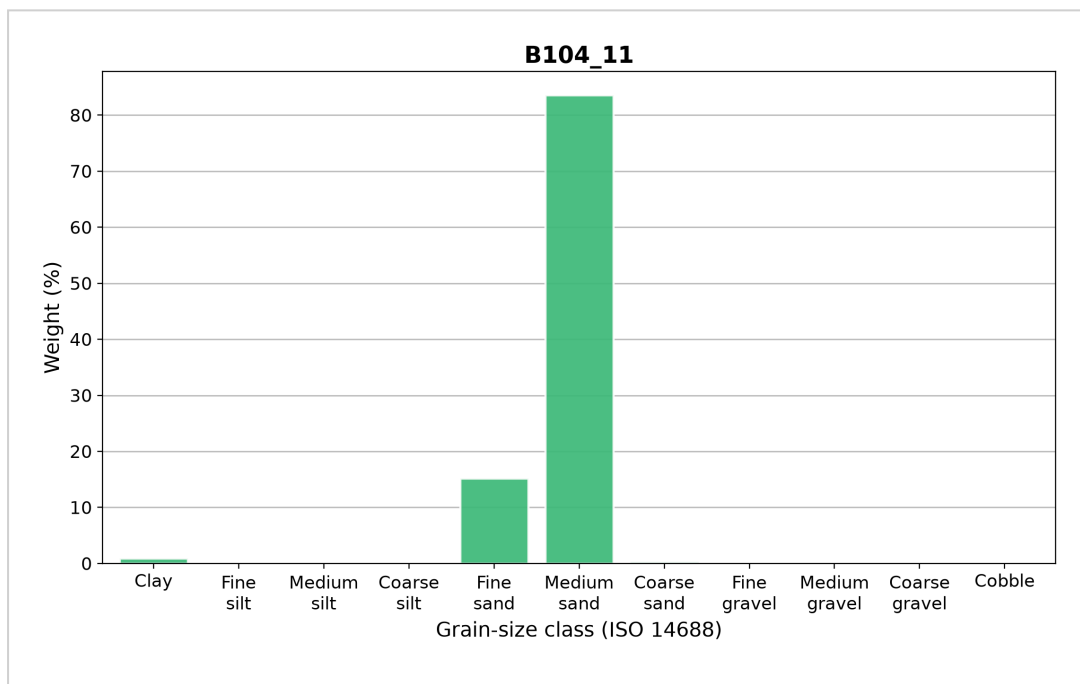
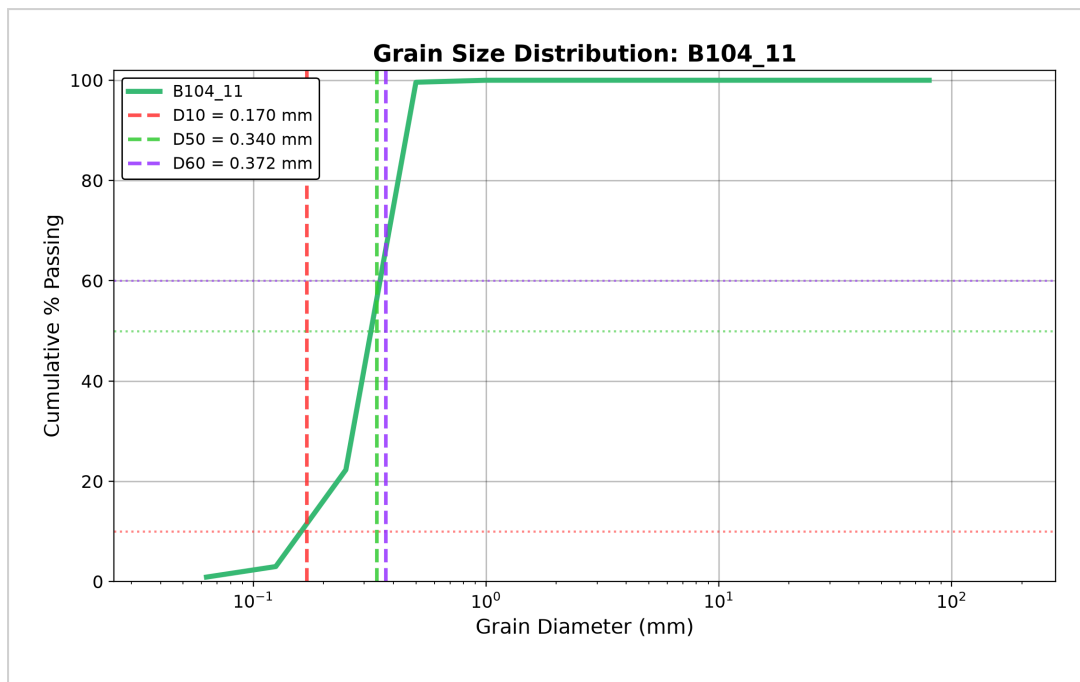


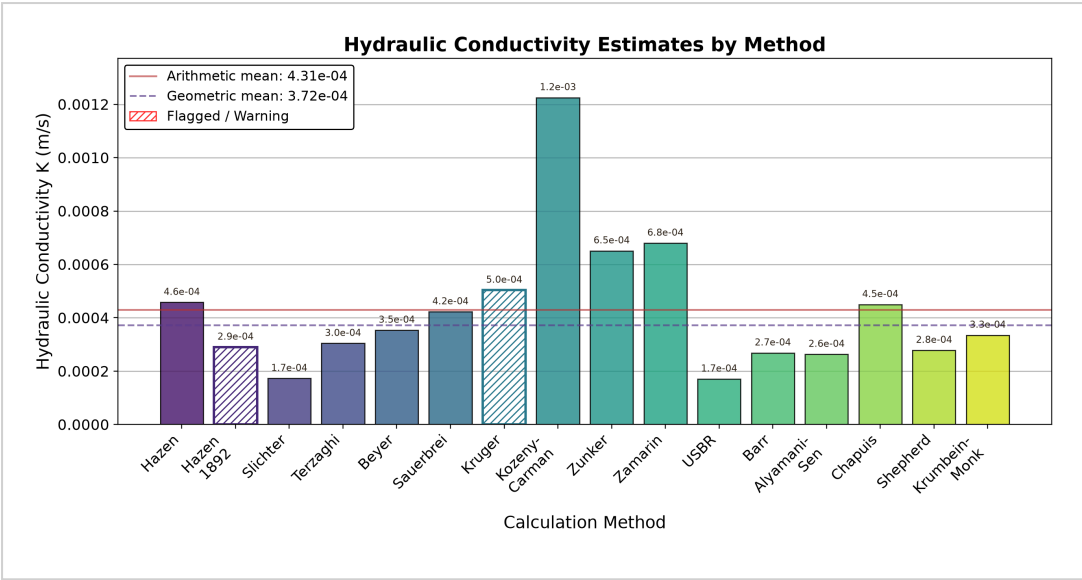
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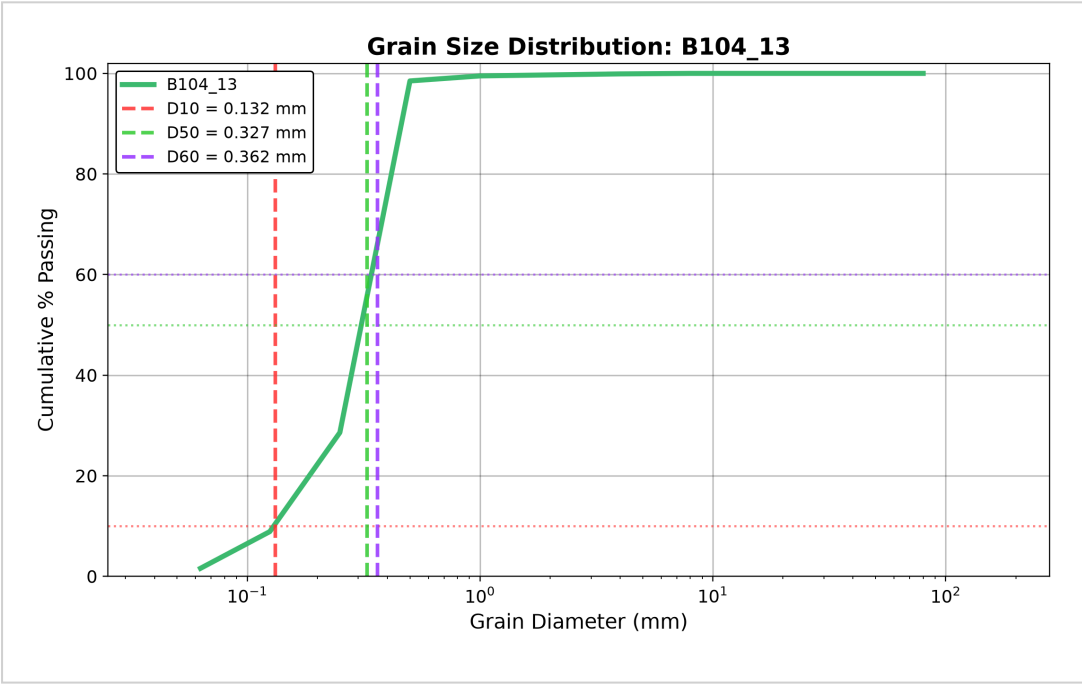


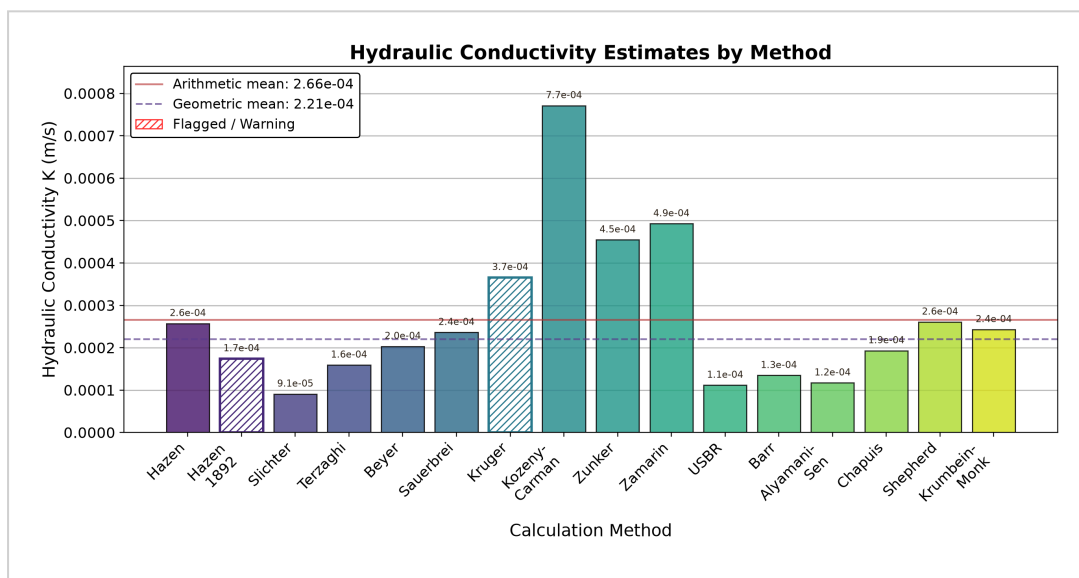
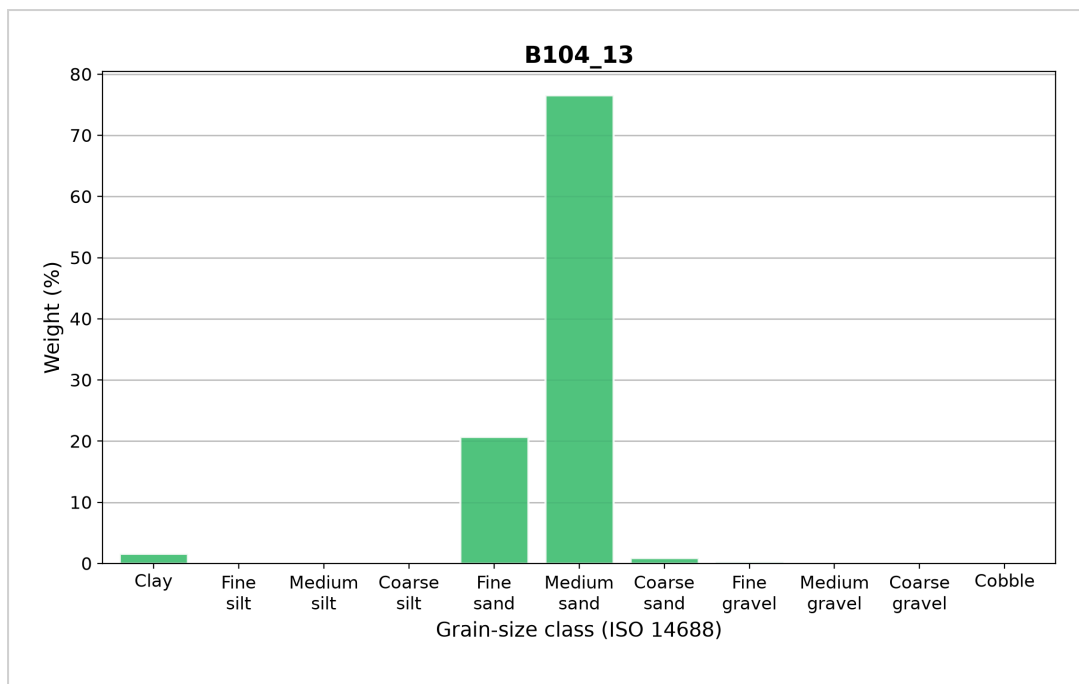
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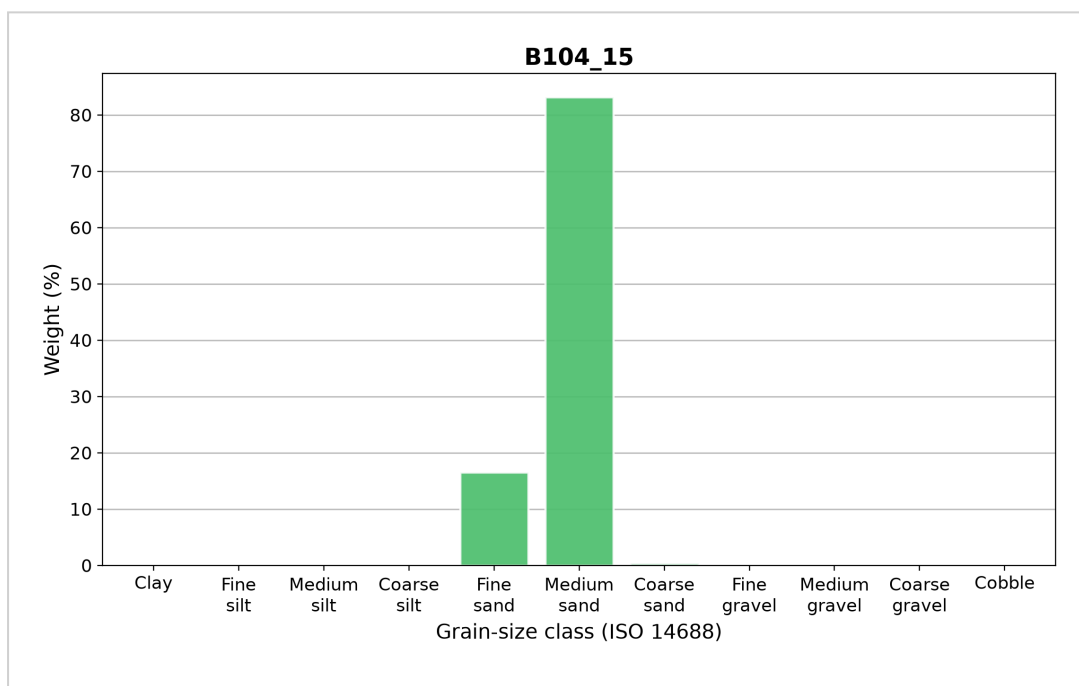
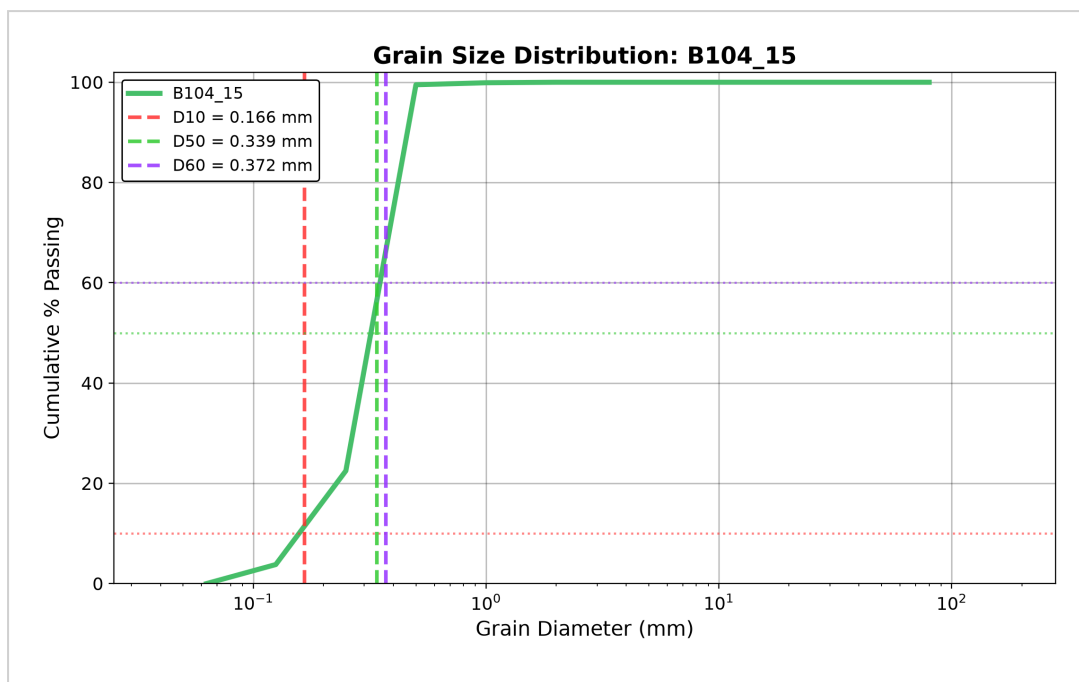


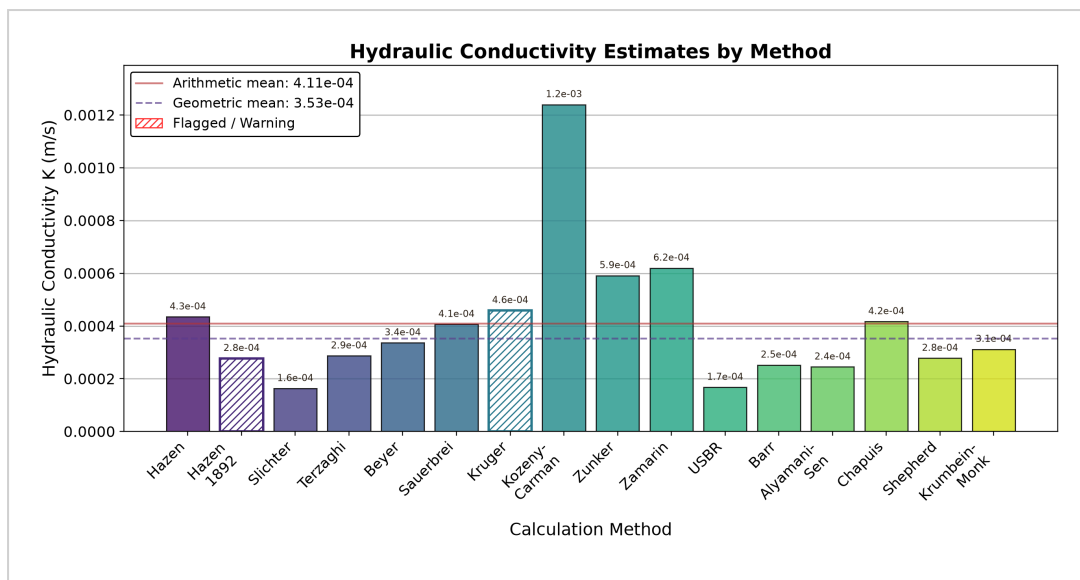
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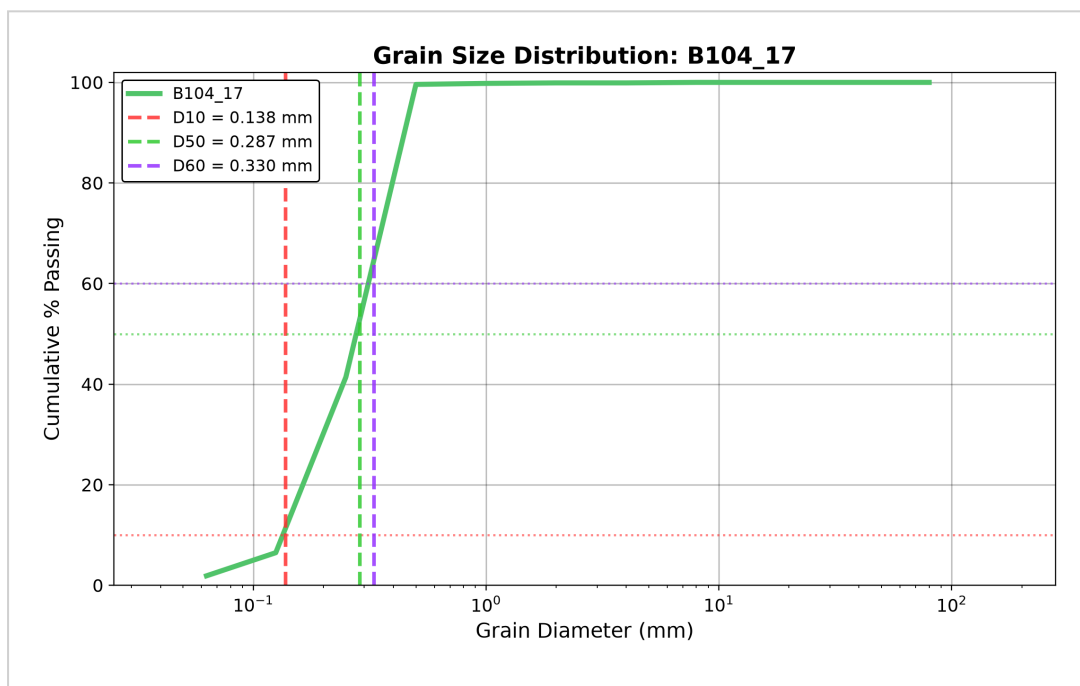


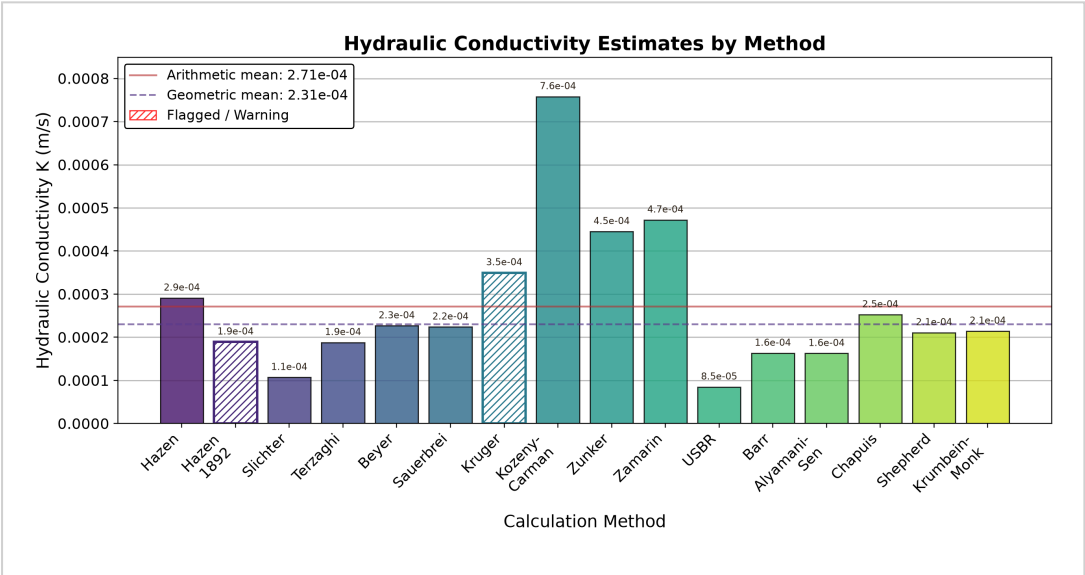
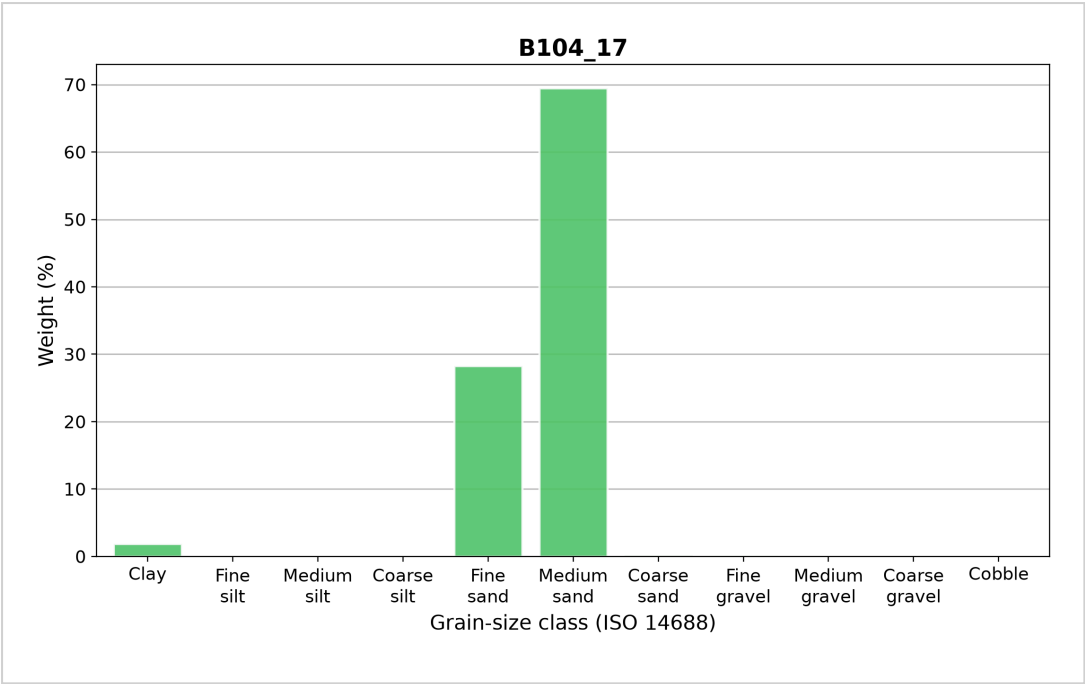
B104_15



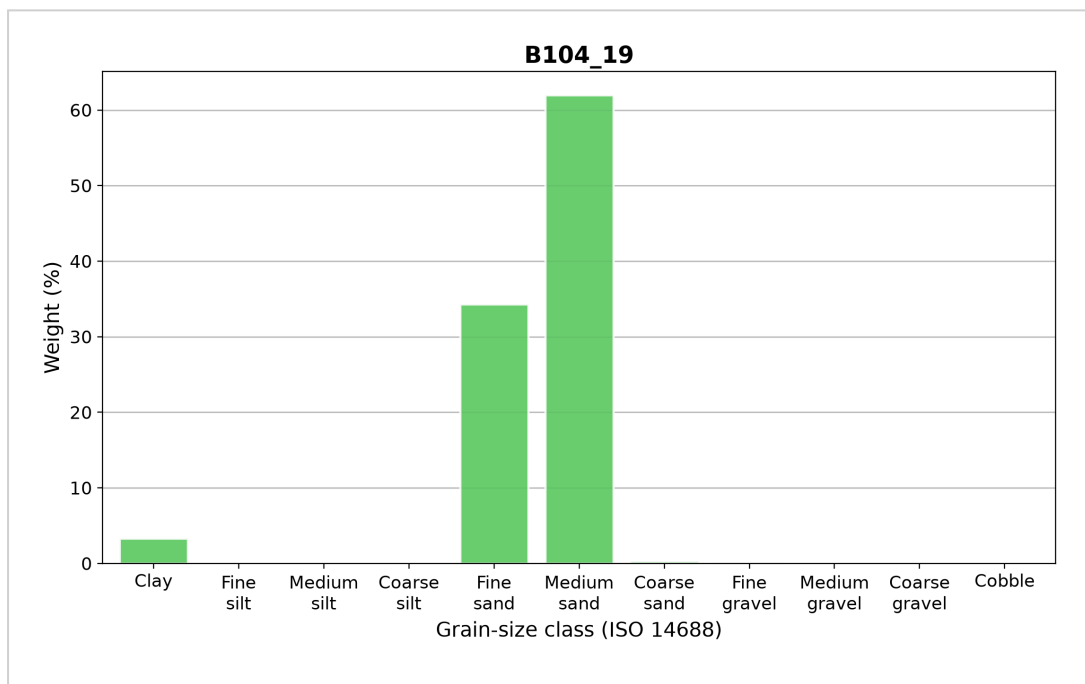
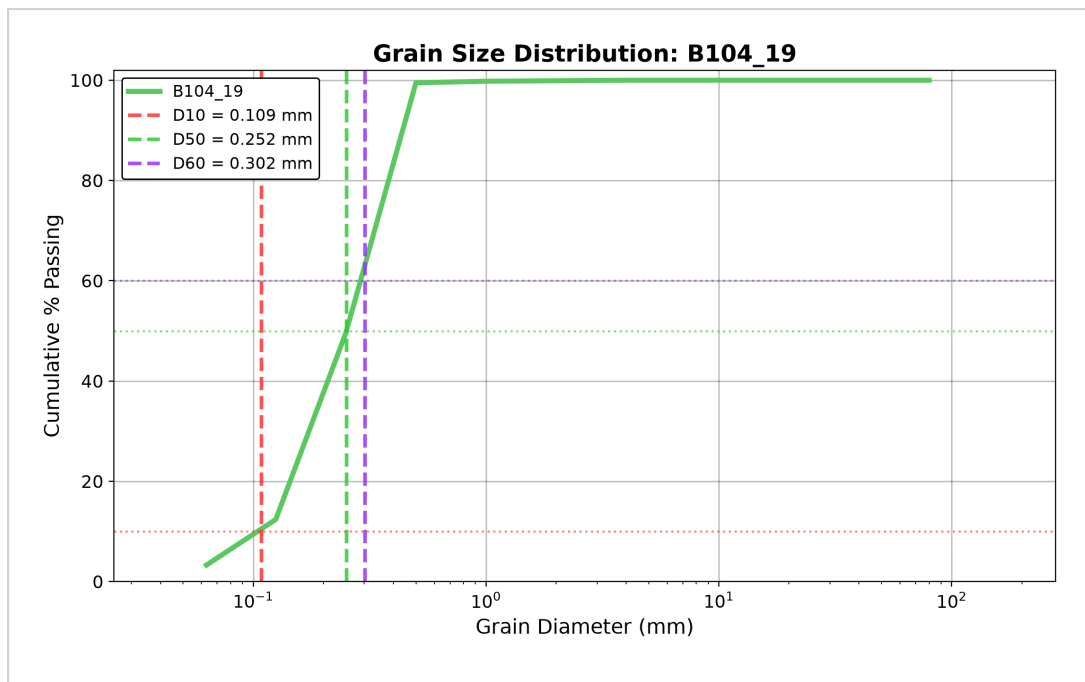


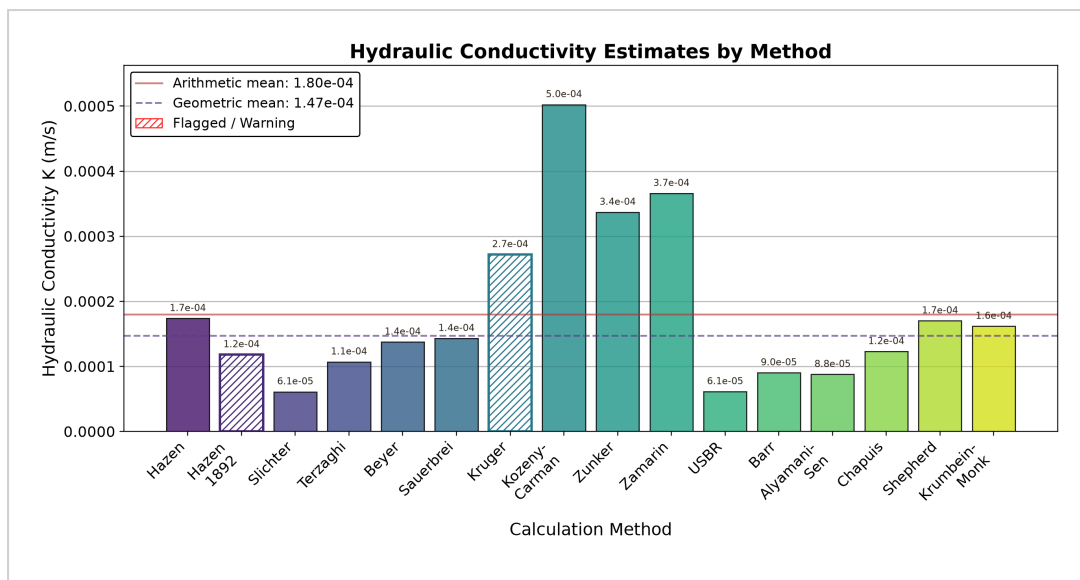
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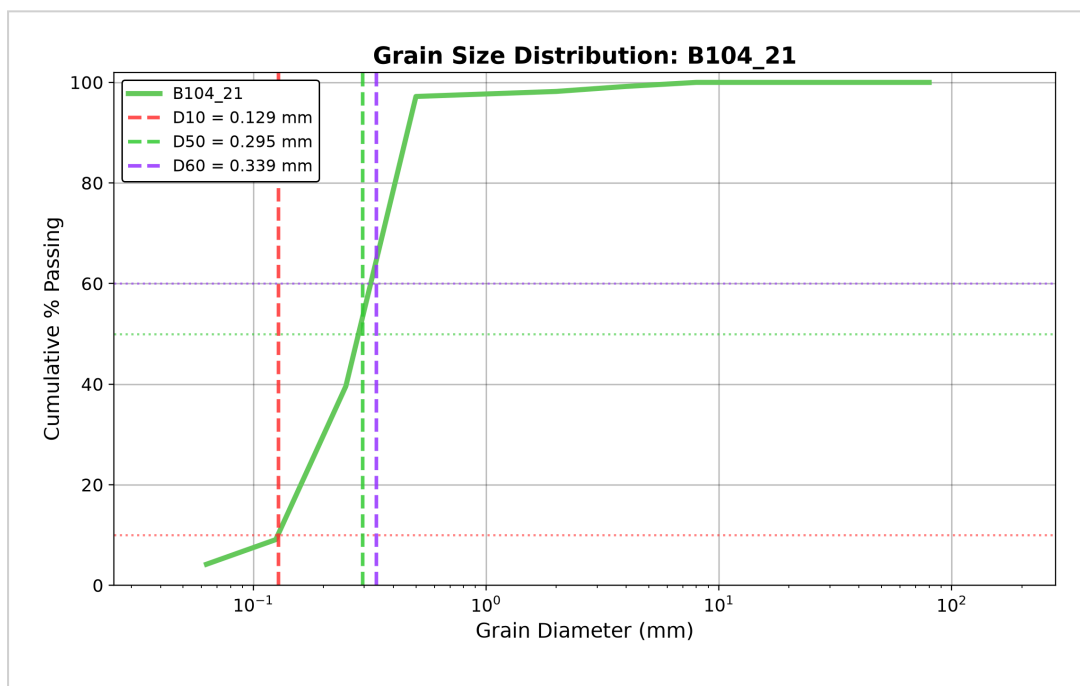


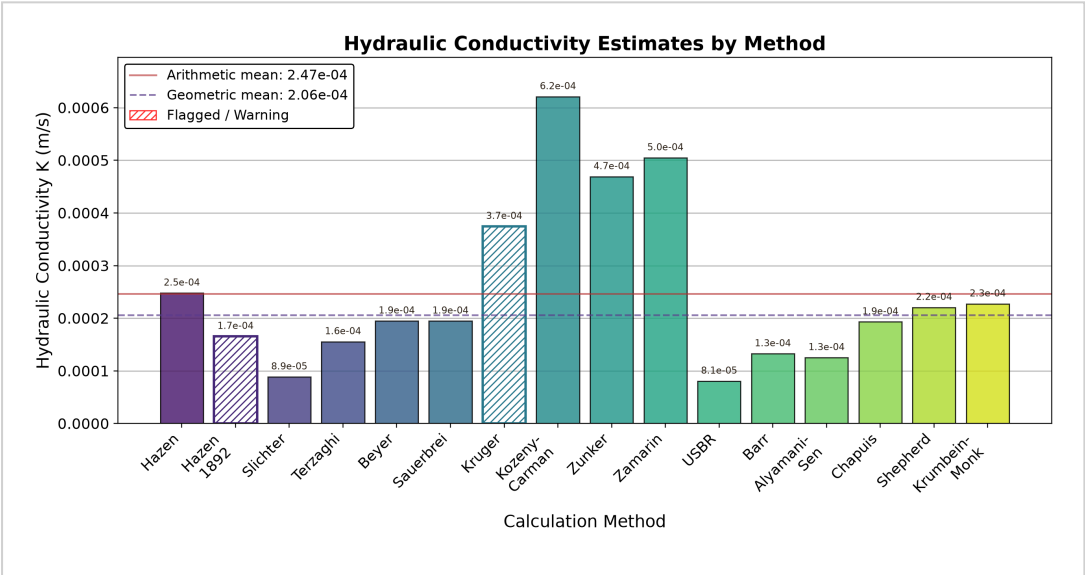
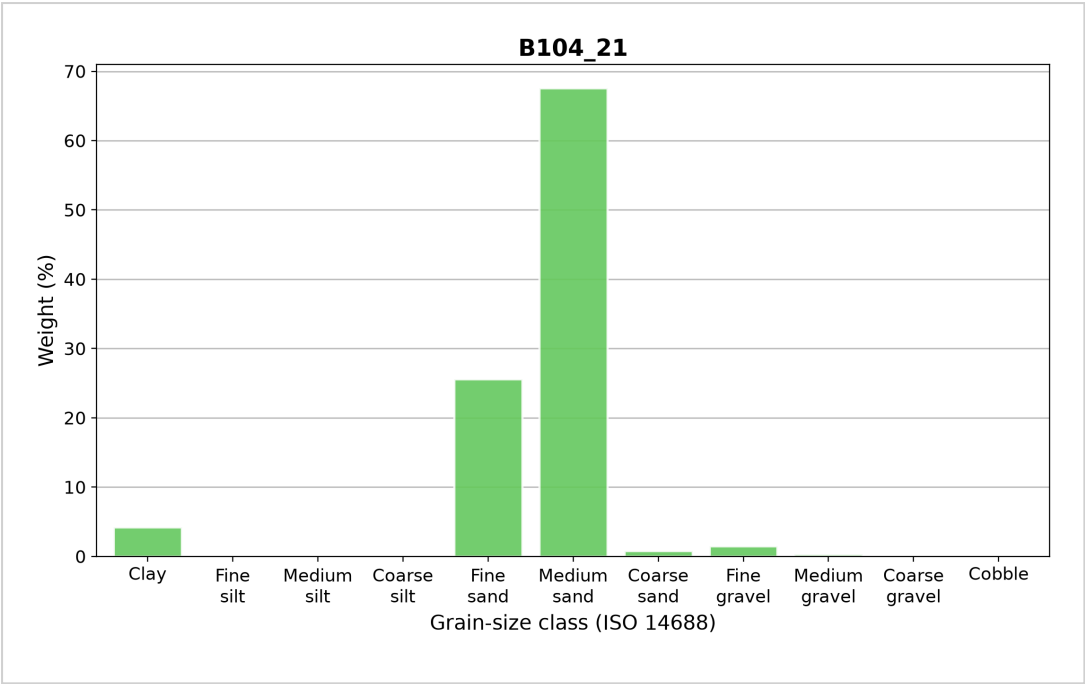
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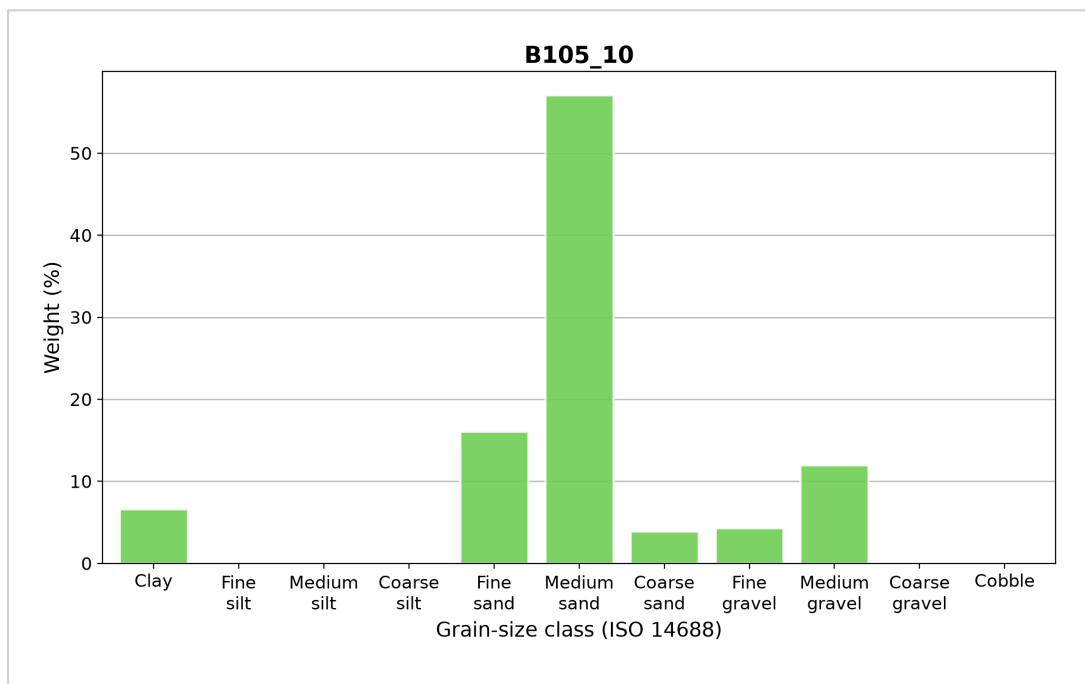
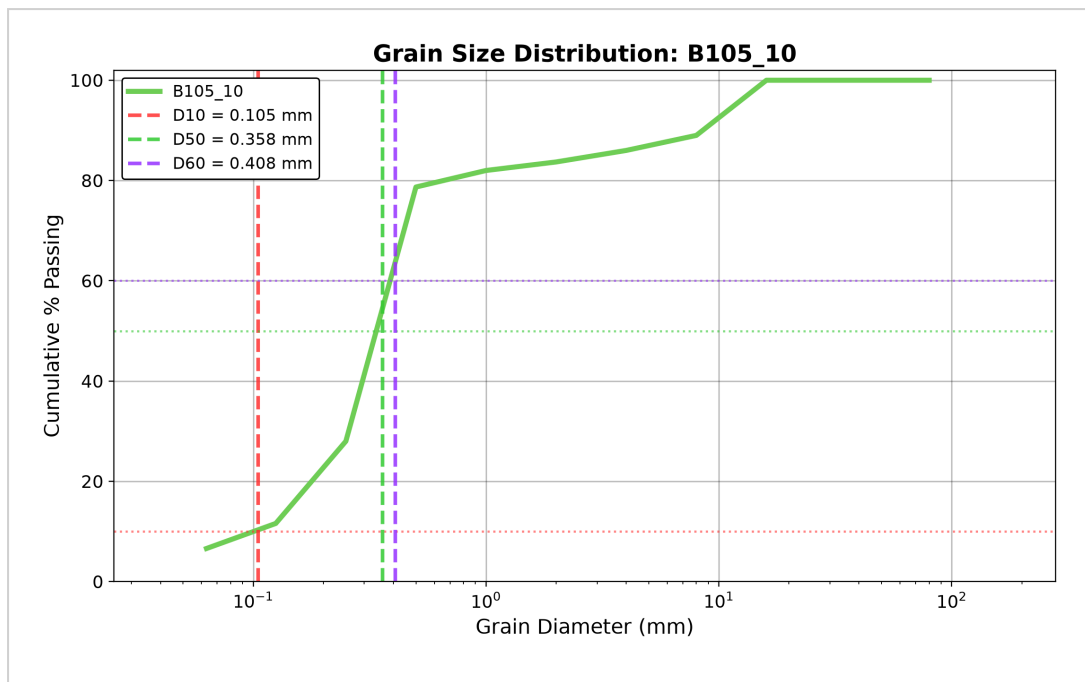


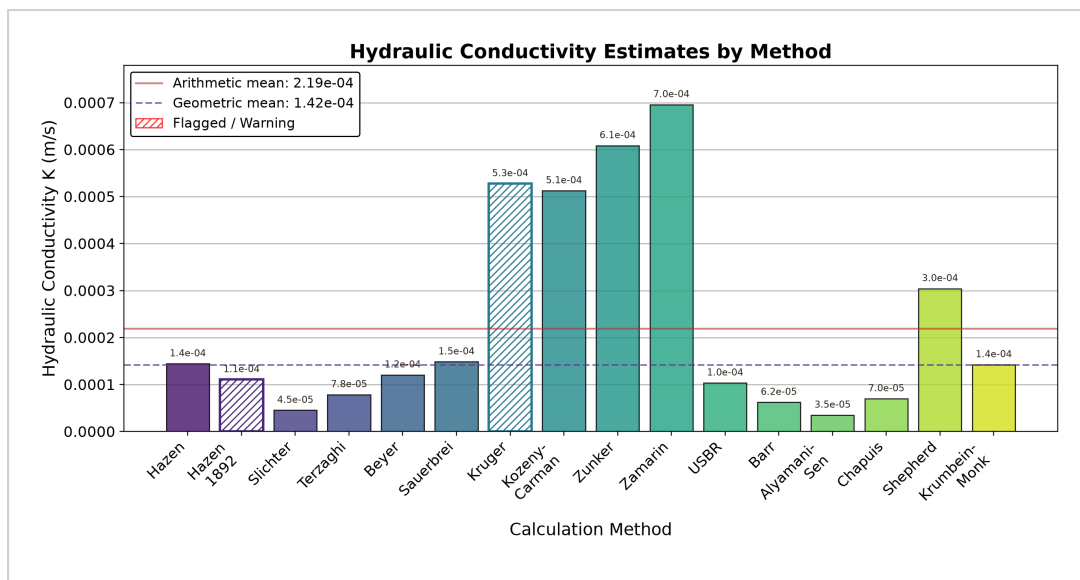
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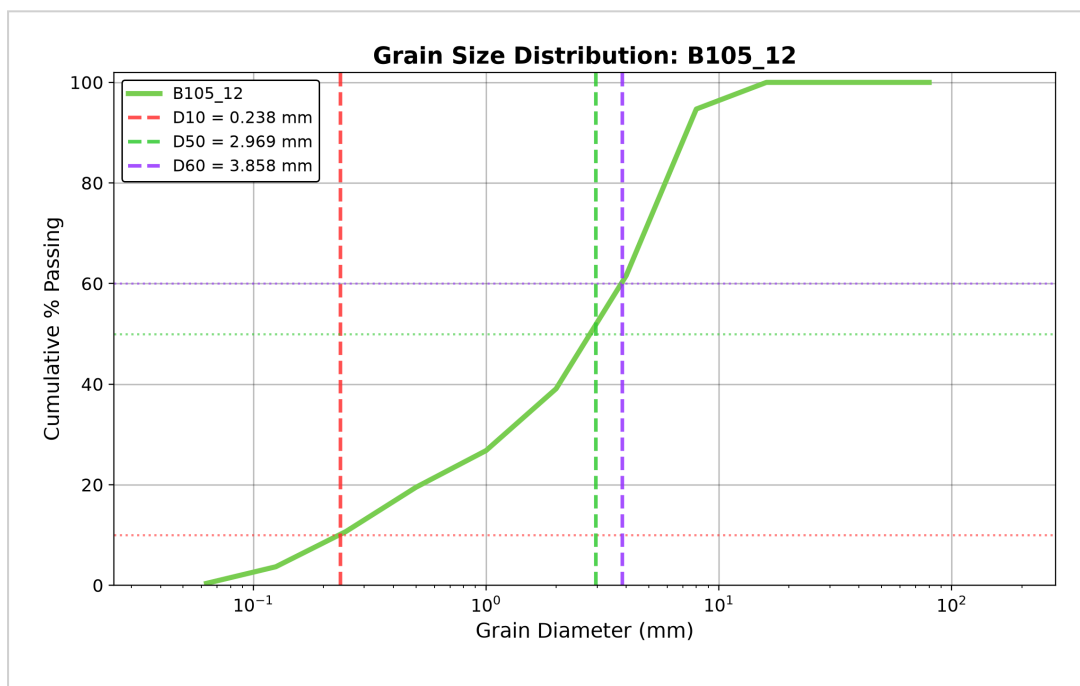


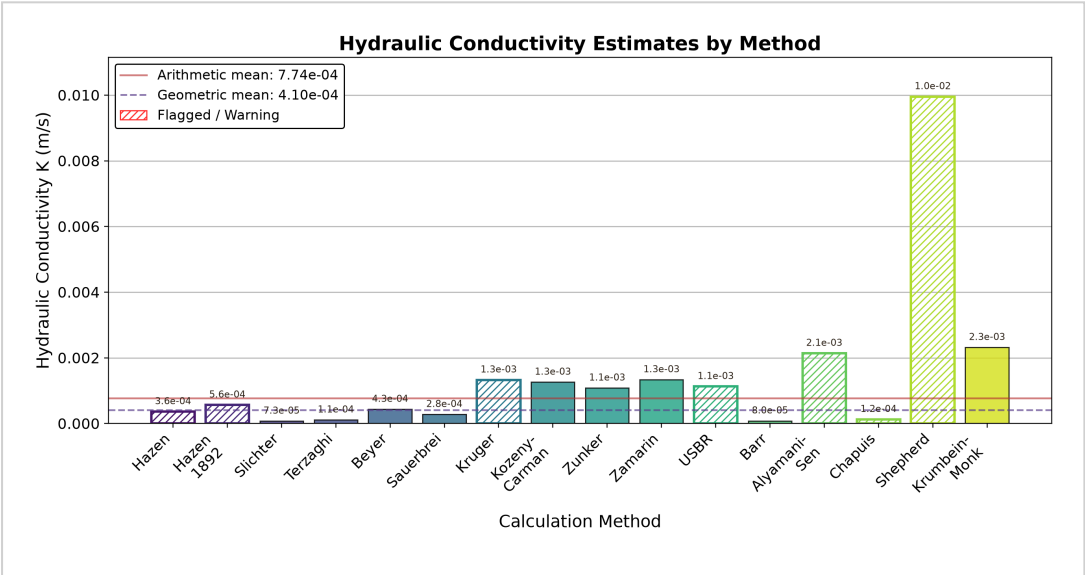
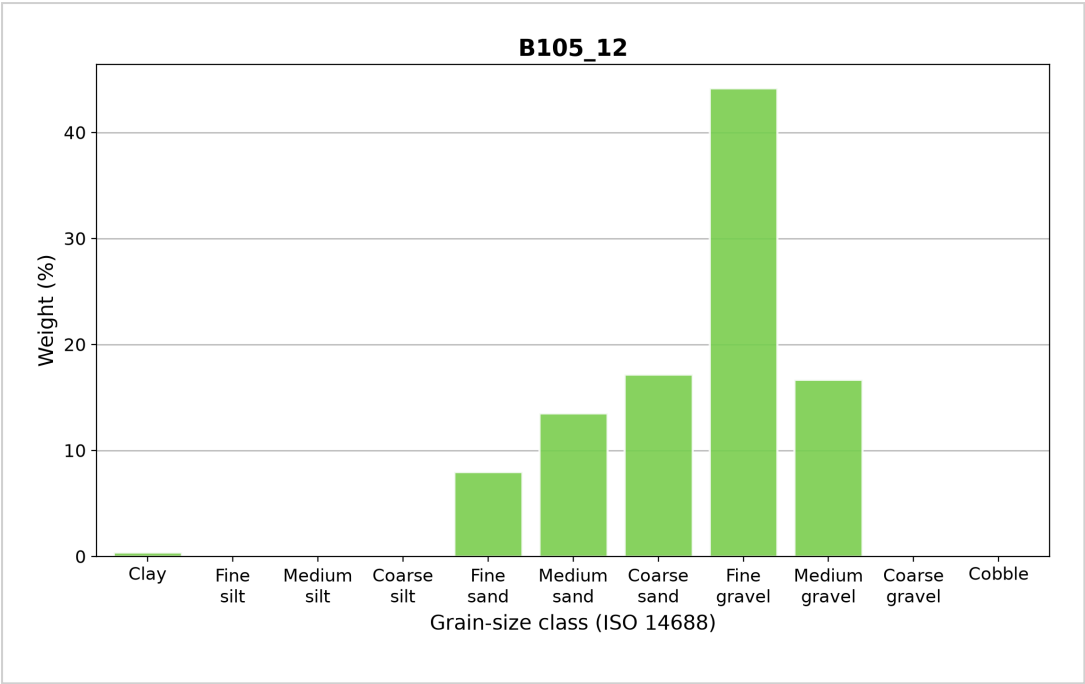
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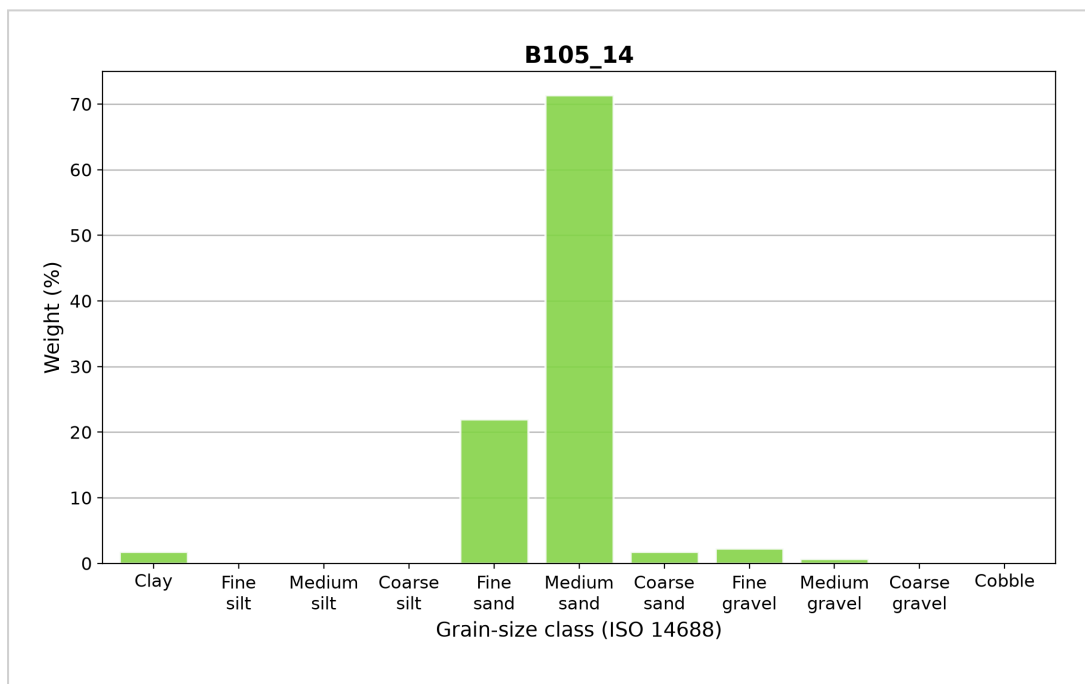
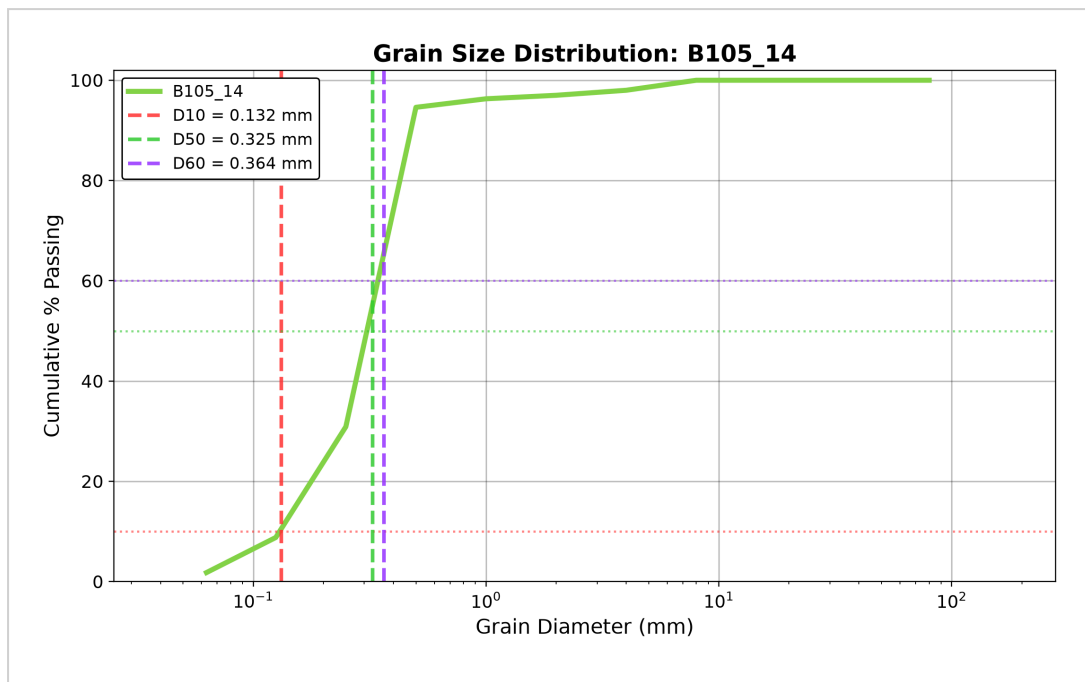


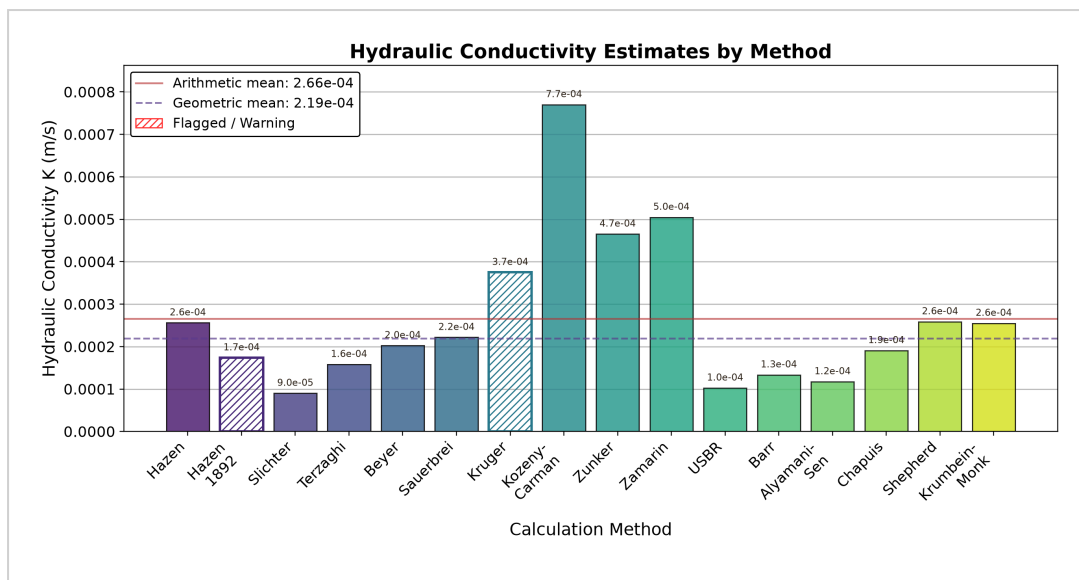
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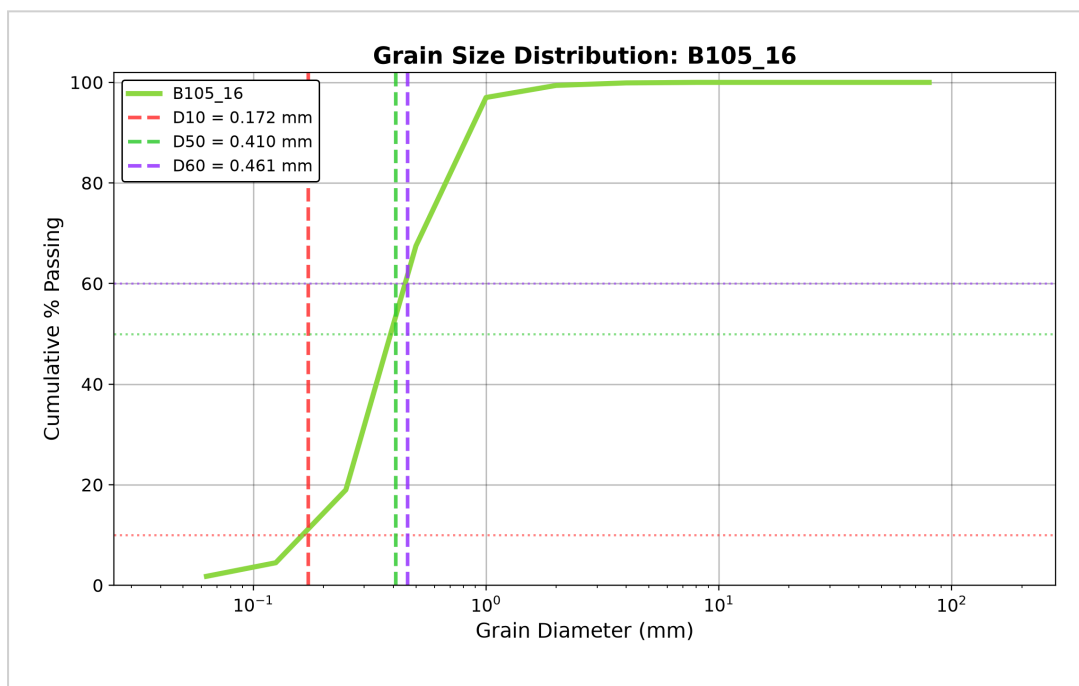


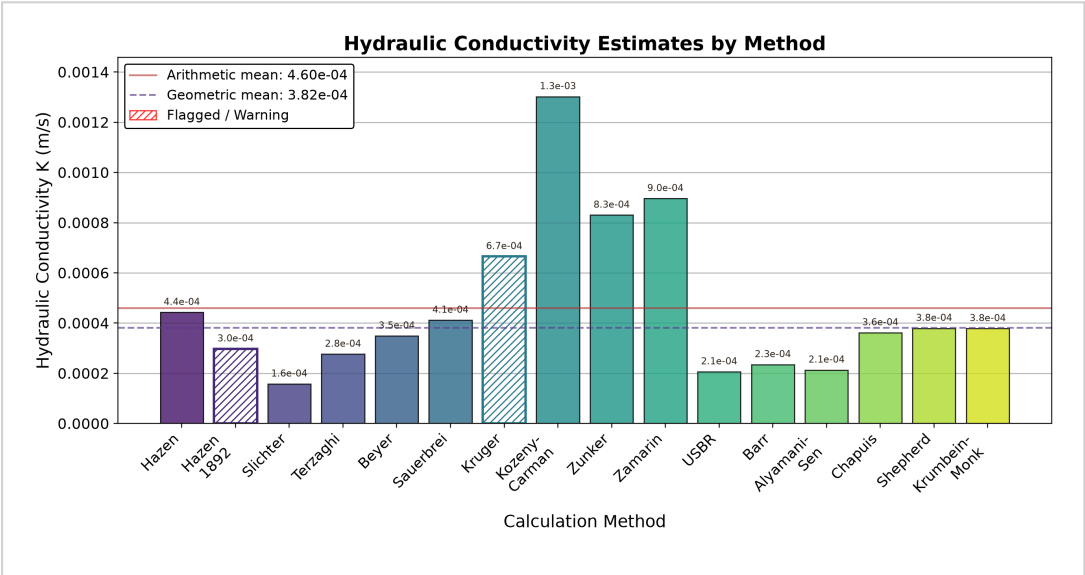
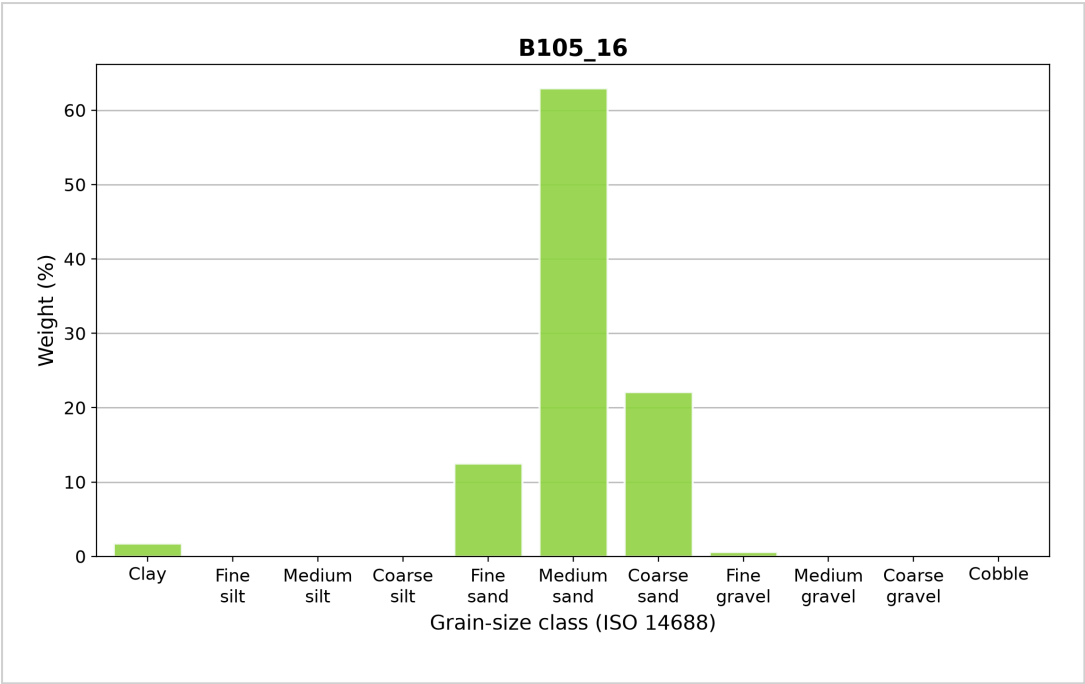
B105_14



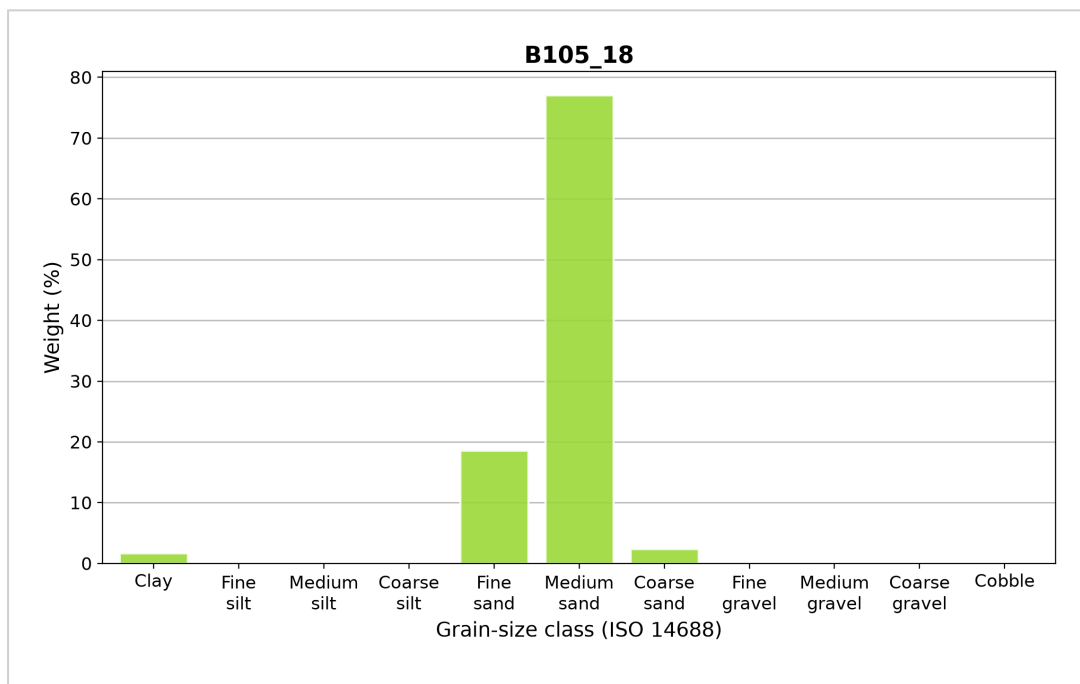
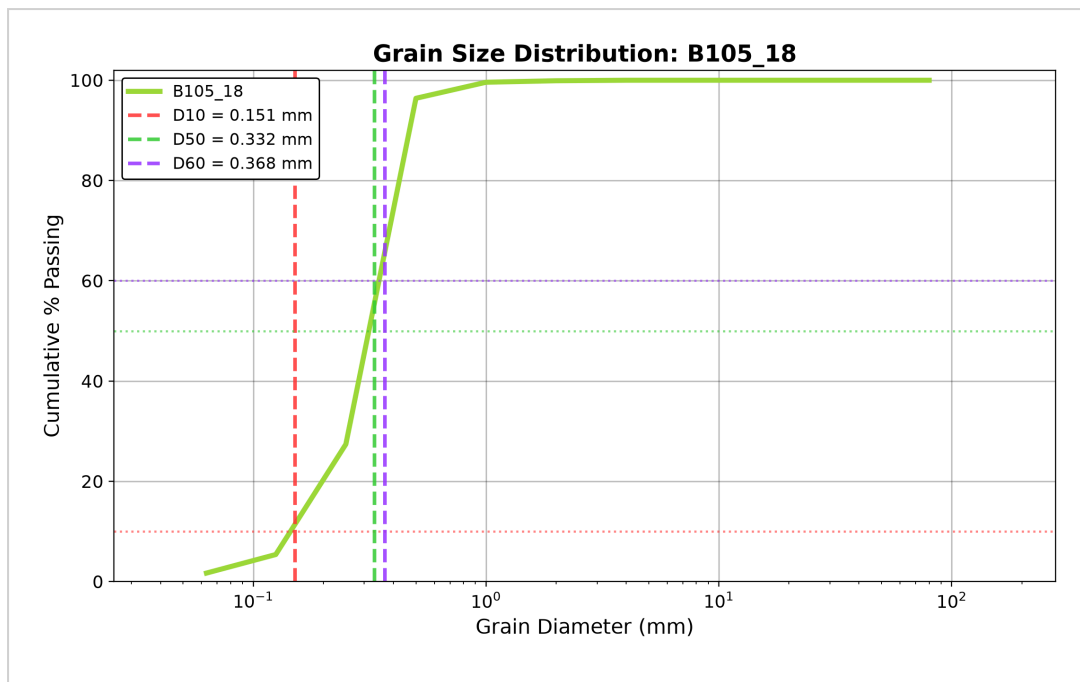


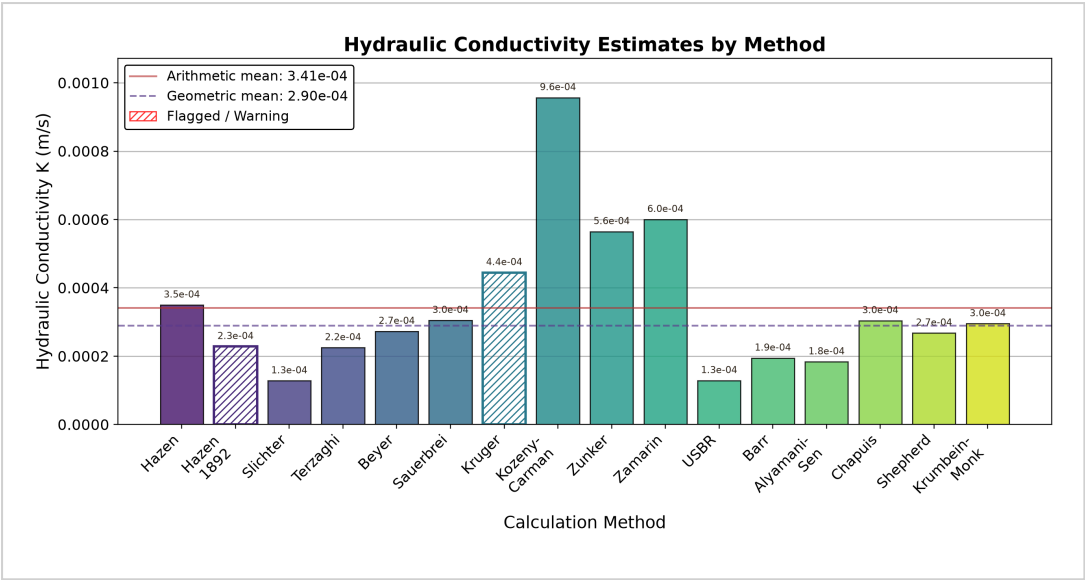
B105_16



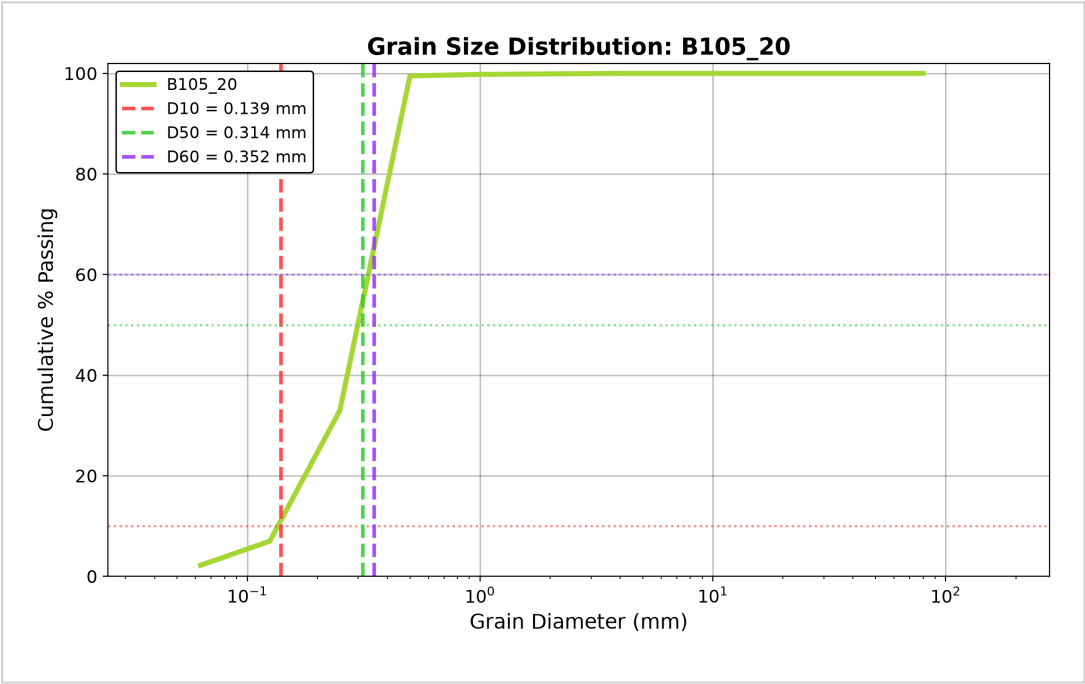


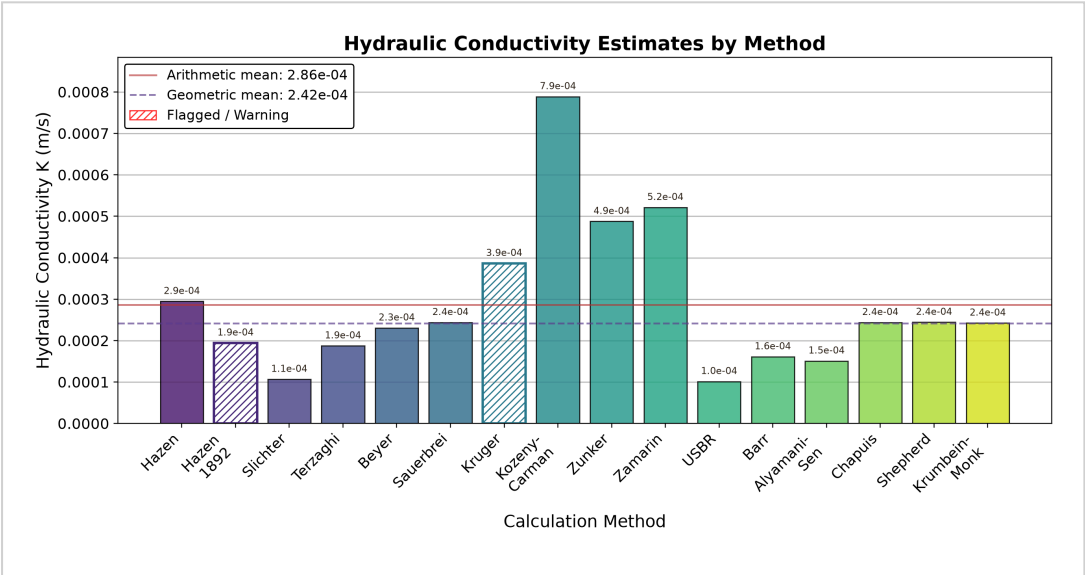
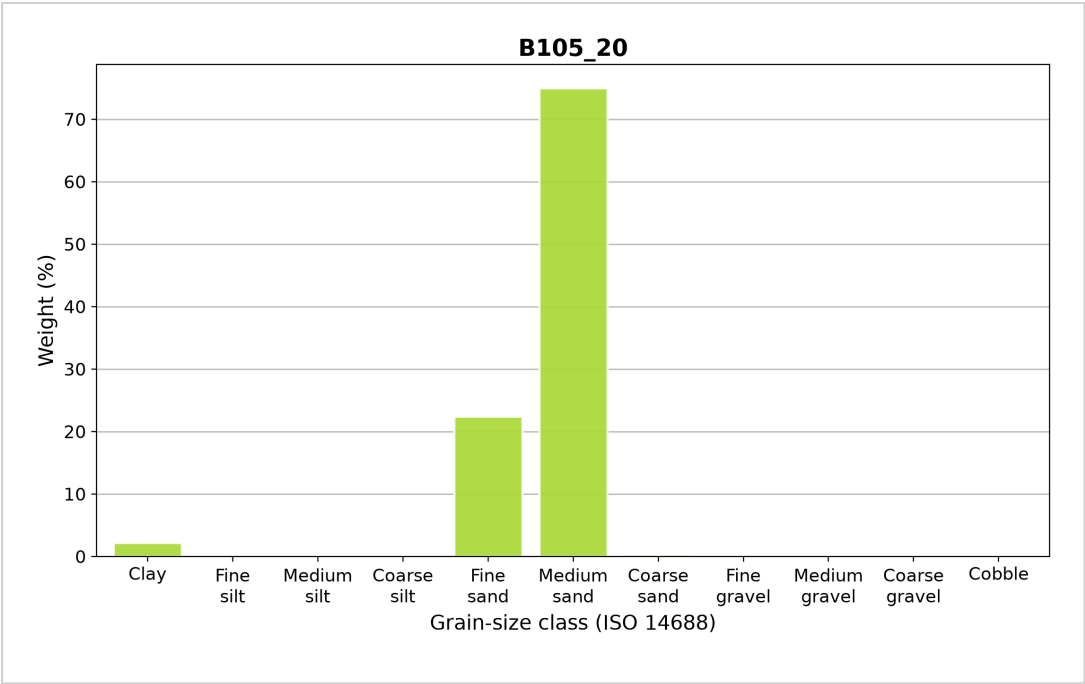
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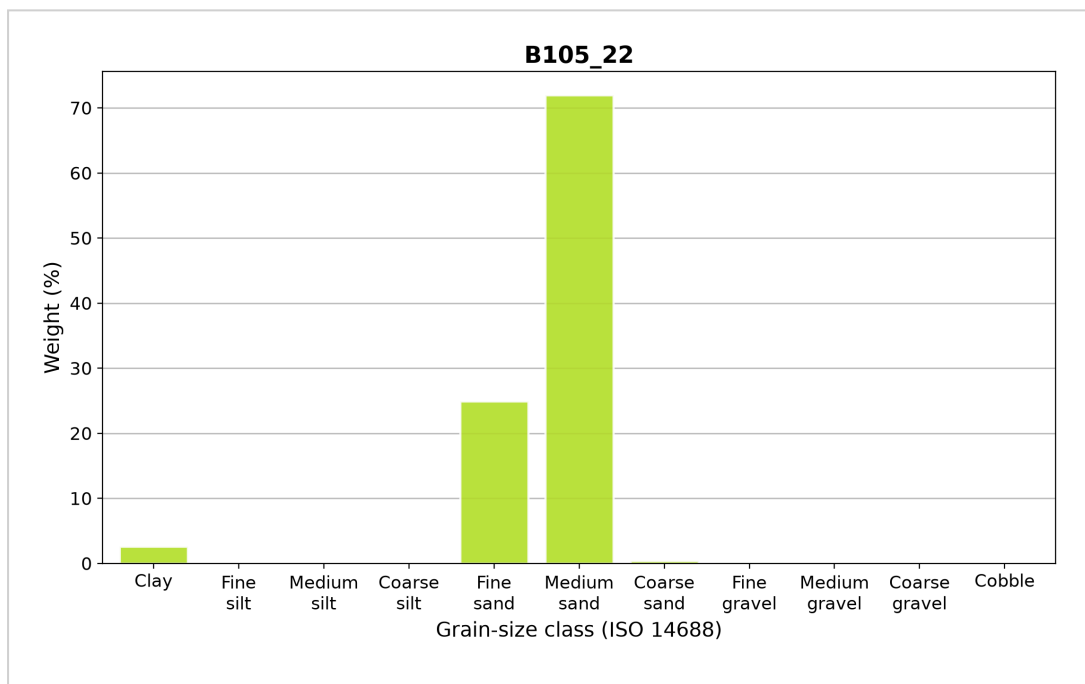
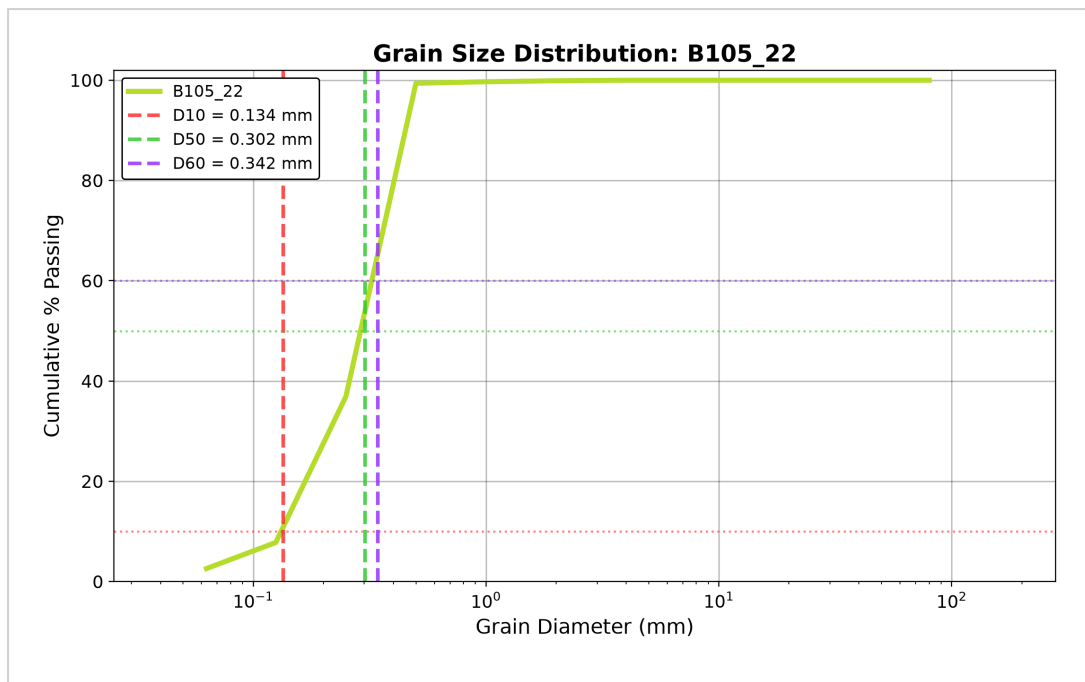


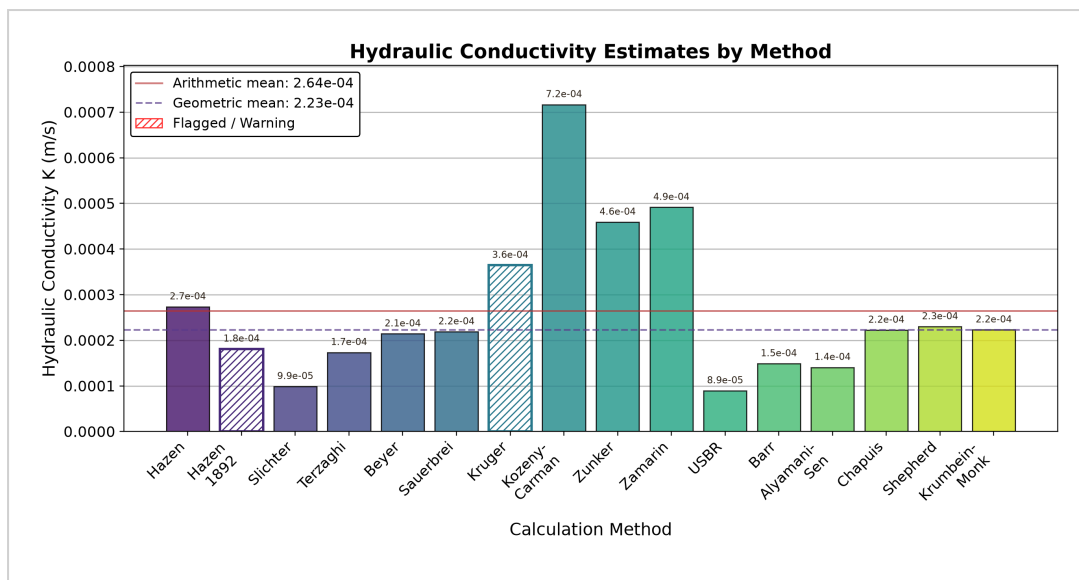
B105_20



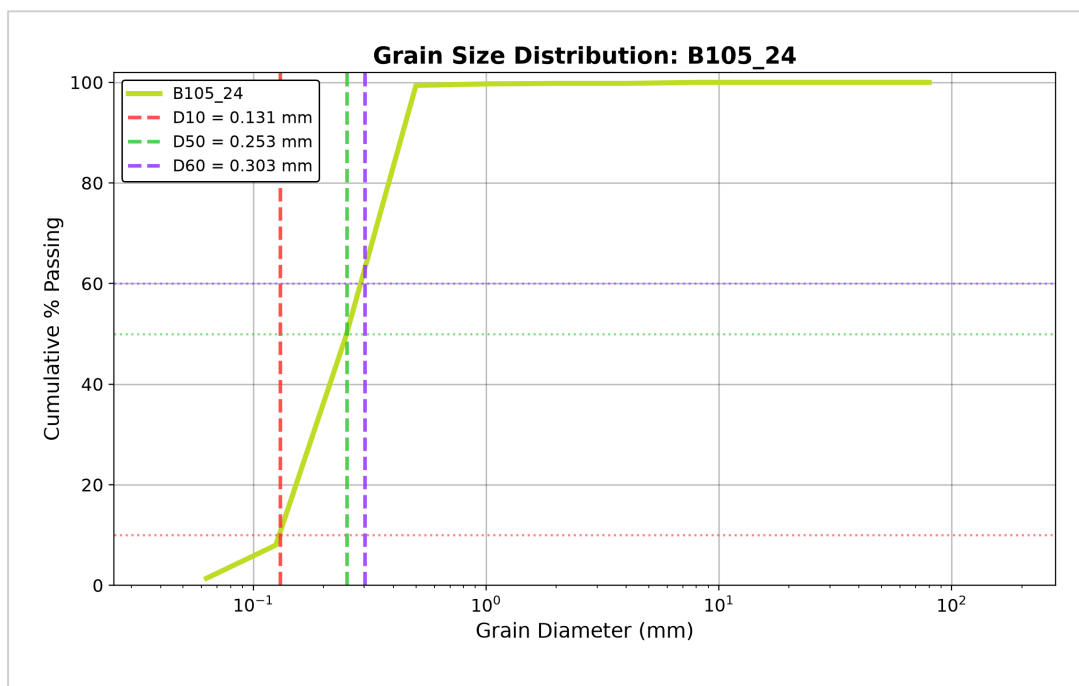


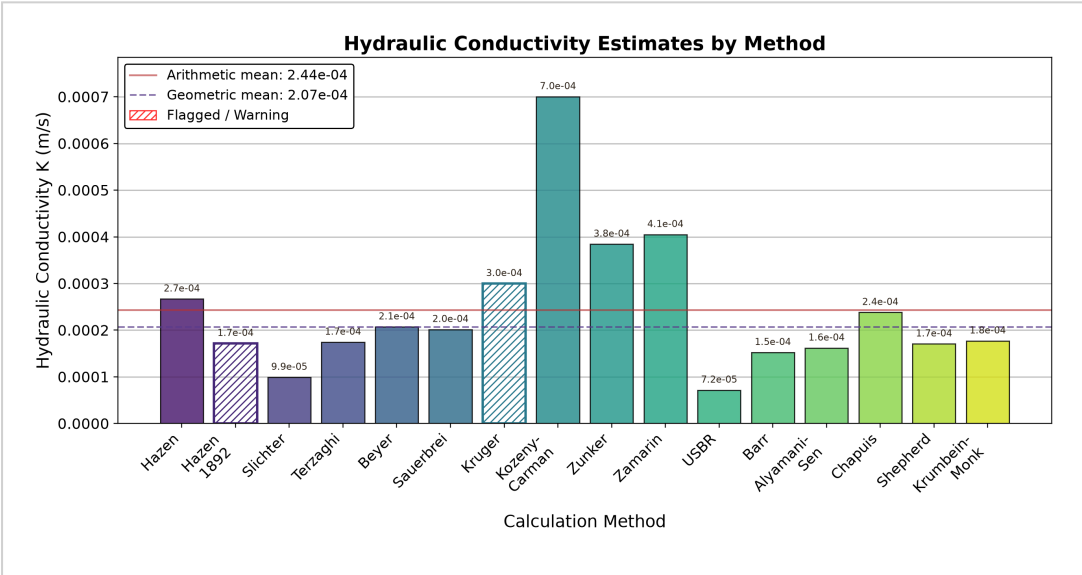
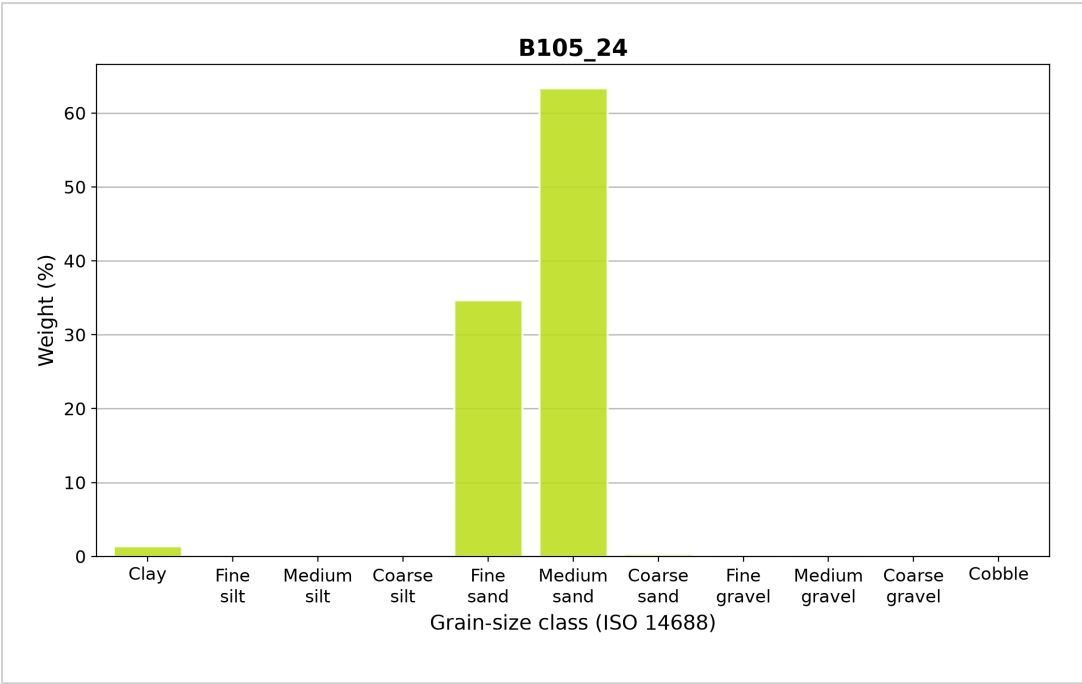
B105_22



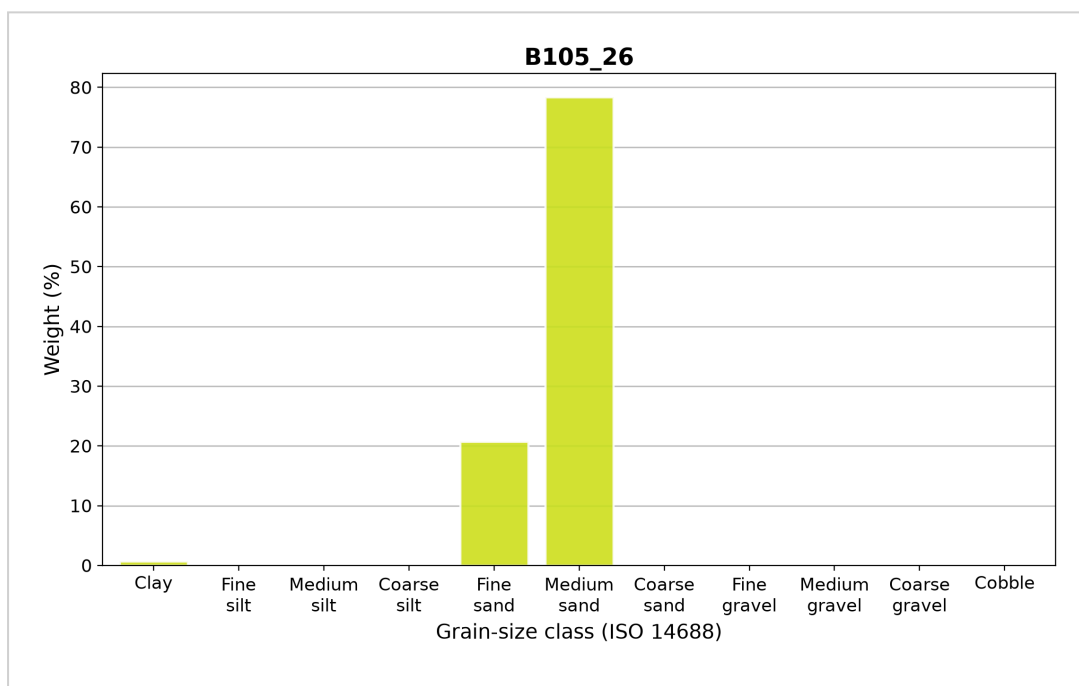
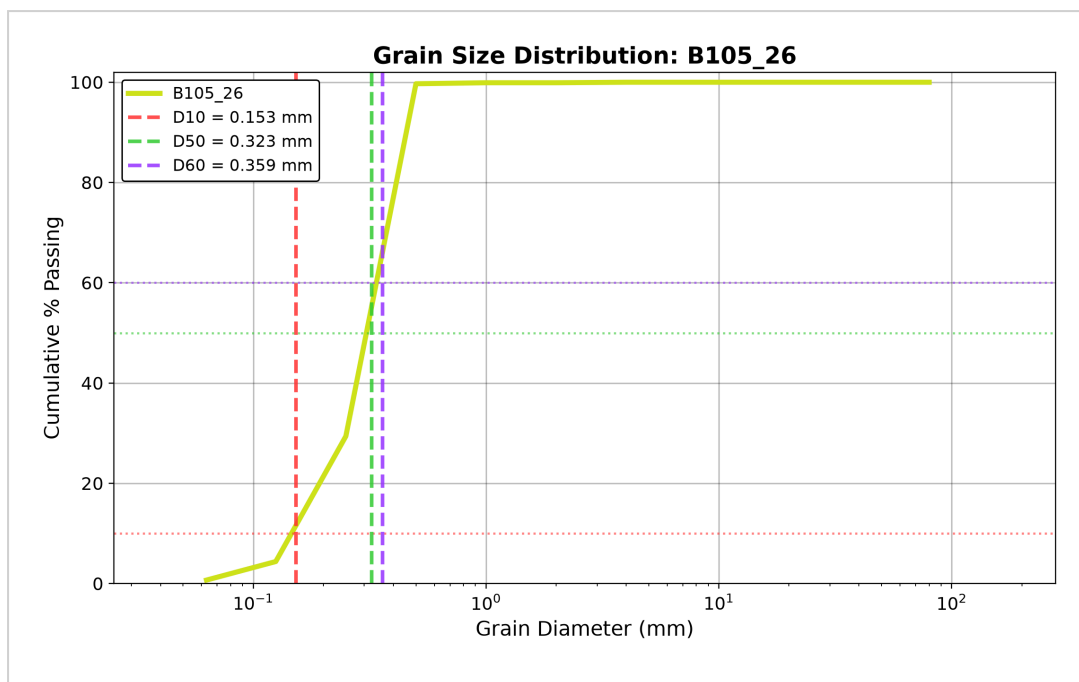


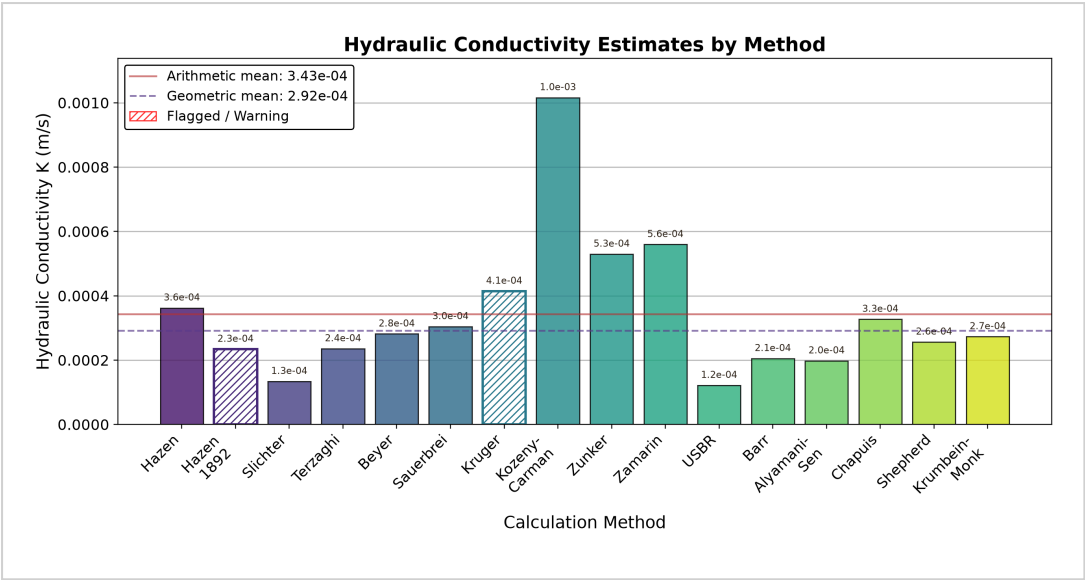
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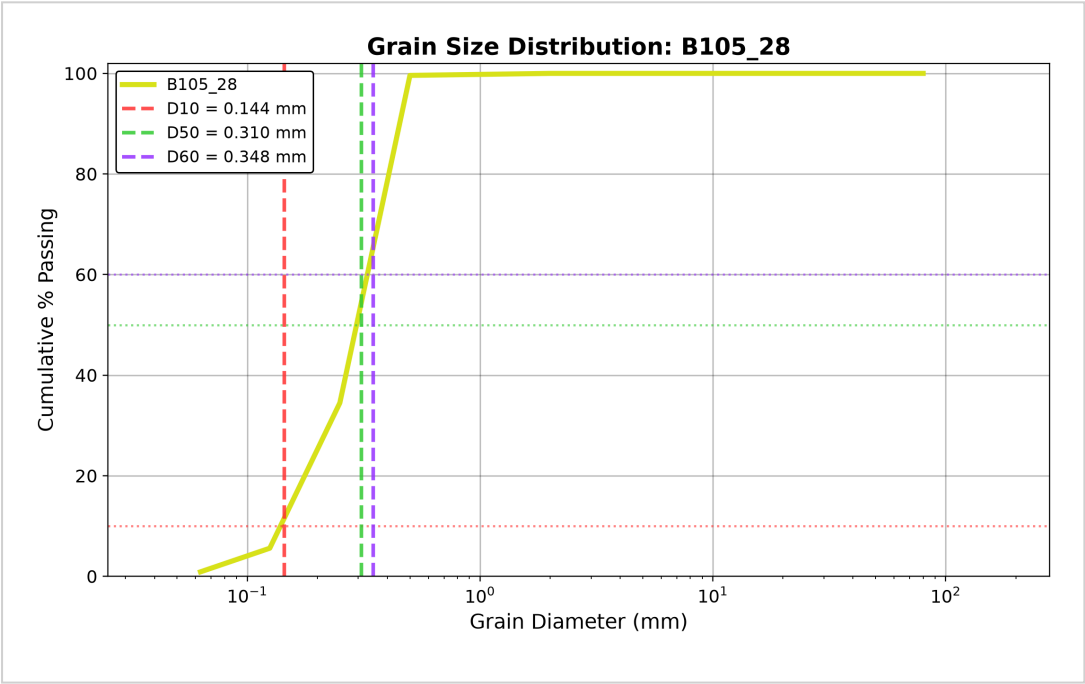


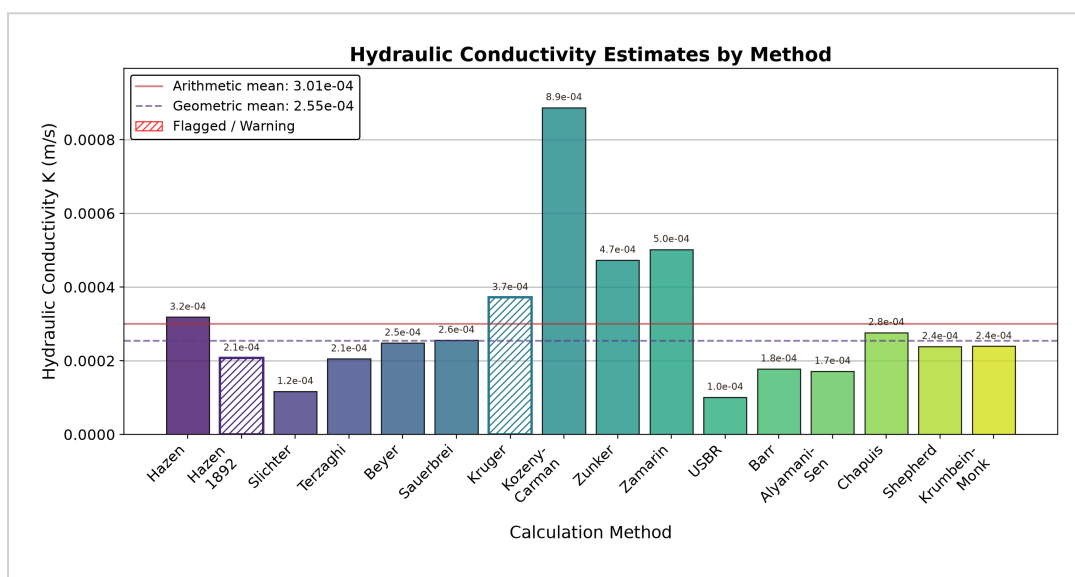
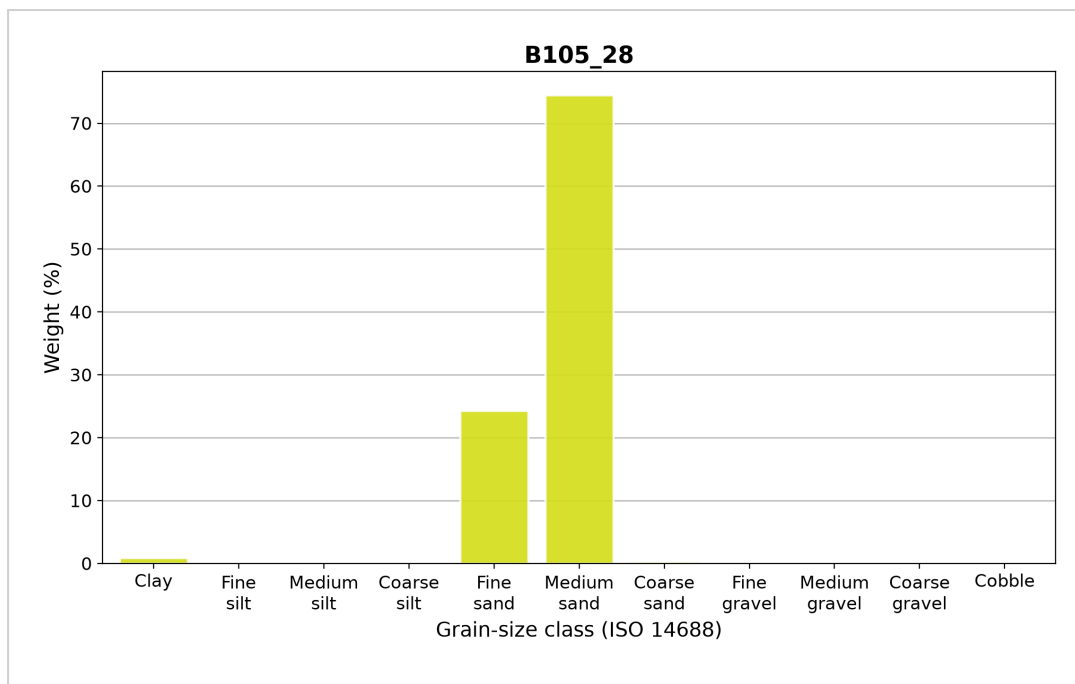
B105_26





B105_28





Interpretation & Discussion

Comparative Analysis

Permeability Characteristics:

- The highest permeability sample is B102_21 with $K = 7.93 \times 10^{-4}$ m/s, classified as High (Clean Sand).
- The lowest permeability sample is B101_13 with $K = 4.96 \times 10^{-5}$ m/s, classified as Moderate (Fine Sand).
- The 16.0-fold difference in permeability reflects the variability in grain size distribution among the samples.

Statistical Overview:

- Average sample K geometric mean: 2.12e-04 m/s
- Standard deviation: 1.35e-04 m/s
- Coefficient of variation: 63.6%